Translational Statistics and Dynamic Nomograms

PhD thesis

by

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Abstract

Translational Medicine, within biomedical and public health research domains, is defined as the convergence of basic and clinical research with the aim to transfer knowledge on the benefits and risks of therapies. The concept of Translational Statistics is proposed to facilitate the integration of biostatistics within clinical research to enhance communication of statistical research findings in an accurate and accessible manner to diverse audiences (e.g. policy makers, patients and the media). The use of appropriate visualisation is central to all areas of statistical research. Providing meaningful graphical representations of data is necessary to identify features about the population from which the data were sampled and may throw up an unsuspected view of the data such as a pattern or unusual observations. Informative graphical representations of statistical models play an important translational role. Static nomograms have been used to visualise statistical models. In this study, we propose the use of dynamic nomograms as a visualisation and translational tool to further aid the communication of the results of a statistical analysis to a non-statistical audience. A visualisation tool for time-to-event data is presented which contains a collection of useful graphical summaries, in particular, the Mean Residual Life function. It includes the classical survival summaries as well as the dynamic prediction for survival function and the mean residual life function as the two attractive alternatives. In theory, most regression-type models presented in the literature could have an accompanying web address to direct the reader to the corresponding dynamic nomogram allowing them to 'interact' with the model to gain insight into the effect of each explanatory variable on the primary response.

Chapter 1

Introduction

1.1 Introduction

Translational Medicine [143, 120] is a framework in the field of clinical and public health research that aims to promote the convergence of basic and clinical research disciplines in order to accelerate the discovery of new diagnostic tools and treatments using a multi-disciplinary 'bench to bedside' approach. According to Fontanarosa and DeAngelis [56], "effective translation of the new knowledge, mechanisms, and techniques generated by advances in basic science research into new approaches for prevention, diagnosis, and treatment of disease is essential for improving health and well-being".

In an analogous fashion, the term 'Translational Statistics' was introduced by Newell et al. [107] where the aim is to facilitate the integration of Biostatistics within clinical research and to enhance communication of research findings in a factual matter to diverse audiences (e.g. policy makers, patients, and the media). Using modern tools of knowledge transfer both clinicians and patients will be provided with tools and summaries to provide a better understanding of the advantages and risks of therapies. As statistical inferential methods become more computational, the models arising can be increasingly complex and difficult to interpret. There is a responsibility on the statistical community to aid in making the results not only reproducible but as understandable as possible, taking into account the target audience.

Model summaries are typically graphical and numerical (e.g. point estimates). The use of appropriate plots and the appropriate use of plots are central to all areas of statistical research and reporting of the subsequent findings. Providing meaningful graphical representations of data is necessary to identify features about the population from which the data were sampled and as graphical representations of data may highlight an unexpected view of the data (e.g. a pattern or unusual observations) that warrants investigation, or aid understanding.
Drawing appropriate graphs is necessary also to provide a way of assessing parts of any assumed statistical model before engaging in formal analysis and is crucial to the presentation and understanding of results and conclusions.
For example, the box-plot is the typical graphical summary used to display a continuous response variable, often as separate groups in studies with a between-subject design. Figure 1.1 displays box-plots for three different simulated responses. In each case, the box-plot suggests that the underlying distribution was a symmetric distribution for each response. The corresponding bean-plot [82] displays the raw data (as rugs), highlights the mean as a reference line and incorporates a smoother to provide a graphical representation of the likely probability density function. Based on the new information it is clear that symmetry is a reasonable assumption but normality is not except possibly for the third response. The sample size is highlighted through the 'rugs' as compared to the box-plot where all that can be extracted is that the sample size is at least 5 in each case.

An extension of the bean-plot is given in Figure 1.2 where smoothed density estimates are given back to back for the between-subject factor of interest, gender. The population of interest is Type 1 diabetics, and the response of interest is an anxiety score relating to their severity of diabetes. The smoothed density plots suggest a skewed distribution for males and females and a possible bimodal distribution for females which may be worth exploring further.

The choice of numerical summaries used is also critical in order to avoid common mistakes which arise when non-statisticians report some statistical results. For example, the (estimated) benefit of a particular treatment is often summarised using a mean or median (for continuous response or time to event data) and reported to

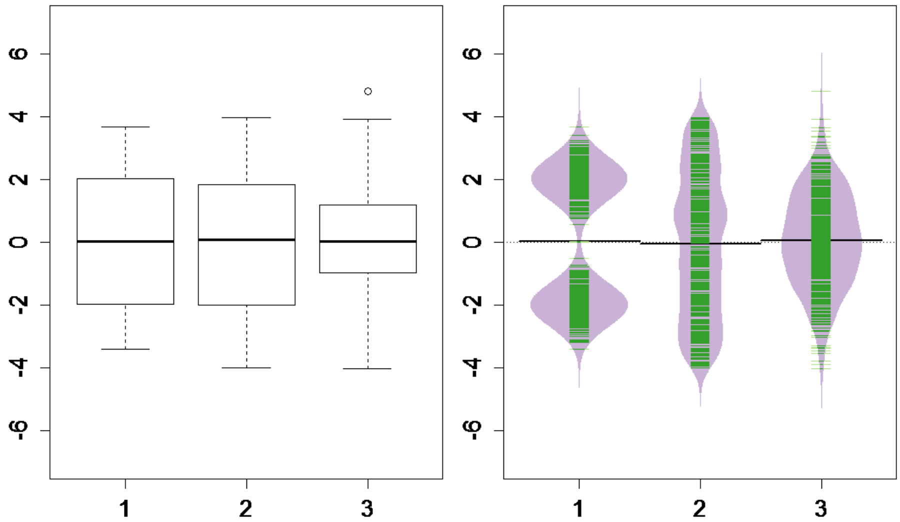

Figure 1.1: Graphical summary of three different continuous responses through a box-plot on the left, and a bean-plot on the right.

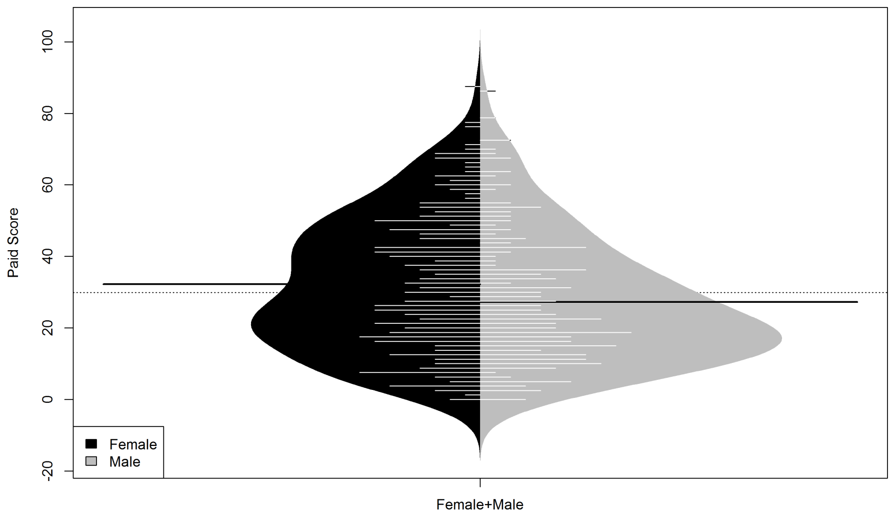

Figure 1.2: Smoothed density estimates of the overall PAID score by gender using a bean-plot.

the patient in this manner. It is quite likely that, at the patient level, this is the treatment effect that they will assume will apply to them when in fact it is highly unlikely to be the case. Providing an interval estimate for the parameter of interest, while including a measure of uncertainty, does not provide any additional information of relevance at the individual level, rather at a policy level (i.e. the effect, in general, of adopting this treatment in the population). One option is to include a tolerance interval, an interval (with an assigned degree of confidence) that is likely to capture a certain proportion of the population of interest.

For example in a particular randomised controlled trial, the estimated difference in the mean improvement (for the continuous response of interest) in the treatment compared to the control arm is 3.09 with a 95% confidence interval of $[0.05, 6.13]$ suggesting a significant treatment effect. The corresponding 95%/95% tolerance interval for the treatment arm ranges from -6.6 to 15.38 (with a mean of 4.39) suggesting that the actual improvement, at the subject level, could range from a lack of improvement of 6.6 units to a positive improvement of 15.4. It could be argued that the information given by the tolerance interval may better translate the results of the trial at the patient level than using an estimate relating to the population mean.

In time-to-event studies (i.e. survival analysis) the median survival is often used as a summary of the survival experience of a patient in the population. It is unlikely however that the median is relevant at a patient level. Stephen Gould in his paper "the median is not the message" [62] recounts his personal experience as a statistician suffering from abdominal mesothelioma, a rare and severe cancer, when given a diagnosis of having eight months left to live. On reading the literature, he discovered that the estimate corresponded to the median survival time from a right-skewed distribution and questioned the practice of reporting the median in isolation to a patient. The median gives an estimate of the survival time for half the cohort in question and is unlikely to apply to any individual.

Another example of this knowledge transfer is in classification problems, typically involving a binary outcome where the usual summary is the odds ratio. Probability and odds are two terms used to describe the possibility of an event happening. Although a high probability can be linked with a high odds value, they are not mathematically equivalent. The probability of an event ranges on the scale of 0 to 1 and (from a frequentist perspective) represents the number of events divided by the total number of possible events. It is also often referred to as 'chance' when talking about positive events and 'risk' for negative events. In contrast, an odd is the ratio of the probability of the event happening to the event not happening. It is typical in clinical research to compare two probabilities as a ratio of their odds often leading to misinterpretation as to what either underlying probability is that formed the ratio of the odds. It has been argued [138] that, when possible, a summary quoting the underlying probabilities is more informative than one based on ratios of odds or indeed of probabilities.

David Spiegelhalter presents one example of a misleading interpretation of probabilities and odds ratios arising from a study of the effect of statins on muscles and joints [57]. This article reports the risk of muscle weakening with and without statins of 87% versus 85%, which translate to odds of $\frac{0.87}{0.13} = 6.7$ and $\frac{0.85}{0.15} = 5.7$; the corresponding odds ratio is, therefore, $\frac{6.7}{5.7} = 1.18$. The risk ratio was $\frac{0.87}{0.85} = 1.02$, a 2% relative change, while the difference in absolute risks was $0.87 - 0.85 = 0.02(2\%)$. The media incorrectly interpreted the odds ratio of 1.18 that the statin in question raises the risk of muscle and joint weakness by up to 20%. Reporting the underlying probabilities alone may have avoided this confusion.

There are various examples in the literature where efforts are made by researchers to facilitate the interpretation of complicated models through the development of alternative ways of summarising population parameters. In epidemiology, the concept of population attributable fraction [50, 54] was introduced as a more easily interpretable measure as it aims to quantify the proportional reduction in disease prevalence that would be achieved if the risk factor could be excluded from the population.

The final result of any statistical analysis is invariably a model summary with estimated (conditional) effects, accompanying standard errors and p-values. For example, data were extracted from a large registry ($n = 6526$) in order to identify the importance of several explanatory variables on the risk of a woman delivering a child by caesarean section compared to a vaginal birth. The primary response is a binary variable indicating if a caesarean section was performed (yes/no). Also, the decision of which explanatory variables to include in the final model (age, BMI, diabetes type, baby's gender, ethnic group, nulliparous, macrosomia, preeclampsia and pregnancy-induced hypertension) was made a-priori by a panel of experts. A logistic regression model was used to estimate the effect of each explanatory variable on the (log) odds of birth by caesarean section (Table 1.1).

The model summary aims to make it clear to the reader as to the effect each explanatory variable (in a fixed effects only model in this example) has on the probability

	Estimate	Odds ratio	Std. Error	z-value	p-value
Intercept	-4.4129	0.012	0.2563	-17.22	<0.0001
Diabetes (gestational)	0.2383	1.269	0.0770	3.10	0.0020
Diabetes (pre-gestational)	1.5968	4.937	0.1457	10.96	<0.0001
Age	0.0495	1.051	0.0064	7.72	<0.0001
BMI	0.0667	1.069	0.0058	11.48	<0.0001
Nulliparous (yes)	-0.4014	0.669	0.0679	-5.91	<0.0001
Macrosomia (yes)	0.0545	1.056	0.0803	0.68	0.4975
Preeclampsia (yes)	0.6985	2.011	0.1672	4.18	<0.0001
PIH (yes)	0.0101	1.010	0.1251	0.08	0.9356
Sex of baby (male)	0.1104	1.117	0.0631	1.75	0.0804
Ethnic (non-european)	0.2770	1.319	0.1098	2.52	0.0117

Table 1.1: The model summary fitted to predict the risk of caesarean section

of the outcome of interest. The p-values are of limited value given the large sample size, the estimates of the regression coefficients are on the log odds scale, and their importance should not be inferred by their magnitude as they are a function of the scale of the corresponding covariate, and may be misleading in the presence of multicollinearity.

This summary might not be the most informative way to present the results of a model to answer the typical questions of interest namely "which explanatory variable has the biggest effect on the probability (and not just on the odds) of having a caesarean section?", or in, "how is the probability affected by changes in modifiable risk factors?".

In the remaining chapters of this thesis new graphical techniques, numerical summaries and a range of software solutions will be presented to aid translation of complex statistical models using the following datasets as illustrative examples.

1.2 Datasets

In this section, six separate datasets are introduced which will be used as illustrative examples to illustrate the methods and applications through out the thesis.

1.2.1 Titanic data

The sinking of the Titanic was one of the greatest shipping tragedies of the 20th century. Data are now available (in the 'PASWR' package in **R** [14]) on the survival outcome of 1309 of the 1324 passengers on the Titanic. The dataset includes passenger information such as age, sex and the ticket class. The primary response 'survival' is a binary response which indicates if the passenger survived the Titanic disaster or not ('No' or 'Yes'). The 'age' variable indicates the age of passengers in years; the ages of children under one year are represented as a fraction of the year. The 'pclass' variable refers to passenger ticket class which includes three levels ('1st' for the upper class, '2nd' for the middle class or '3rd' for the lower class). The gender of passengers is recorded in 'sex' ('male' or 'female'). The primary question of interest is to explore the effects of 'age', 'pclass' and 'sex' on the probability of survival.

Characteristics		Total passengers	Survived	Not survived
Pclass		1309		
	1st n(%)		200 (61.9%)	123 (38.1%)
	2nd n(%)		119 (43.0%)	158 (57.0%)
	3rd n(%)		181 (25.5%)	528 (74.5%)
Sex		1309		
	female n(%)		339 (72.7%)	127 (27.3%)
	male n(%)		161 (19.1%)	682 (80.9%)
Age	mean (SD)	1046	28.92 (15.06)	30.55 (13.92)

Table 1.2: Descriptive statistics for Titanic data covariates

In Table 1.2 summary statistics of each explanatory variable by survival is shown; it includes the count and percentage for factors ('pclass' and 'sex') and the mean and standard deviation (SD) for 'age'.

The proportion surviving appears to depend on age, gender and ticket class. Novel graphical techniques will be introduced in Chapter 2 to explore this underlying relationship further.

1.2.2 Horseshoe crabs data

The Horseshoe crabs data comprises information measured on 173 female horseshoe crabs, each of which had at least one male crab attached to each of them in their nest. The study investigated factors affecting the number of male partners in addition to the female's primary partner ('Satellites'). The explanatory variables include the female crabs width ('Width'), a binary factor indicating whether the female has dark colouring ('Dark') and a binary factor indicating whether the female had good spine condition ('GoodSpine'). The response variable is the number of male partners in addition to the female's primary partner ('Satellites'). The data are available in the in the `glm2` package in **R** [96] derived from Agresti (2007) [5]. These data will be used to illustrate methods to best understand the role of a covariate on a response variable which follows a Poisson distribution.

1.2.3 Lung cancer data

The Lung cancer data in the `survival` package [131] includes time to death of 228 patients with advanced lung cancer. The study aims to investigate the effect of explanatory variables on the survival of lung cancer patients. The explanatory variables are age ('age'), gender ('sex'), the amount of weight loss in last six months ('wt.loss') and the ECOG performance score ('ph.ecog' which is a categorical variable varying from 0 = *good* to 5 = *dead*). The data is a random sample from a larger registry from a study on the advanced colorectal or lung cancer patients [93]. These data will be used to illustrate novel methods to summarise and visualise time to event data.

1.2.4 Ragweed data

Ragweed is a plant, prevalent throughout North America, that produces pollen which often causes allergic reactions. The 'Ragweed' dataset [128] were recorded during the 1993 ragweed season in Kalamazoo, Michigan. This study aims to predict the ragweed pollen level ('ragweed') using the 'temperature', 'rain', wind speed forecast ('wind') and pollen season day ('day.in.seas'). These data are used as an example in Rupert and Wand's text on semiparametric regression [122] and will be used in this thesis as an illustrative example of how to translate results from a semi-parametric regression using a transformed response variable.

1.2.5 Bronchopulmonary dysplasia (BPD) data

The Bronchopulmonary dysplasia (BPD) data presented in the 'Applied Survival Analysis: Regression Modeling of Time to Event Data' book by Hosmer and Lemeshow (2008) [76]. The primary goal of the study is to determine factors which predict the length of time low birth weight infants (< 1500 grammes) with bronchopulmonary dysplasia (BPD) were treated with oxygen [76].

A total of 78 infants are included in the study, with 35 receiving surfactant replacement therapy and 43 not receiving this treatment. The time-to-event variable is defined as the total number of days (we converted it into months for simpler interpretation) the infant required the supplemental oxygen therapy, where a censored observation corresponds to an infant who was still on oxygen at the time of their last follow-up visit. The data are available on the Wiley's FTP website `ftp.wiley.com/public/sci_tech_med/survival`, and will be used in this thesis to illustrate the use of the dynamic prediction and the mean residual life functions as alternative summaries.

1.2.6 Sleep to Lower Elevated Blood Pressure: A Randomized Controlled Trial (SLEPT)

A randomised-controlled, two-group, parallel, blinded, single-centre, Phase II trial of 134 participants was designed to assess the likely change in mean 24-hour ambulatory systolic blood pressure between baseline and 8-week follow-up. Hypertension and sleep quality (i.e. deprivation) were the co-primary outcome measures because of their association with both poor sleep quality and cardiovascular disease [98].

The population of interest were subjects with hypertension (systolic blood pressure, 130 to 160 mm Hg; diastolic blood pressure, $<$110 mm Hg) and poor sleep quality in a community setting. A multi-component online sleep intervention consisting of sleep information, sleep hygiene education and cognitive behavioural therapy was assigned to the intervention group while the control arm was given a standardised cardiovascular risk factor and lifestyle-education session (usual care). These data will be used as an illustration of how dynamic graphics can be embedded into a scientific publication.

1.3 Thesis Structure

This thesis aims to extend the concept of translational statistics by introducing new numerical, graphical and model summary tools. The journey through all statistical analyses is very similar; it starts with a research question, a suitable design is employed, data are collected, graphical and numerical summaries are generated, an appropriate statistical model is fitted, and the results are written up often in different ways depending on the audience (e.g. study report or as a statistical paper). The post design and data collection stages can be classified into three principal steps namely *Visualise*, *Model* and *Translate*. The remainder of this thesis has these three steps as section headings with relevant chapters appearing under each section in addition to a 'conclusions and further work' chapter.

Chapter 2: Visualise

Visualisations comprise any graphics, images or animations which help to deliver a message. In this chapter, an overview of the various visualisation techniques typically used in statistics together with some examples of the modern advances in dynamic visualisations will be presented. This chapter will motivate the importance of using well-designed graphics, where we will consider static and interactive graphics. The structure of a static graphic will be discussed, and some modern graphical packages introduced such as 'Tableau' software and '`plotly`', '`flexdashboard`' and '`shiny`' packages in **R** to produce interactive graphics. Finally, the use of interactive graphics will be illustrated using several examples from the datasets introduced.

Chapter 3: Model

Chapter 3 ('Model') presents a succinct summary of the generalised linear model framework and the Cox proportional hazards model using examples of models with a continuous, binary, count and time-to-event responses. Specific attention will be paid to summarise time-to-event data. The survival function will be reviewed as it is the standard summary of time-to-event data for decades and suggestions made to make it more easier to translate. The dynamic prediction of survival and the mean residual life functions will be introduced as new alternative summaries.

Chapter 4: Simulate

In Chapter 4, we explore and present methods to estimate the uncertainty in the mean residual life function, in particular, the use of computational inference and resampling techniques. The results of a comprehensive simulation study will be presented and discussed to identify the suitability of the proposed approach.

Chapter 5: Translate

Chapter 5 ('Translate') will focus exclusively on translational tools, in particular, when summarising time-to-event data and visualising statistical models. Dynamic nomograms will be introduced as a novel tool to use as a translational tool to communicate the results of a statistical analysis to a non-statistical audience. The '`DynNom`' package written in **R** [118] will be presented which allows the creation and deployment of a dynamic nomogram in a user-friendly manner. By ways of illustration, dynamic nomograms will be used to visualise all the illustrative datasets including examples of modelling continuous, binary, count and time-to-event responses.

Chapter 2

Visualise

2.1 Introduction

Visualising data through the use of statistical graphics is a "transdisciplinary field informed by scientific, statistical, computing, aesthetic, psychological and sociological considerations" [38]. Data visualization is a general term used for building graphics, images, charts or animations to communicate a message. It is used by many disciplines such as science, education, engineering, medicine, etc. The diagrams, information graphics, concept maps, graphs, flow charts are some examples of well-understood visualisation tools.

A well-formatted graph should be easily and quickly comprehensible compared to summaries of data presented as numbers in a table. John Wilder Tukey (1915-2000), summarised the true purpose of graphical display in his paper [134] as:

1 Graphics are for the qualitative/descriptive—conceivably the semiquantitative—never for the carefully quantitative (tables do that better).

2 Graphics are for comparison—comparison of one kind or another—not for access to individual amounts.

3 Graphics are for impact—interocular impact if possible, swinging-finger impact if that is the best one can do, or impact for the unexpected as a minimum—but almost never for something that has to be worked at hard to be perceived.

4 Finally, graphics should report the results of careful data analysis—rather than be an attempt to replace it. (Exploration—to guide data analysis—can make essential interim use of graphics, but unless we are describing the exploration process rather than its results, the final graphic should build on the data analysis rather than the reverse.)

Graphics are an essential part of the process of data analysis. It is unlikely nowadays that graphics are created by hand, which was not the case up to 40 years ago. For example, Figure 2.1 is a photograph from the Gossett archive in the Guinness Store House in Dublin of a graph created by Gossett to investigate the effect of a particular marketing campaign on the sales in Scotland compared to Northern England. The variable represented is the percentage of sales of the corresponding month of the previous year coloured by region with reference markers for the time the advertisement campaign began. No doubt a lot of time, thought and effort went into the creation of such a graphic.

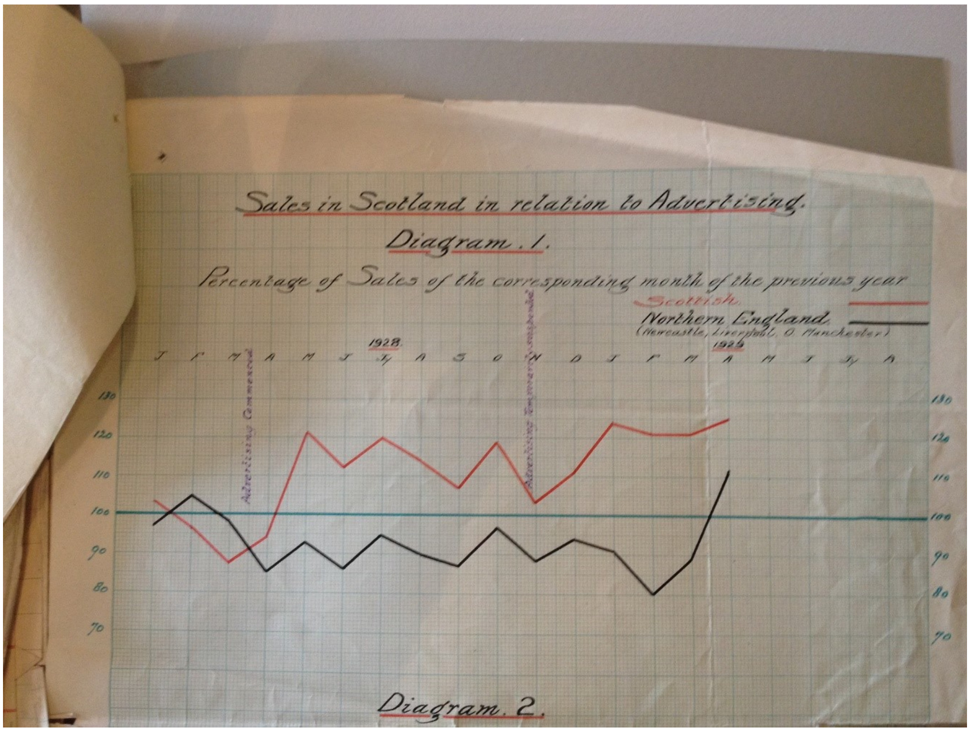

Figure 2.1: A handmade graphic of Guinness sale in Scotland in relation to advertising

Regardless of the method used to create a visualisation, the key objective remains the same; to give new insights into data by conveying the complex information clearly and efficiently. The correct use of statistics (the discipline) is needed to guide as to what should be plotted while cognisant of the intended audience.

For example, when analysing a continuous response from a two-sample paired problem a labelled scatter-plot with a line of equality allows the reader to gain an understanding of the likely treatment effect and whether the effect depends on the baseline. The data presented in Figure 2.2 relate to a sample of 30 osteoporotic women where the aim was to investigate whether or not an exercise regime can significantly improve, on average, balance over a two-month period of exercise over any natural improvement of controls.

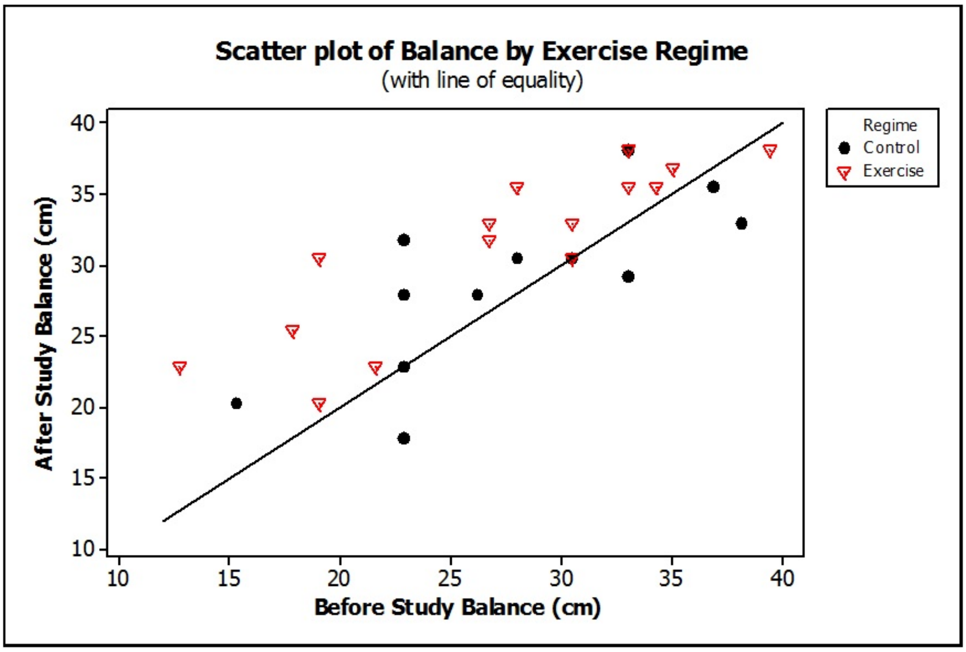

Figure 2.2: Visualising a two-sample paired continuous response

Figure 2.2 shows clearly that most of the Exercisers improved their Balance over the study. It appears that there is, in general, a slight improvement for the Controls. Both regimes tend to show changes that are, in general, parallel to the line of equality thus suggesting that there is a consistent and constant change not dependent, in any sense, on the magnitude of the subjects Balance. This graphical representation suggests that the use of the simple improvements (i.e. After Balance Before Balance) may give an adequate summary of the effect of the interventions in question.

An example of the use of graphics where the audience are statisticians is to represent a variance-covariance matrix as a heatmap in order to identify if a simpler structure (e.g. AR(1), compound symmetry) may be justified.

The four heatmaps in Figure 2.3 visualise the variance-covariance matrix from a linear mixed model for a design with five times follow-up a measurement (Y) for 100 patients assigned to one of 10 therapists nested in three different hospitals. The heatmap aims to visualise the assumed correlation structure for the underlying model; unstructured in this case.

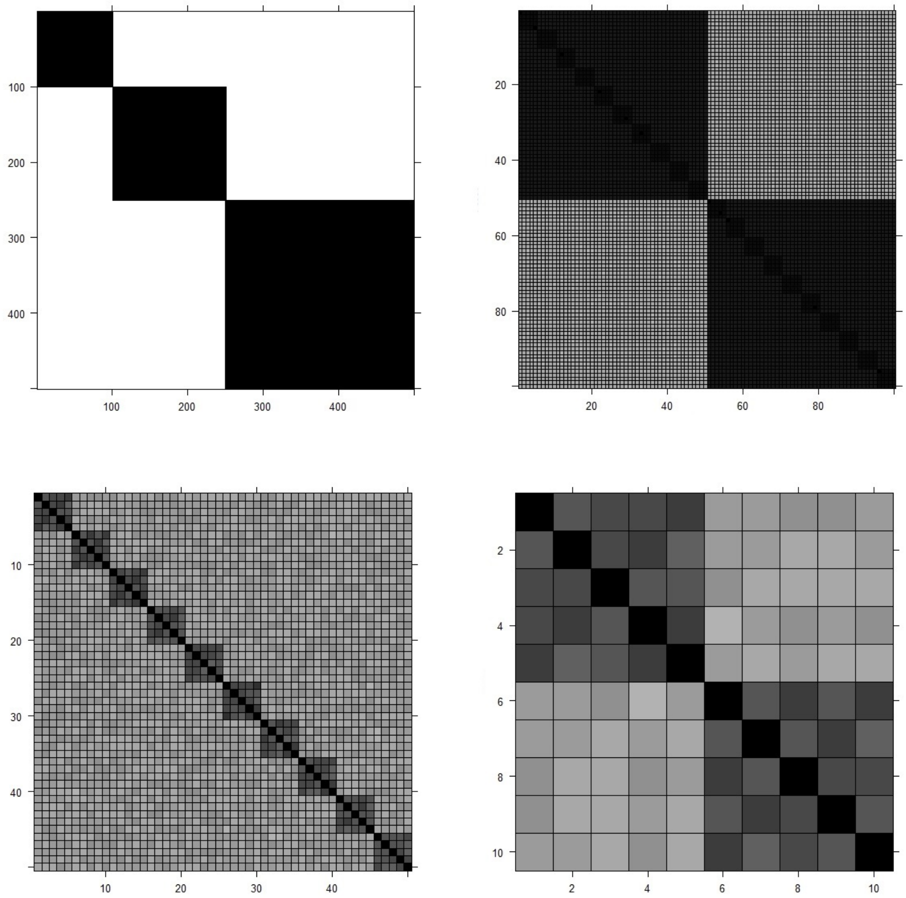

Figure 2.3: The heatmap of a variance-covariance matrix from a linear mixed model

The top left plot shows that Y is highly correlated within hospitals. The top right plot highlights the variance-covariance matrix of two of the therapists in the first hospital which apparently indicates strong correlation within therapists. There is also a structure within each of 10 patients treated with the first therapist as shown in the bottom left plot. The last plot on the bottom right visualises the variance-covariance of the first two patients indicating that follow-up times do not follow a specific structure and they also differ for each patient.

An example of the misuse of graphics is given in Figure 2.4. The left hand-side graph displays the mean and uncertainty in the response across four categories suggesting that levels A and B are comparable and levels C and D are comparable while the right hand-side, a clearly more informative graph, displays the same data as box-plots with the individual data overlaid. Despite the well-documented concerns of using mean and error bar-plots [72], these graphs are still quite popular in many domains.

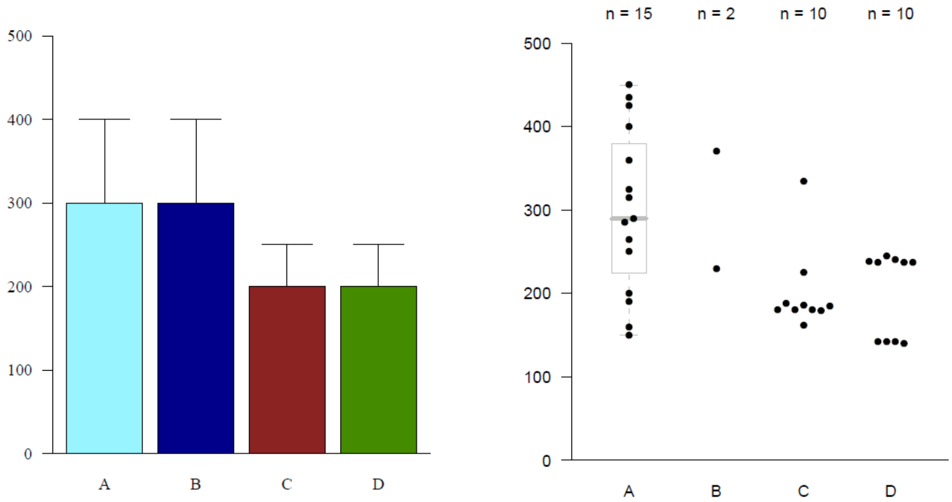

Figure 2.4: An example of a misused graphic

The most significant developments in data visualisation were in the 1980s and 1990s through the invention of hardware and software to generate computer graphics. A series of books appeared that have had an enormous impact in the field namely 'Exploratory Data Analysis' [135], 'Graphical Methods for Data Analysis' [27], 'The Visual Display of Quantitative Information' [133], 'The Elements of Graphing Data' [36] and 'The Grammar of Graphics' [147]. The latter book builds on earlier work by Bertin [17] to present a general framework (i.e. a grammar) for the generation of graphics.

More recently, the 'Handbook of Data Visualization' book [32] offers a collection of chapters written by experts including various views of data visualisation and several recent texts [35, 142] giving general principles and useful advice on creating informative graphics. Antony Unwin in his recent text; 'Graphical Data Analysis with R' [136], aims to show (through a series of cases studies) the power of graphical displays of data using various examples in **R**.

The emergence of software (e.g. 'xgobi' and 'ggobi') to generate interactive graphics has enabled the user to interact directly with the components of a graphic. This gives users the chance to 'take control' of the graph and explore the data being represented to study specific features, or use animation to follow patterns over time. Despite these advances, static graphics remain the primary graphical method to display data in presentations, articles and reports. In this chapter, a brief overview of the use of static graphics will be given in Section 2.2 and interactive graphics in Section 2.3, using classical and modern statistical tools to visualise data. Examples of these techniques and new methods for visualisation will be given using the illustration datasets introduced in Section 1.2.

2.2 Static graphics

There are many software programs and packages available to generate static graphics. Excel is still a popular tool for such endeavours; however, it includes a limited range of statistical graphics (e.g. drawing a histogram is difficult in Excel). Dedicated statistical software packages such as SAS, SPSS and Minitab provide a range of static graphics while the open source **R** software provides a framework to create basic to complex static graphics through the many packages available to visualise data. For example, Hadley Wickham's popular **R** package '`ggplot2`' [144], built on the principles of Grammar of Graphics [147] has arguably become the 'standard' for the generation of graphs in **R**.
An example of syntax needed to create a scatterplot of Age (in years) against BMI (Figure 2.5) from the slept trial data introduced in Section 1.2.6 is given below.

```
> library(ggplot2)
> ggplot(data=dat, aes(BMI, Age)) + geom_point()
```

The scatterplot suggests that there is little evidence of a relationship between age and BMI in the sample. The underlying grammar of graphics framework is best seen by partitioning the code into its layered grammar components. The layers for Figure 2.5 are:

- Frame: Age*BMI

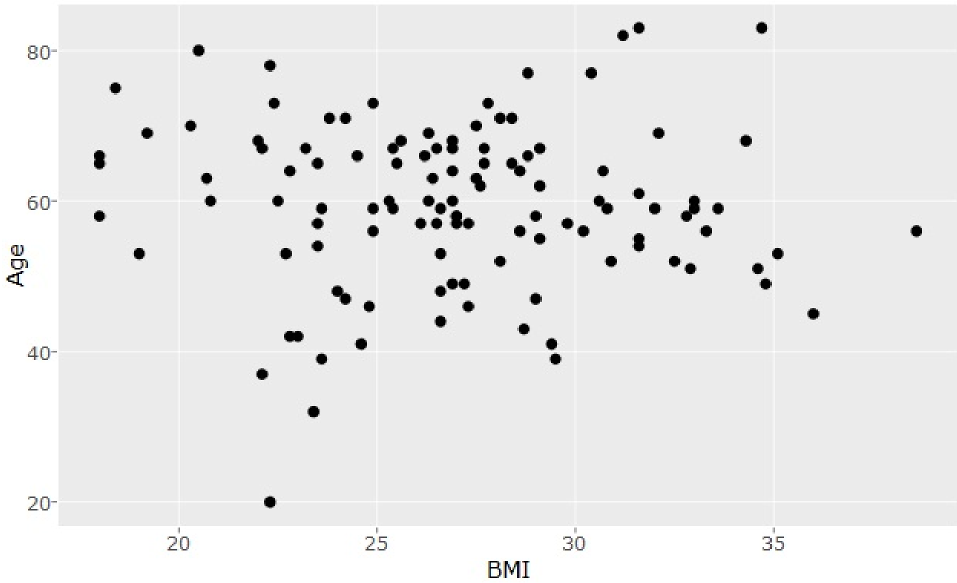

Figure 2.5: A scatterplot of Age vs BMI

- Graph: point()

This simple framework can be extended to other graphics with more complicated structures. For example, the above plot might have one or more layer components, e.g. the inclusion of a smoother to indicate the likely functional form between the covariate and the response. Therefore, a well-known graphic can be broken into its components of layered grammar, those components then can be modified separately depending on the graphical component required.
The `ggplot2` package is a widely used graphics package in **R** which implements the layered grammar of graphics framework through the use of the following components:

1 Layer,

2 Scale,

3 Coordinate system,

4 Faceting.

A 'Layer' creates the objects on a plot and includes five essential parts; Data, Mapping, Statistical transformation, Geometric object and Position adjustment. A graphic may include multiple layers which are related to one another and share common features.

Mapping defines the role variables play in the plot (e.g. assign variables to axes). A Statistical transformation (stat) transforms data, for instance, generating a smoother function to summarise the relationship between variables. Geometric objects (geoms) control the type of plot; they include point and text (0 dimensions), path and line (1 dimension and 2 dimensions). For example, a point geom will generate a scatterplot, while a line geom will generate a line plot. Each geometric object can only represent specific visual attributes (aesthetics), for example, a point geom has a position, colour, shape and size aesthetic. Position adjustment is also used to adjust the location of elements of the plot. It is useful to avoid overlaps or to reduce overplotting.

'Scale' handles mapping data to aesthetic attributes, so that every aesthetic needs at least one scale. A 'Coordinate system' (coord) uniquely define the location of objects onto the plot and controls the way axes and grid lines are displayed. The Cartesian coordinate systems are most commonly used, but semi-log coordinate or polar coordinate are other popular systems. 'Faceting' splits data into subsets to visualise on the same plot.

The aim of the 'ragweed' data introduced in Section 1.2.4 is to forecast the ragweed pollen level as a function of the temperature, rain, wind speed forecast and pollen season day. The **R** code below will generate a boxplot and the scatterplots shown in Figure 2.6 needed to investigate the marginal relationships between these explanatory variables and the (transformed) response; the square root of ragweed.

```
> library(SemiPar)
> library(ggplot2)
> p1 <- ggplot(data=ragweed, aes(x=rain, y=sqrtragweed)) +
+     geom_boxplot()
> p2 <- ggplot(data=ragweed, aes(x=temperature, y=sqrtragweed)) +
+     geom_point() + geom_smooth()
> p3 <- ggplot(data=ragweed, aes(x=wind.speed, y=sqrtragweed)) +
+     geom_point() + geom_smooth()
> p4 <- ggplot(data=ragweed, aes(x=day.in.seas, y=sqrtragweed)) +
+     geom_point() + geom_smooth()
```

```
> grid.arrange(p1, p2, p3, p4, ncol=2)
```

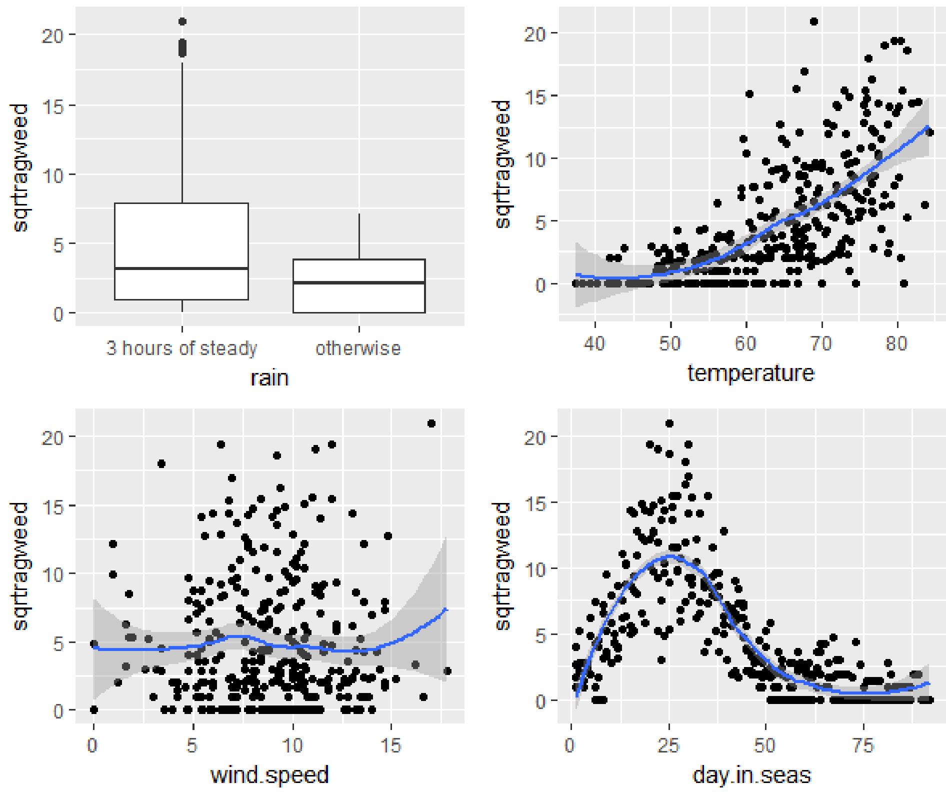

Figure 2.6: Relationships between the square root ragweed and the other variables

The boxplot in the top left-side of Figure 2.6 show that we could expect more ragweed pollen grains (ragweed level) if the following day had at least three hours of steady rain. Also, the temperature of the following day seems to have a strong positive effect on the square root of ragweed level; however, the wind speed forecast for following day looks not to affect it. The plot at the right bottom displays a non-linear relationship of the square root of ragweed with the pollen season day which is nicely visualised using a smoother (e.g. there appears to be an optimal value of the day in season at 25 days).

One challenge of using static graphics is when we try to portray more than two variables in a single plot. However, using different aesthetic mapping would ease this difficulty. One example is the use of transparency in the lightness of the line as a function another variable (also known as alpha-blending) which could represent a notion of accuracy (or even use shaded regions).

For example, the classical display for the time-to-event data is a plot of the estimated survival function (typically using the Kaplan-Meier estimator) as a line over time.

The level of uncertainty is rarely displayed (i.e. accompanying confidence intervals are not displayed on the graph) in order to remove clutter. One option is to use the thickness of the line to represent the diminishing number of observations over time. In order to do this, the 'dat.alpha' function in Appendix A.1 was written to fit a survival curve and an accompanying data frame which includes a variable indicating the number of subject at risk. Alpha blending can now be applied to this variable. As illustrated below (Figure 2.7) for the BPD data.

```
> library(ggplot2)
> dat.p <- dat.alpha(BPD$ondays, BPD$censor, factor=BPD$surfact)
> ggplot(data=dat.p) +
+     geom_step(aes(x=time, y=surv, alpha=n.risk,
+     color=as.factor(Factor), group=Factor)) +
+     theme(legend.title=element_blank()) +
+     guides(alpha=FALSE, title=NULL)
```

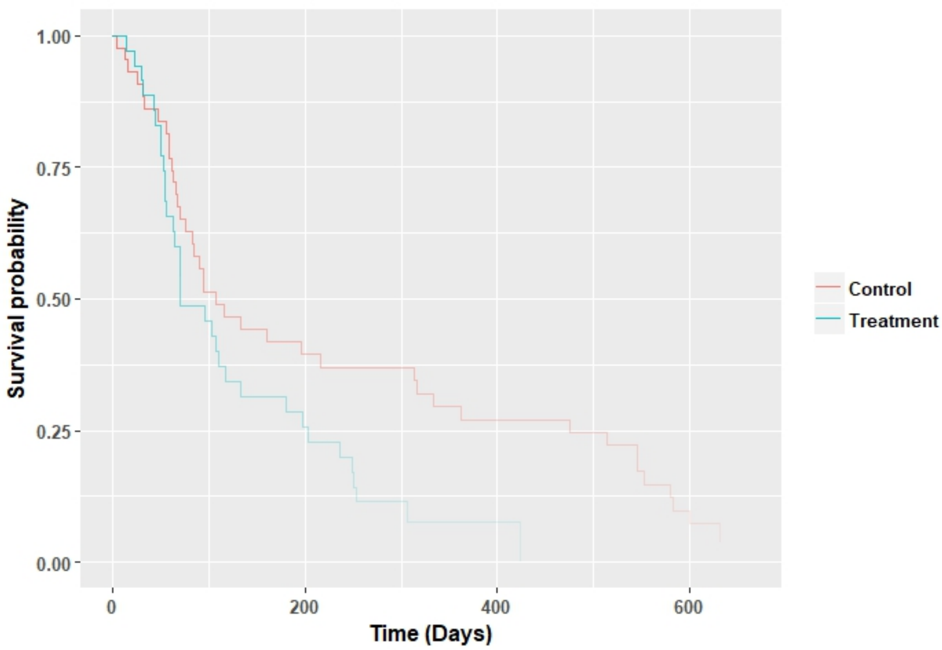

Figure 2.7: The use of alpha-blending for the survival estimate of the BPD data

2.3 Interactive graphics

The ability to create interactive and dynamic graphic methods is greatly increasing the potential of data visualisation as it allows a user to interact and manipulate graphical objects and statistical aspects of a graph directly. In the two recent decades, a large number of interesting ideas to present dynamic, interactive data visualisation has been brought together (see [**132**, **151**]). Most of these ideas and methods are described in "Visual statistics: seeing data with dynamic interactive graphics" by Young [152]. The simplest advantages of using interactive graphics are querying values and reformatting graphics (panning, zooming or focusing) which give a complete visualisation experience by controlling how and what is needed to be displayed and linked brushing which joins several related graphics for the same dataset.

One of the first software programs in this area was PRIM-9 which was developed by Fishkeller, Friedman and Tukey [55] to use dynamic graphics to explore multivariate data. PRIM-9 allowed picturing, rotation, isolation and masking which helped users to see multi-dimensional graphics from different angles and identify hidden structures in multivariate data. Nowadays, there are several commercial software packages available designed to provide interactive graphics focused on data visualisation (e.g. 'Qlik' and 'Tableau') or data analysis (e.g. 'JMP' and 'Mondrian'). The '`iplots`' [137], '`rggobi`' [88], '`ggvis`' [29], '`plotly`' [126] and '`shiny`' [28] packages also offer interactive graphics in **R**.

The '`ggplot2`' package is still one of the most powerful tools to visualise data in **R**; however, the emergence of new packages focused on animation and dynamic visualization provide an impressive set of free innovative visualisation tools.

Examples of using Tableau, **R** packages '`plotly`' and '`shiny`' follow to compare and contrast the capabilities of each. Tableau is a powerful (and expensive) standalone commercial software tool, while '`plotly`' and '`shiny`' packages are free, open source popular packages for generating interactive graphics in **R**.

2.3.1 Tableau

Tableau is a software tool built on a data visualisation languages developed recently to produce interactive graphics [70]. According to the Tableau's Chief Scientist (Pat Hanrahan), "Tableau Software was founded on the idea that analysis and visualisation should not be isolated activities but must be synergistically integrated into a visual analysis process" [71]. Tableau is marketed as a modern data visualisation tool in the business intelligence area as it promises a fast and easy way for data visualisation compared to the traditional approaches in this field. It allows users to quickly connect, visualise, and share data by creating and sharing dashboards using minimum programming skill. Users can just drag and drop data into the system to create interactive graphs, charts, maps. The sharing option in Tableau public where they can collaborate with other team members.
Figure 2.8 shows a screenshot of a basic example of using Tableau to create a stacked bar chart for the Titanic dataset in Section 1.2.1. In this worksheet a horizontal bar chart of 'Age' by 'survived' is produced by dragging 'Age' to Rows shelf, the sum of 'Number of Records' to the Rows shelf and 'Survived' to the Color. The other options such as Size, Labels, Shape and Detail are also available to add more dimensions into the graph. The filter option adds interactivity by letting users select data to highlight their area of interest.

Central to Tableau is the concept of the dashboard, an amalgamation of several graphs linked and displayed on a single page. Figure 2.9 shows a screenshot of a dashboard of bar-chart, pie-charts and tables generated in Tableau to visualise several aspects of the Titanic dataset. Summaries of Age, Gender and Class ticket and the survival proportions within selected levels of explanatory variables are presented using interactive bar-chart, pie-charts and tables. The published version of this dashboard in the Tableau public is available on https://amir-jalali.com/apps/titanic.

The interaction between age, gender and ticket class on the probability of survival becomes clear when interacting with the dashboard. For example, about 86% of females passenger older than 40 survived compared to just 16% for the corresponding males.

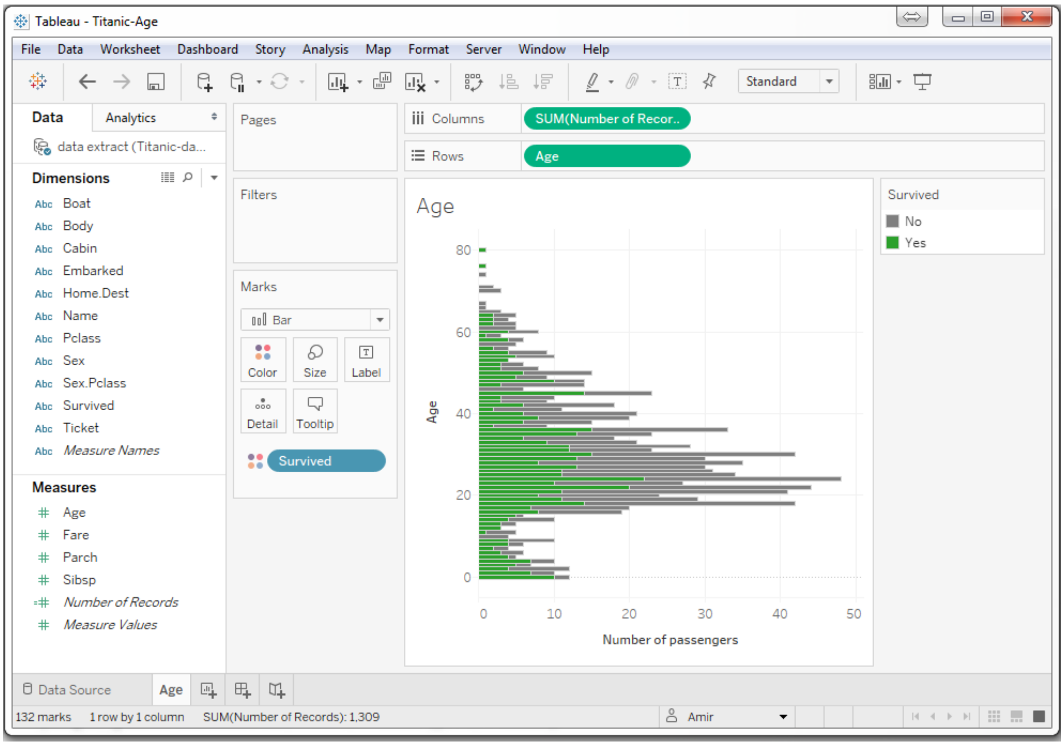

Figure 2.8: A screenshot of building a bar chart in Tableau software

2.3.2 plotly

plotly [126] is an online data visualisation tool which aims to provide a web-based interface for users to create interactive graphics using popular software such as Python, **R** or MATLAB by providing graphical options such as zooming, brushing, tooltips etc.

For example, the '`ggplotly`' wrapper function in the `plotly`'s **R** package will provide interactivity for graphical objects created with '`ggplot2`' by translating '`ggplot2`' objects to interactive plots which can be easily shared online. The **R** code below was used to generate a dashboard for the Crabs data in Section 1.2.2 using '`plotly`' as displayed in Figure 2.10.

```
> library(ggplot2)
> library(plotly)
> library(glm2)
> dat <-data(crabs)
> ply1 <- ggplotly(ggplot(dat, aes(x=Satellites)) +
+    geom_histogram(aes(y=..density..),
```

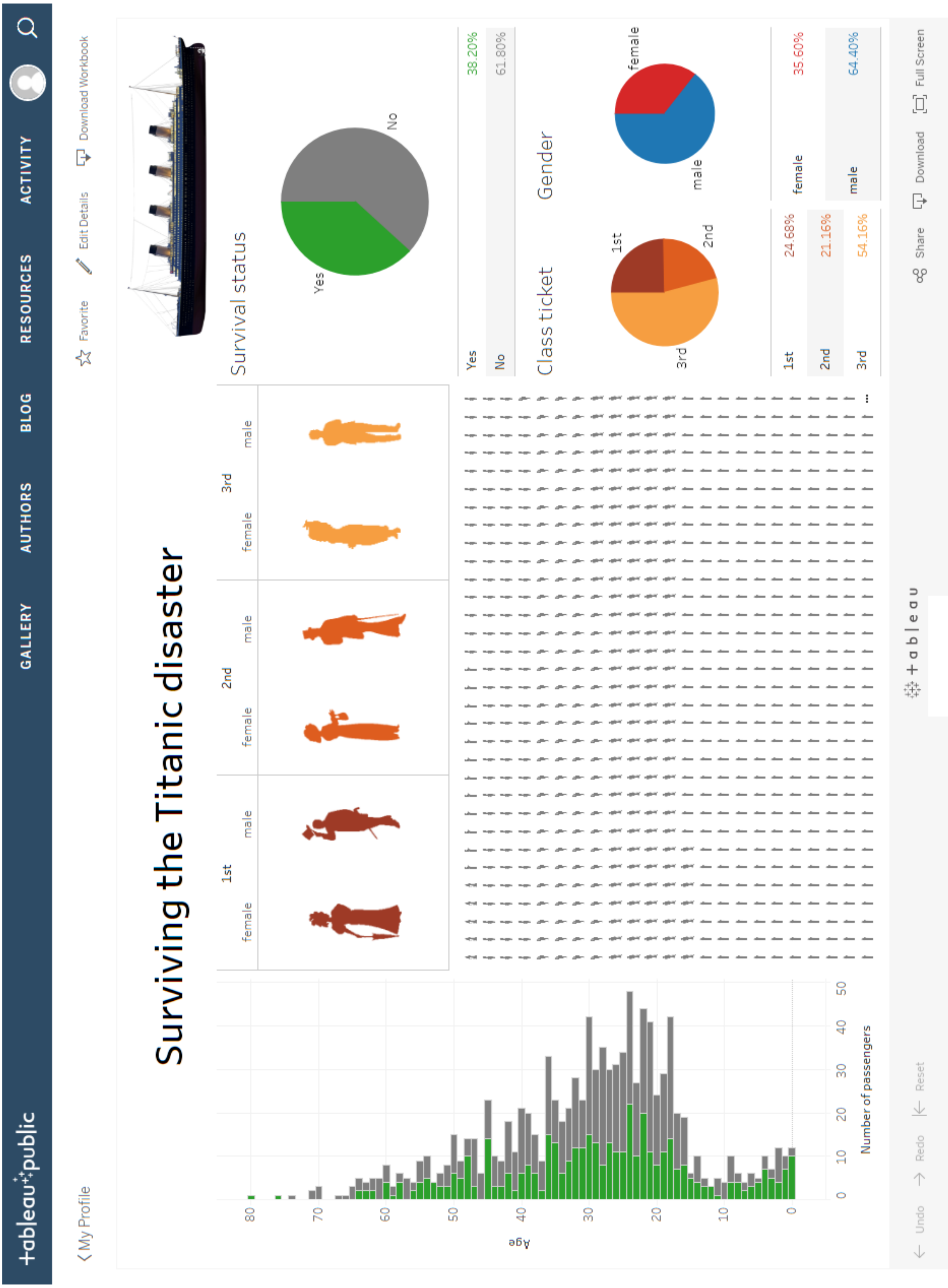

Figure 2.9: A screenshot of the Titanic data visualisation dashboard using Tableau

```
+    binwidth=density(crabs$Satellites)$bw) +
+    geom_density(fill="red", alpha=0.2),
+    tooltip=c("Satellites", "density"))
> ply2 <- ggplotly(ggplot(dat, aes(x=Width, y=Satellites)) +
+    geom_point() + geom_smooth())
> ply3 <- ggplotly(ggplot(dat, aes(x=Dark, y=Satellites, fill=Dark)) +
+    geom_boxplot(alpha=0.8) + theme(legend.position="none"))
> ply4 <- ggplotly(ggplot(dat, aes(x=GoodSpine, y=Satellites,
+    fill=GoodSpine)) +
+    geom_boxplot(alpha=0.8) + theme(legend.position="none"))
### create multiple ggplotly
> subplot(ply1, ply2, ply3, ply4, nrows=2)
```

The screenshot of the dashboard shown in Figure 2.10 was published online, and it is available to interact with at https://amir-jalali.com/apps/crabs. It includes the density plot of the primary variable ('Satellites') and also a scatter-plot and box-plots to visualise the relationship between the explanatory variables ('Width', 'Dark' and 'GoodSpine') with 'Satellites'.

The histogram/density plot of 'Satellites' in the top left-side of Figure 2.10 show that about 35% of the female horseshoe crabs have no additional male partners. However, some of them have up to 15 male partners in addition to their primary partner. The marginal effects of exploratories and the 'Satellites' are also displayed graphically. For example, there is no strong relation to whether the female crabs have a good spine condition or have a dark colouration. Besides, a small positive relationship between 'Width' and 'Satellites' can be observed, in other words, the wider crabs are expected to have more male partners.

2.3.3 flexdashboard

The '`flexdashboard`' **R** package [8] is one of the easiest ways to build dashboards (in HTML) by packaging the collection of visualisations of interest (created in **R**) using the **R** authoring framework 'R Markdown'.

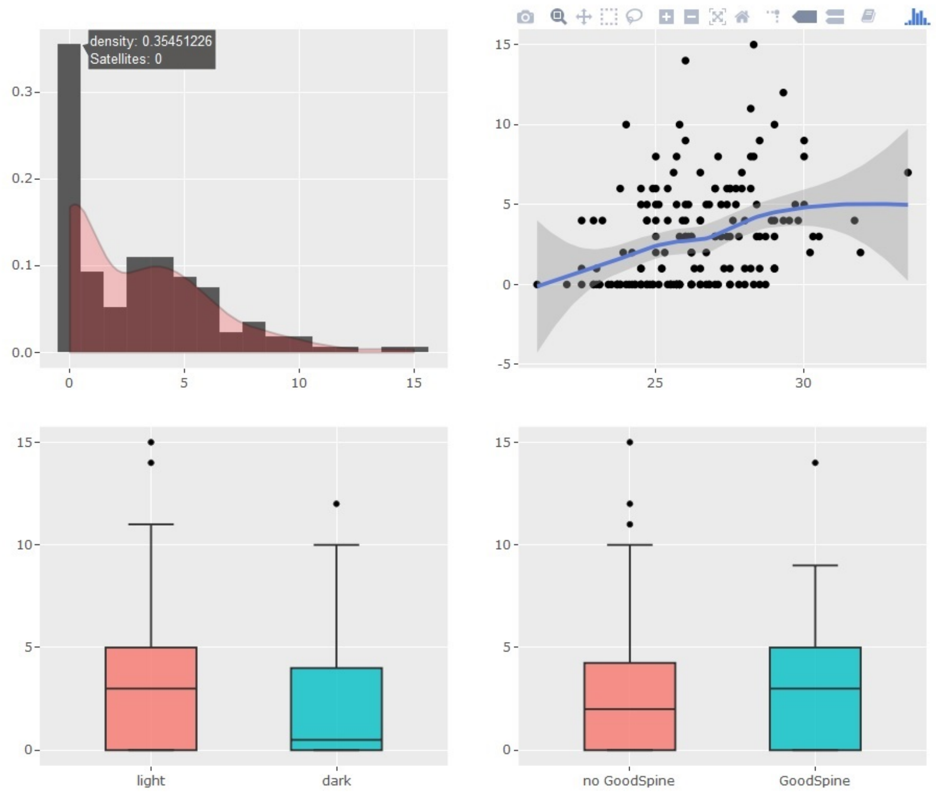

Figure 2.10: A screenshot of a dashboard for Crabs data using plotly

A Flex Dashboard template is included as a predefined R Markdown document template when the '`flexdashboard`' package is installed. Building a dashboard using this package is quite straightforward by placing the relevant **R** code into chunks. For example, the code below was used to create the simple dashboard for the Slept trial data in Section 1.2.6 which is shown in Figure 2.11.

```
---
title: "flexdashboard demo"
output:
flexdashboard::flex_dashboard:
orientation: columns
vertical_layout: fill
---
```{r setup, include=FALSE}
library(flexdashboard)
library(ggplot2)
```
```

```
```

Column {data-width=650}

```{r}
dat <- read.csv("Continuous_response_RCT.csv", header=T)
```

```{r,fig.width=11, fig.height=6}
ggplot(dat, aes(Baseline, Response)) + geom_point(aes(colour = Sex)) +
guides(colour = guide_legend(title = NULL)) +
facet_grid(.~dat$Group) +
geom_abline(aes(intercept = 0, slope = 1))
```

The mean difference for Female is:

```{r}
mean(dat$Response[dat$Group=="Treatment" & dat$Sex=="Female"]) -
mean(dat$Response[dat$Group=="Control" & dat$Sex=="Female"])
```

The mean difference for Male is:

```{r}
mean(dat$Response[dat$Group=="Treatment" & dat$Sex=="Male"]) -
mean(dat$Response[dat$Group=="Control" & dat$Sex=="Male"])
```
```

The published version of this dashboard is available on https://amir-jalali.com/apps/flexdemo. Moreover, using ‘`plotly`’ along with ‘`flexdashboard`’ allows the user to build interactive dashboards (with limited functionality) as HTML documents. However, interactive graphics which require more functionality are best created using the ‘`shiny`’ package.
```

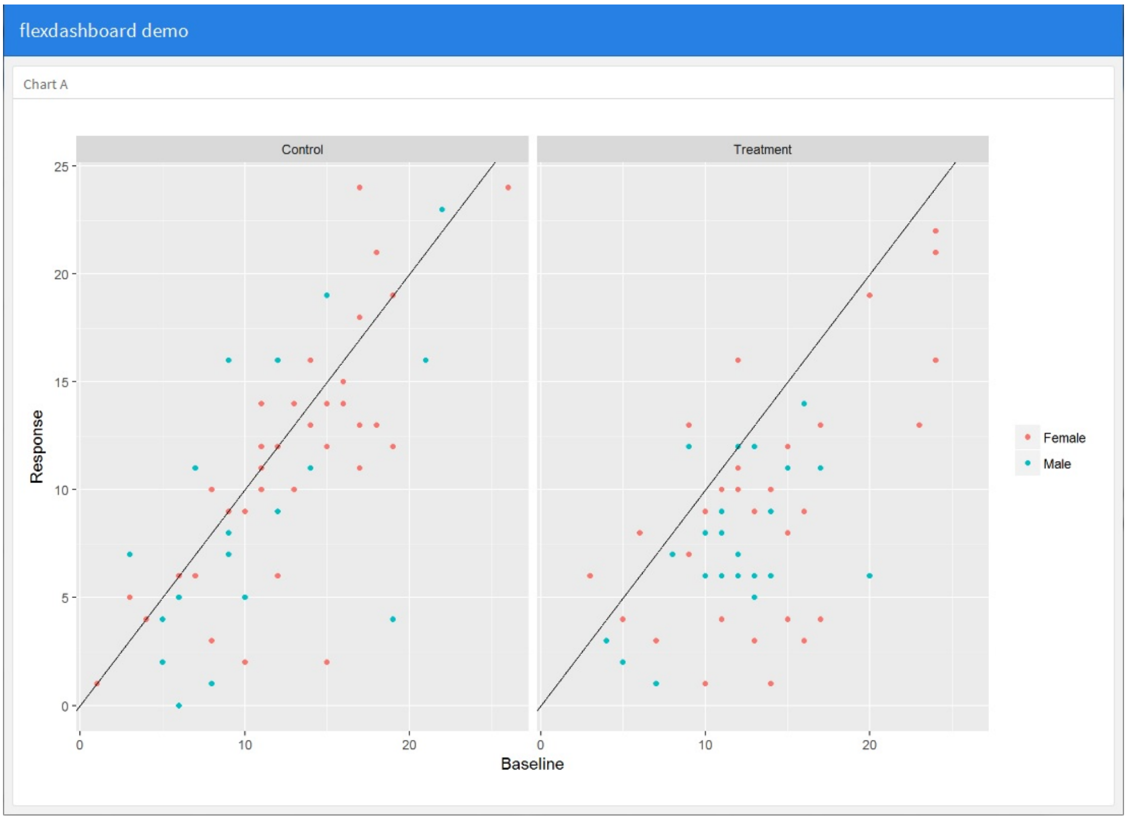

Figure 2.11: A screenshot of the dashboard created for the Slept trial data using the ‘`flexdashboard`’ package

2.3.4 shiny

‘shiny’ is an **R** Package to generate interactive HTML content to display **R** functions visually without the need to learn HTML. The `shiny` package is a framework for making **R** code a visual object where the arguments needed for the functions in question can be entered on a web page without the necessity to have **R** running in the background.

A shiny application consists of two main components; a user-interface script (a source file called ‘ui’) and a server script (‘server’). The shiny UI controls the layout and appearance of the application, while the shiny Server specifies all the **R** actions. In other words, one script is commanding all the HTML and one goes beyond the **R** code to manipulate the data, produce output (e.g. graphs, tables or model summaries) and customise the analysis for any specific needs. An example of ‘ui’ and ‘server’ files needed to make a shiny application is shown in the **R** code below. It makes a very simple shiny app to draw a histogram for a required number of samples generated from a standard normal distribution.

```
## app.R ##
> runApp(list(
+    ui <- bootstrapPage(
+    numericInput('n', 'Number of obs', 100),
+    plotOutput('plot')
+    ),
+    server <- function(input, output) {
+    output$plot <- renderPlot({ hist(rnorm(input$n)) })
+    }
+    ))
```

The power of shiny arises from the ability to use reactive expressions. A reactive expression is an **R** object which gets inputs, applies **R** code to take the required actions and returns another **R** object of interest. This object will keep track of the reactive values and automatically update whenever the inputs changes. A reactive function is a general function which makes reactive expressions, for example, reactive functions used to input values (e.g. 'selectInput' or 'numericInput' functions), read datasets (e.g. 'reactiveFileReader' function), perform calculations and analysis in **R** and return results such as graphs, tables or summaries (e.g. 'renderPlot', 'renderTable' and 'renderText' functions). Some HTML tags are also available to pass to the 'ui' to customise it, in a way which is not provided by the usual layout options. For example, they include tags$b (to creates bold text), tags$em (to emphasise) and tags$script (to add a script such as javascript). A list of 110 useful functions to build these HTML tags in shiny is available by shiny::tags.

The shiny application displayed in Figure 2.12 was created to visualise the Slept trial data in Section 1.2.6. It includes an interactive scatter plot which represents the effect of each explanatory variable (baseline, age, BMI, adherence and resting pulse) on the response (or change score in response) where a smoothed curve could be fitted to highlight this effect. The effect of other explanatory variables can also be included when adding colour and size or by panelling the graph by a factor. In

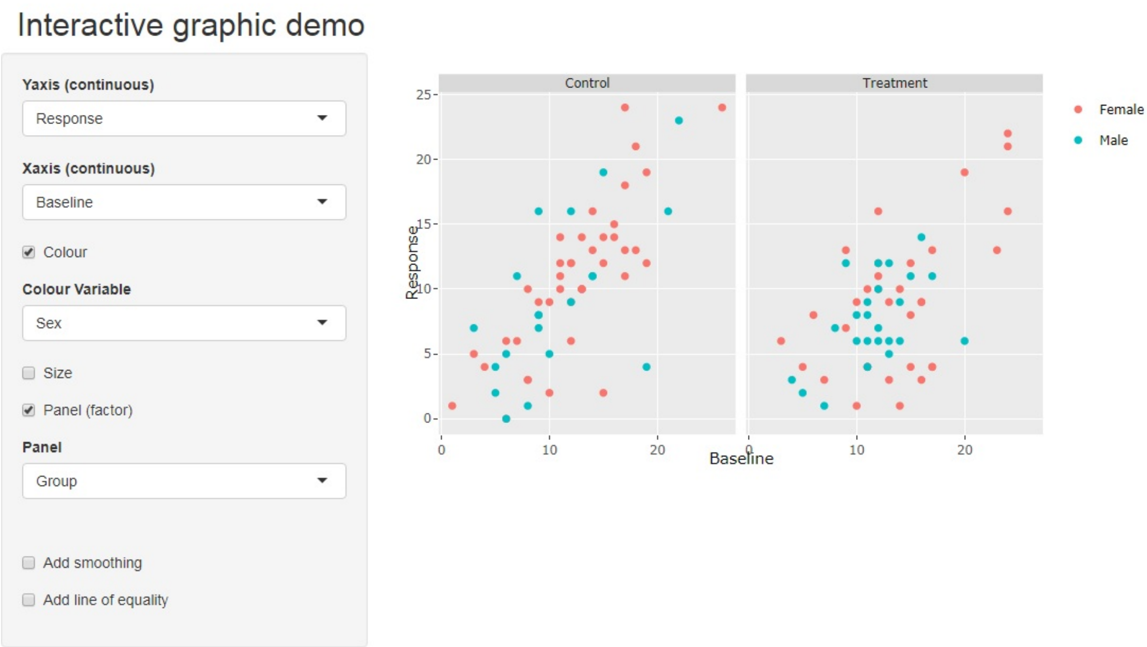

Figure 2.12: A screenshot of the shiny application to visualise the Slept trial data

the case where the change score is the variable of interest, the line of equality is also worthwhile to be included as a reference line. This application is available on https://amir-jalali.com/apps/intgraph; the ‘ui’ and ‘server’ code used to create this application are given in Appendix A.4.

‘shinydashboard’ is a recent extension to shiny which contains all the interactive features of shiny while providing more themes, layouts and widget functionality. The main difference with shiny is in the ‘ui’ code, where some minor modifications are needed. The ‘ui’ shinydashboard requires three components namely a header, a sidebar, and a body as follows:

```
## ui.R ##
> ui <- dashboardPage(
+     dashboardHeader(),
+     dashboardSidebar(),
+     dashboardBody()
+     )
```

The ‘dashboardHeader’ function is similar to ‘headerPanel’ in shiny and contains the application title and an option to include a list of arguments such as messages, notifications and tasks. The 'dashboardSidebar' is also similar to the 'sidebarPanel' and contains menu items used as input controls that can be passed to the body of the dashboard. Lastly, the 'dashboardBody' is used to navigate between the shiny content which will be generated on the server.

Figure 2.13: A screenshot of the shiny application using shinydashboard to visualise the Slept trial data

A similar application to the shiny application given in Figure 2.12 using 'shinydashboard' was created to visualise the Slept trial data. This application is pictured in Figure 2.13 and is available on https://amir-jalali.com/apps/shinyddemo, also the 'ui' and 'server' code used to create this application are given in Appendix A.5. The 'flexdashboard' could also be built on top of 'shiny' to design interactive dashboard and applications. This allows users to create dynamic dashboards in **R**. This takes the effort out of learning shiny for publishing a graphic on the web. However, the 'flexdashboard' is ultimately an HTML document which provides interactivity using embedded JavaScript. In contrast, 'shiny' and 'shinydashboard' provide more functionality, but they require a server behind them to execute **R** code.

2.4 Summary

In this chapter the role of data visualisation was discussed, a brief history of the advances in data visualisation given, in particular, the grammar of graphics framework for generating static graphics. The importance of generating 'audience appropriate' graphics was discussed and examples were given.

Static and dynamic graphics were discussed and their merits compared. Modern methods of generating interactive graphics were introduced together with examples of published visualisations.

Tableau, with its ability to quickly drag-and-drop variables from a loaded database, was introduced as one example of modern data visualisation software. Popular visualisation packages in **R**, such as '`ggplot2`' and '`plotly`' were discussed which allow the creation of customised graphics in **R** while other packages such as '`shiny`', '`flexdashboard`' and '`shinydashboard`' were presented as powerful tools to create interactive dashboards by combining the benefit of the analytic power of **R** along with the functionality of Javascript and HTML. The ability to code in **R** is a requirement to generate a Shiny object/app while no coding knowledge is needed to use Tableau. However, the ability to build interactive web applications in **R** (for free) is clearly attractive. The use of well-designed graphics is hugely important in the world of small, medium and big data. Although interactive graphics play an important role, especially in understanding big data, the typical representation of data is through a static graph. Recent developments in data analysis and statistical languages (i.e. advanced **R** packages and modern business intelligent software) means that it is no longer difficult to create graphics which the user can interact with. Dashboards have become not only fashionable but also essential in the world full of reporting and data analytics.

In Chapter 5 new approaches will be proposed to blend static and dynamic graphics (for data and models) in a novel manner to make additional graphics available in an accessible manner for the reader.

Chapter 3

Model

3.1 Introduction

All statistical methods introduced in this chapter investigate the relationship between a variable of interest (known as the *predictand*, *response* or *dependent variable*) and a group of other observed variables which are called *predictor*, *explanatory*, or *independent* variables. The response or explanatory variables are considered as *discrete* (also called *categorical*) and *continuous.* For instance, examples of discrete variables are blood type, eye colour or the number of patients arriving in an emergency department, and examples of continuous variables are age, weight and time to death of the lung cancer.

The modelling process of the statistical theories presented in the following sections were summarised by Dobson [44] in the following steps:

1 Specify a model equation to link the response and the explanatory variables;

2 Estimate the model parameters;

3 Check the model adequacy;

4 Make inferences; for example, testing hypotheses and calculating confidence intervals for the parameter of interest.

In this chapter, a summary of various classical statistical modelling techniques will be presented, and the most relevant aspects will be discussed in detail. We begin with an overview of Generalised Linear Models, including a discussion on confidence intervals for regression parameters as well as prediction intervals. Logistic and Poisson regression models will be used to illustrate the modeling of the Titanic and Crabs datasets introduced in Chapter 1. In the second part, we elaborate on regression models for time-to-event data, in particular, the Cox proportional hazards model. The survival and hazard functions will be reviewed as the most popular summaries of time-to-event data. The comparison of two survival distributions will also be considered, using the log-rank test as well as various graphical approaches. Finally, the dynamic prediction for survival and mean residual life functions will be introduced as useful alternative functions.

3.2 Generalised Linear Models

The Linear Regression model is the simplest statistical method to investigate the linear effects of some explanatory variables on a response variable. The linear regression model in its more general form is defined as:

$$Y = X\beta + e \tag{3.1}$$

where $Y = [y_1, y_2, \ldots, y_n]$ is a $n \times 1$ vector of independent observations from the response variable, $\beta = [\beta_1 \ldots \beta_p]$ is an unknown $p \times 1$ vector of model parameters and X is a $n \times p$ matrix called the design matrix with its $\mathrm{i^{th}}$ row being the vector of explanatory variables for the $\mathrm{i^{th}}$ observation ($x_i^T = [x_{i1} \ldots x_{ip}]$). The vector of model parameter β can be estimated using the least squares method as:

$$\hat{\beta} = (X^T X)^{-1} X^T Y \tag{3.2}$$

The Generalised Linear Model (GLM) are more flexible by providing a unified framework for modelling a wide variety of responses including continuous, categorical and count variables. It was first introduced by McCullagh and Nelder [**104**, **97**]. The

GLM models have been studied by many other authors such as Agresti [4], Dobson [44], Madsen [95] and James [78].
The Generalised Linear Model has two components:

1. The probability distribution of the response variable;
2. The equation linking the mean of the response variable to a linear combination of the explanatory variables.

The Generalised Linear Model assume that the response variable has a probability distribution from a family of density functions called *the exponential family*. Consider a single random variable Y whose probability distribution depends on the vector of parameters θ. The distribution belongs to the exponential family if it has the following form [44]:

$$f(y;\theta) = \exp[a(y)b(\theta) + c(\theta) + d(y)] \tag{3.3}$$

where $a(.)$ and $d(.)$ are known functions of y and $b(.)$ and $c(.)$ are known functions of θ. The most common distributions such as the Normal, Binomial, Poisson, Exponential and Gamma distributions can be expressed in the form given by (3.3) and therefore they all belong to the exponential family. The distribution is said to be in linear canonical form if $a(y) = y$ and $b(\theta)$ is called the natural parameter of the distribution [91].
The GLM model focuses on modelling a particular function of the mean response variable ($E(Y) = \mu$) which is usually linked to the natural parameter of the response distribution. This function ($g(.)$) must be a monotone, differentiable transformation of μ and is called the *link function* (η):

$$\eta = g(\mu) \tag{3.4}$$

Taking all this into consideration, the GLM model can be formulated as:

$$g(\mu) = X\beta \tag{3.5}$$

The identity, logit and logarithm functions are examples of the link function which are used as the standard link for when the response variable is assumed to have Normal, Binomial or Poisson distribution, respectively. The most common link functions for various response families are summarised in Table 3.1.

Distribution	Support of distribution	Link function $\eta = g(\mu)$	$\mu = g^{-1}(\eta)$
Normal	$\mathbb{R} : (-\infty, \infty)$	μ (identity)	η
Bernoulli Binomial Multinomial	Integer: $\{0, 1\}$ Integer: $\{0, 1, \ldots, n\}$ Set of integers: $\{0, 1, \ldots, n\}$	$\log(\frac{\mu}{1-\mu})$ (logit)	$\dfrac{\exp(\eta)}{(1+\exp(\eta))}$
Poisson	Integer: $\{0, 1, \ldots\}$	$\log(\mu)$ (logarithm)	$\exp(\eta)$
Exponential Gamma	$\mathbb{R}_+ : (0, \infty)$	$-\mu^{-1}$ (reciprocal)	$\frac{1}{\eta}$

Table 3.1: Common link functions for GLM models

The models are usually fitted via the Maximum Likelihood (ML) method. This means that a parametric likelihood function will be maximised as a function of β using the distribution from the exponential family and the link function. The log-likelihood function for a $n \times 1$ vector of observations $y = [y_1, \ldots, y_n]$ from an exponential family distribution can be expressed as:

$$l(\beta; y) = \sum_{i=1}^{n} a(y_i) b(\beta_i) + \sum_{i=1}^{n} c(\beta_i) + \sum_{i=1}^{n} d(y_i) \tag{3.6}$$

The first derivative of $l(\beta; y)$ respect to β is called the score function ($U = \frac{\partial l(\beta;y)}{\partial \beta}$) which is needed to obtain the ML estimate for the GLM parameters. Hence, the ML estimate ($\hat{\beta}$) is the solution of the equation $U(\beta) = 0$ which can be solved using Newton-Raphson in the cases where no closed form is available.

Almost all standard statistical software such as **R**, SAS, Minitab, SPSS, GLIM, STATA can fit GLM models. The excellent book on "Statistical modelling in R" by Aitkin et al. [7] includes useful technical theory related to GLM, with special attention to their application in **R**.

Binary and count data are two of the most common responses and they will be discussed in 3.2.1 and 3.2.2, where examples of models fitted to the Titanic (see 1.2.1) and Crabs (see 1.2.2) datasets are given as illustration.

3.2.1 GLM for binary responses

A binary variable is a categorical variable with only two levels, such as male/female, passenger alive/dead, or treatment succeeds/fails. The binomial distribution which is the probability distribution used to analyse binary responses has the following form:

$$f(y;\pi) = \binom{n}{y}\pi^y(1-\pi)^{n-y}$$

where $\pi \in [0,1]$ is the probability of success and $y = 0,\dots,n$.
The binomial distribution is part of the exponential family defined in (3.3) as follows:

$$f(y;\pi) = \exp\left(y\log\left(\frac{\pi}{1-\pi}\right) + n\log(1-\pi) + \log\left(\binom{n}{y}\right)\right) \tag{3.7}$$

The natural parameter ($b(\pi) = \log(\frac{\pi}{1-\pi})$) is the logit function which is also the canonical link function for the binomial distribution. The mean of the binary outcome is the probability of success π so the logit function is a reasonable choice as the link function. The GLM model can be written as (3.8) to model the proportion of successes in each subgroup ($\pi = p(Y=1|X)$) defined in terms of the observed explanatory variables.

$$\log\left(\frac{\pi}{1-\pi}\right) = X\beta \tag{3.8}$$

This model is also known as the logistic regression model. The maximum likelihood method can be used as the standard approach to estimate the model parameters (β).
As an illustrative example, a logistic regression model is fitted to the Titanic data to investigate the factors that affect the probability of surviving the Titanic disaster (π). The explanatory variables included are 'age' (in years), 'pclass' (ticket class: 1st, 2nd or 3rd) and 'sex' (male/female). The model summary is presented in Table 3.2.

According to Table 3.2, all the predictors in the model (pclass, sex and age) are found to have significant effects on the probability of passenger survival (π). The estimated coefficients show that the best chance of surviving, overall, is within the young females in first-class. In general, the chance of a passenger survival is higher

	Estimate	Odds ratio	Std. Error	z-value	p-value
Intercept	3.5221	33.855	0.327	10.781	<0.0001
age	-0.0344	0.966	0.006	-5.433	<0.0001
pclass (2nd)	-1.2806	0.278	0.226	-5.678	<0.0001
pclass (3rd)	-2.2806	0.101	0.226	-10.140	<0.0001
sex (male)	-2.4978	0.082	0.166	-15.044	<0.0001

Table 3.2: The logistic model summary fitted to the Titanic data

in females and within the higher ticket classes. Also, this chance is reduced as the age of passenger increases.

The logistic regression model displayed in Table 3.2, can be represented in the linear form in (3.9) or in probability form in (3.10):

$$\log(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}) = 3.52 - 0.03\ age_i - 1.28\ pclass_{(2nd)i} - 2.29\ pclass_{(3rd)i} - 2.50\ sex_{(m)i} \quad (3.9)$$

$$\hat{\pi}_i = \frac{\exp(3.52 - 0.03\ age_i - 1.28\ pclass_{(2nd)i} - 2.29\ pclass_{(3rd)i} - 2.50\ sex_{(m)i})}{1+\exp(3.52 - 0.03\ age_i - 1.28\ pclass_{(2nd)i} - 2.29\ pclass_{((3rd)i)} - 2.50\ sex_{(m)i})} \quad (3.10)$$

It is worthwhile to study the odds ratio of each predictor to have a better understanding of the variable effect sizes. The odds ratio of a one-unit increase in a continuous explanatory variable (X_j) can be calculated by exponentiating the regression coefficient, i.e. $\exp(\beta_j)$ (in the case of a categorical explanatory variable the ratio is obtained in the same way but with respect to a chosen reference level). Modelling the log odds is mathematically attractive, but it is argued that the interpretation may be difficult to understand [3, 119]. For instance, consider again the Titanic example where the odds ratio for gender (male/female) is $e^{-2.4978} = 0.082$. This shows a much lower odds of surviving for male passengers compared to females, with the same age and ticket class. The gender effect can be better translated if it can be represented on a probabilistic scale. Odds and probability are often (many times mistakenly) used interchangeably, and large or small odds values inadvisedly interpreted as trivial (in absolute term) without reference to the underline probability. For example, the odds of surviving a 25 year old female in 3rd class is 1.45 compared to 0.12 for the corresponding male, while the corresponding probabilities were 0.59 and 0.11 respectively.

3.2.2 GLM for count responses

Count variables are another typical type of response which takes the non-negative integer values $(0, 1, \ldots)$. They represent the number of events occurring randomly in a given interval of time or space, e.g. the number of births per hour, the number of deaths due to AIDs in Ireland per year or the number of bacteria seen in a petri dish. The Poisson distribution is commonly used to analyse count data and has the following form:

$$f(y; \lambda) = \frac{\lambda^y e^{-\lambda}}{y!} \quad y = 0, 1, 2, \ldots$$

where $\lambda > 0$ is its expected value.

A key property of the Poisson distribution is the equality of the mean and the variance (i.e. $E(Y) = \mathrm{var(Y)} = \mu$). The Poisson distribution is part of the exponential family defined in (3.3) as follows:

$$f(y; \lambda) = \exp\left(y \log(\lambda) - \lambda - \log(y!)\right)$$

The natural parameter $(\log(\lambda))$ is the logarithm function which is the canonical link function for the Poisson distribution. The count mean is positive so the logarithm function is a reasonable choice as the link function. The GLM model can be formed as (3.11) to model the number of counts conditional on the observed explanatory variables X.

$$\log(\lambda) = X\beta \tag{3.11}$$

This model is also known as the log-linear regression model. The maximum likelihood method can be used as the standard approach to estimate the model parameters (β). The regression coefficient β_j represents the expected change in the log of the mean response per one-unit increase in the j^{th} continuous explanatory variable (x_j).

The 'Crabs' dataset in 1.2.2 is used here as an example of Poisson regression model fitted to investigate the relationship between the width, colour and spine condition of female crabs on the number of their male partners (defined as 'Satellites'). This dataset has been previously used as an illustrative example in the 'Categorical data analysis' book by Agresti [6]. The summary of the fitted Poisson regression model is given in Table 3.3.

	Estimate	exp(β)	Std. Error	z-value	p-value
Intercept	-2.8202	0.06	0.5707	-4.94	<0.0001
Width	0.1492	1.16	0.0207	7.20	<0.0001
Dark (yes)	-0.2652	0.77	0.1024	-2.59	0.0096

Table 3.3: The Poisson regression model summary fitted to the Crabs data

The model summary in the Table 3.3 shows 'Width' and 'Dark' have significant (at the 5% level) effects on the number of male partners of female crabs. Overall, the wider, light colour female crab has more expected number of male satellites. The Poisson regression model displayed in Table 3.3, can be represented in the linear form in (3.12) or in the mean response scale in (3.13):

$$\log(\hat{\lambda}_i) = -2.820 + 0.149\ Width_i - 0.265\ Dark_{(yes)i} \tag{3.12}$$

$$\hat{\lambda}_i = \exp(-2.820 + 0.149\ Width_i - 0.265\ Dark_{(yes)i}) \tag{3.13}$$

The coefficients in (3.13), for example, indicate that the expected number of satellites for a female crab with a dark colour ('Dark'=yes) will be multiplied by $e^{-0.265} = 0.77$ times the one with a light colour ('Dark'=no). In other words, the female crabs with dark colour will be expected to have 23% less male partners. Once more note that the effect of explanatory variables could be better translated if they could be represented on the mean response scale. For example, a dark colour female crab with width = 30 is expected to have 5 additional male partners ($\hat{\lambda}_1 = 5.21$) while a light colour one with the same width is expected to have 4 ($\hat{\lambda}_2 = 3.99$). This issue will be revisited in Chapter 5.

3.2.3 Confidence intervals

In the previous sections, GLM models were introduced where parameters are estimated using maximum likelihood estimation. In addition to point estimation, it is worthwhile to identify the uncertainty of the estimate by a range of plausible values for the unknown parameter (known as interval estimate). The point estimation provides the most reasonable estimate for the quantity of interest while the confidence interval provides a range of values for a fixed level of confidence. A $1-\alpha$ confidence interval for the parameter of interest θ can be expressed as:

$$P(L < \theta < U) = 1 - \alpha \tag{3.14}$$

where L and U are functions of explanatory variables that define the lower bound and upper bound of the interval, respectively.

The parameter θ or a function of it which is independent of the parameter is needed to compute the confidence limits (L and U). This function is called a 'pivotal quantity'. Confidence intervals, in most cases, are symmetric sets around the estimated parameter; however, they might be non-symmetric depending on the distribution of pivotal quantity. A symmetric confidence interval can be written in the form of (3.15):

$$[\hat{\theta} - se(\hat{\theta}) * C, \hat{\theta} + se(\hat{\theta}) * C] \tag{3.15}$$

where $se(\hat{\theta})$ is the standard error of the estimate, and C is a critical value that relates to the pivotal quantity distribution. This critical value measures the fraction of standard error needed to be added or subtracted from the estimated parameter to obtain the desired confidence level.

As an example, assume a random sample of size n from a random variable X with normal distribution where $E(X) = \mu$ and $V(X) = \sigma^2$, and consider building a confidence interval for the mean of the population μ. The pivotal quantity in this case would be:

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Note that Z is a function of X and μ but its distribution is independent of μ. Then using (3.14), a 95% confidence interval for μ is calculated as follows:

$$\begin{aligned} P\left(L_X < \mu < U_X\right) &= 0.95 \\ \Longrightarrow P\left(L_X^{'} < \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} < U_X^{'}\right) &= 0.95 \\ \Longrightarrow \begin{cases} U_X = \bar{X} + 1.96\frac{\sigma}{\sqrt{n}} \\ L_X = \bar{X} - 1.96\frac{\sigma}{\sqrt{n}} \end{cases} \end{aligned}$$

The width of the confidence interval is a function of sample size, the standard error and the confidence level. Due to the equation (3.15), wider confidence intervals can be a result of a:

- smaller number of observations,
- higher confidence level, and
- larger standard error.

Confidence intervals for GLM parameters

Confidence intervals can be used to measure the uncertainty of the estimates of GLM parameters. Assume the GLM model in (3.5) and consider that $\hat{\beta}$ is the ML estimate of the vector of regression parameters β. The variance-covariance matrix of the $\hat{\beta}$ is calculated as:

$$\Sigma = (X^T W X)^{-1} \tag{3.16}$$

where W is a $n \times n$ diagonal matrix with elements $w_{ii} = \frac{1}{V(Y_i)}(\frac{\partial\mu_i}{\partial\eta_i})^2$ [44]. The sampling distribution of the maximum likelihood estimator $\hat{\beta}_j$ $(j = 1, \ldots, p)$ is approximately normal for large samples:

$$u_j = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma_{jj}}} \sim Z \tag{3.17}$$

where σ_{jj} is the j$^{\text{th}}$ entry of the diagonal of the variance-covariance matrix in 3.16. The distribution is exact for normally distributed response variables since $\hat{\beta}$ in (3.2) is a linear combination of the observations Y_i [44]. Therefore, the $(1-\alpha)*100\%$ confidence interval for β_j can be computed as:

$$\hat{\beta}_j \pm z_{1-\frac{\alpha}{2}}.se(\hat{\beta}_j) \tag{3.18}$$

where the standard error of the j$^{\text{th}}$ parameter is calculated by the j$^{\text{th}}$ entry of the diagonal of the variance-covariance matrix in (3.16) ($se(\hat{\beta}_j) = \sqrt{\sigma_{jj}}$). In the case that there is a scale parameter ϕ to estimate (e.g. σ for normal response), the sampling distribution of $\hat{\beta}$ is defined as follow:

$$u_j = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\phi}\sigma_{jj}}} \sim t(n-p) \tag{3.19}$$

Hence, the t-score ($t_{(n-p;\ 1-\frac{\alpha}{2})}$) needs to be used as the critical value instead of the z-score in (3.18) to compute the confidence interval in this case.
As an example, consider the logistic regression fitted to the Titanic example in Table 3.2. There is no scale parameter to estimate and the coefficient for 'age' is estimated as $\hat{\beta}_{age} = -0.0344$ with the standard error $se(\hat{\beta}_{age}) = \sqrt{\sigma_{22}} = 0.006331$. Therefore, a 95% confidence interval for the 'age' using (3.18) is computed as:

$$\begin{aligned} &\hat{\beta_{age}} \pm z_{(0.975)}.\sqrt{\sigma_{22}} \\ \approx\ &[-0.0468, \ -0.0220] \end{aligned}$$

The confidence interval above suggests that age in years has a significant negative effect on the probability of survival as described in the previous sections. In other words, young people had more chance to survive the Titanic disaster.

Prediction intervals

One of the key points when fitting regression models is to have accurate predictions when estimating the mean response conditional on a set of given explanatory variables. Once the model is fitted, and parameters are estimated, it is quite straightforward to predict the response variable based on a set of explanatory variables using the estimated model equation. However, it is desired that the predicted quantities are accompanied by measures of precision [97].

An approximate $(1-\alpha)*100\%$ confidence interval for the mean response can be obtain using:

$$\hat{\eta}_i \pm u_{1-\frac{\alpha}{2}}.\sqrt{\hat{\phi}h} \tag{3.20}$$

where $\hat{\eta}_i = x_i\hat{\beta}$ is the linear prediction, $u_{1-\frac{\alpha}{2}}$ is the proper critical value depending on the significance level and $h = x^T\Sigma x$. For some distributions such as the binomial, Poisson and exponential, there is no scale parameter to estimate ($\phi = 1$) and the z-score can be used as the critical value in (3.20), but in the cases where there is a scale parameter to estimate (e.g. σ for normal responses), the t-score will be used as the critical value.

In contrast to this, the prediction interval for the linear prediction of a new observation ($\eta^* = x^*\beta$) is defined as:

$$\hat{\eta}^* \pm u_{1-\frac{\alpha}{2}}.\sqrt{\hat{\phi}(1+h^*)} \tag{3.21}$$

where $h^* = x^{*T}\Sigma x^*$.

The confidence interval for the mean response will quantify the uncertainty around the mean response estimate for a given set of explanatory variables while the prediction interval for a single new observation will quantify the uncertainty around the prediction of a future observation. The prediction interval is wider than the confidence interval due to the additional variation in the single subject rather than the average over many subjects.

The corresponding intervals to the (3.20) and (3.21) in terms of the fitted values ($\hat{\mu}$) are obtained when applying the inverse link transformation $g^{-1}(.)$ to the confidence limits. This transformation will produce confidence intervals on the same scale as the response variable which is best for interpretation.

Once more consider the logistic model fitted to the Titanic dataset in Table 3.2, where we are interested in the probability that 25 years old male passengers with 1st class ticket (x = [age=25, pclass="1st", sex="male"]) will survive. The linear prediction will be estimated approximately as $\hat{\eta} = 0.164$ when applying x in (3.9) so that the survival probability would be approximate as $\hat{\pi} = \frac{\exp(\hat{\eta})}{1+\exp(\hat{\eta})} = 0.54$. There is no scale parameter to estimate ($\phi = 1$) and $x^T \Sigma x = 0.0325$. Therefore, the 95% confidence interval for the linear prediction ($\eta = x\beta$) is obtained as:

$$\hat{\eta_i} \pm z_{(0.975)}.\sqrt{h} \;\approx\; [-0.1887,\; 0.5175]$$

Similarly, the 95% prediction interval for the linear prediction for the new observation ($x^* =$ [age=25, pclass="1st", sex="male"]) is obtain as:

$$\hat{\eta_i^*} \pm z_{(0.975)}.\sqrt{h^*} \;\approx\; [-1.8271,\; 2.1560]$$

However, the confidence interval and the prediction interval above can be transformed back to the probability scale as [0.453, 0.627] and [0.139, 0.896] respectively, using the inverse logit function ($\mu = \frac{\exp(\eta)}{1+\exp(\eta)}$).

3.3 Modelling time-to-event data

Time-to-event data refers to a type of outcome that measures the time from an origin time-point until an event happens. It is also called the survival time or failure time. As an example within medical content, the outcome of interest might be the death from a particular disease; the time to recurrence of an illness after an initial recovery; or the time until the patient is cured. The event of interest could be positive such as discharge from the hospital, negative such as death due to breast cancer, or even neutral such as cessation of breastfeeding.
Time-to-event data have two important characteristics which make the analysis of these data different from other response variables [44]:

- The time variables are non-negative and often skewed distributions with long tails;
- Survival times for some of the subjects might not be known completely (e.g. someone survived beyond the end of the study), sometimes called *censoring*.

In general, censoring might occur due to following reasons:

- ⊠ *Study termination*: the event does not occur before the study ends;
- ⊠ *Loss to follow up*: the unit of observation withdrew from the study for any reason during follow up;
- ⊠ *Competing risks*: some other events such as death from another cause occurs which prevents the subject from being observed.

Censoring is classified into the following three categories depending on whether the event happens:

- after the censored time (right censoring);
- before the censored time (left censoring);
- between two censored times (interval censoring).

The most common method of censoring is right censoring where the event of interest occurs at an unknown time-point after the censored time. This type of censoring could happen randomly during the study time (patient dropout), or due to the study termination. In the latter case, the latest event-free time-point is considered for the censoring time (i.e. the most recent information about the subject's survival status). In general, the right censoring is classified in three following categories:

Type I censoring: The length of study is set in advance. As a result, those who did not experience the event before the end of the study will be censored.

Type II censoring: The number of events (k) is set up in advance. As a result, the study will end after k events occur, and the rest of observations will be considered as censored.

Random censoring: A censoring time and an event time are assumed for each observation. Then, the observation is recorded as censored if the censoring time occurs before the event time.

As previously stated, random censoring can occur randomly at any time during the study, while for Type I and Type II censoring, the censoring happens at the end of the study time. In medical studies, it is most common to have a mixture of random censoring and Type I censoring.

A censored observation is an incomplete observation, not a missing value; it contains partial information about the event time [145]. This means that ignoring the censored observations will lead to a biased estimate of the parameter of interest. Therefore, the main challenge in survival analysis is to account for this partial information throughout the analysis. A survival outcome is formed of two variables; a continuous variable representing the observed survival time, and a binary variable (status) indicating whether the observed time corresponds to the event time or the censoring time. The status variable is frequently coded as 1 if the event occurred and 0 if censoring occurred.

In most approaches in survival analysis, the censoring time and the event time are assumed to be independent. This assumption helps to easily deal with the censoring mechanism as is described later. Let T and C be two independent variables denoting the time-to-event and censoring time respectively. Consider the pair (X, δ) where X is the minimum observed time and $(X = \min(T, C))$ and δ is the censoring indicator (called status). The cumulative distribution function for the positive range of $\mathbb{R}$ determines as:

$$F(t) = P(X \leq t) \tag{3.22}$$

The cdf at time t represents the probability that the event has occurred before time t.

3.3.1 Survival function

In survival analysis, unlike the analysis of other types of outcomes, it is common to summarise the survival times using the survival function (and not just using a numerical summary like the mean or the median). The estimates can be obtained parametrically and non-parametrically. The survival function is the complement of the cumulative distribution function $F(t)$, which in the medical context represents the event-free survival experience of an individual. The survival function $S(t)$ is the probability that a subject survives (the event) beyond a particular time t which is calculated by:

$$S(t) = P(X > t) = 1 - F(t) \tag{3.23}$$

Therefore, the survival function is monotone decreasing (i.e. it heads downward as t increases; i.e. $S(a) \geq S(b)$ if time $a < b$), and it ranges from 0 to 1. Figure 3.1 shows an example of a survival function.

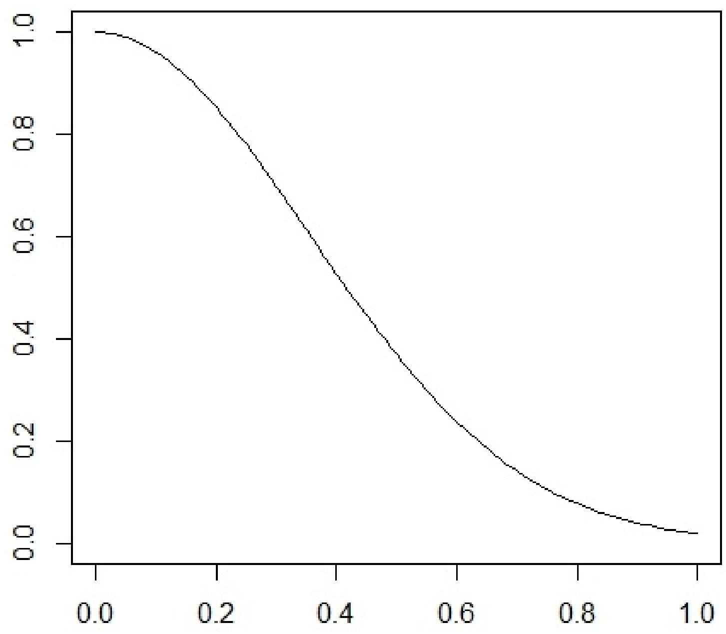

Figure 3.1: The plot of a survival function

Due to the skewed nature of the time distributions, the median is typically used as the preferred summary with time-to-event data. The estimated median survival time t_m can be calculated as the solution of the following expression:

$$S(t_m) = \frac{1}{2} \tag{3.24}$$

As mentioned previously, the usual way of summarising survival data is by using the survival function, which in the absence of censoring can be easily estimated using the empirical survival function as:

$$\hat{S}(t) = \frac{1}{n}\sum_{i=1}^{n} I(t_i \geq t) \tag{3.25}$$

where t_i are the observed survival times and I is the indicator function that takes

1 if the expression in brackets is true and 0 otherwise. So $\hat{S}(t)$ is the observed proportion of subjects alive at time t.

In the presence of censoring, the empirical estimator does not provide a reliable estimate of the survival function. In this case, it can be computed using a parametric method, assuming a specific distribution for X, or using a non-parametric approach such as the Kaplan-Meier estimator explained later in this chapter.

Hazard function

One way of characterising the time distribution is by using the hazard function. It provides the instantaneous failure rate for the event occurring in the next unit of time, conditional on the subject not having experienced the event up to time t.

The hazard function plays a significant role when modelling the effect of covariates/factors on survival time in survival analysis. $h(t)$ denotes the hazard function at time t which is defined as:

$$h(t) = \lim_{dt \to 0} \frac{P(t \leq X \leq t + dt | X \geq t)}{dt} \tag{3.26}$$

The hazard function can be written as:

$$h(t) = \frac{f(t)}{S(t)} = \frac{F^{'}(t)}{1 - F(t)} \tag{3.27}$$

The cumulative hazard function H(x) is is a non-decreasing function of t defined as

$$H(t) = \int_{0}^{t} h(x)dx \tag{3.28}$$

The cumulative hazard function can be written in terms of the survival function by simply substituting the hazard function in (3.27) in (3.28) and computing the integral as:

$$H(t) = -\log(S(t)) \tag{3.29}$$

The survival function can be represented in terms of the cumulative hazard function as:

$$S(t) = e^{-H(t)}$$

The survival function demonstrates the probability that an event will not happen while the hazard function is the opposite; it focuses on events happening. It plays a significant role in analysing time-to-event data as the effects of covariates or factors on the survival time can be modelled through the hazard function. We will return to this in more detail in Section 3.3.2.

Survival function estimation

The survival and hazard functions can be estimated using parametric methods which are well studied in the literature (see [89, 90]). The most commonly used distributions for survival outcome include the exponential, Weibull, gamma, or lognormal. Estimation of the parameters can be obtained by maximising the likelihood function usually using the numerical approach; however, there is not always enough information about the underlying survival distribution, particularly the tail behaviour.
The likelihood function of the n observed pairs $(t_1, \delta_1), \ldots, (t_n, \delta_n)$ defined by expression (3.30) is the product of the density functions, where the contribution of the censored observations are through the survival function,

$$L(t, \delta) = \prod_{i=1}^{n} f_X(t_i)^{\delta_i} S_X(t_i)^{1-\delta_i}, \tag{3.30}$$

where $f_X(t)$ is the density function of X and $S_X(t)$ is the survival function of X. The likelihood function comprises two parts; one part includes the event times and the other one includes the censored times where the censoring mechanism is assumed to be independent of the event times.
In contrast to the parametric methods, the Kaplan-Meier and Nelson-Aalen estimators are two well-known non-parametric techniques to estimate the survival function. The Kaplan-Meier (KM) estimator proposed by Kaplan and Meier in 1958 [83] is

defined at time t as:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{e_i}{r_i}\right) \tag{3.31}$$

where e_i is the number of events at time t_i and r_i is the number of subjects at risk at t_i (i.e. those who are alive just before t_i). The notion of the risk set [12] in survival analysis is key to incorporate the censoring information to the analysis. The KM estimator can be thought of as the multiplication of the estimated survival probabilities at every single event time until t. In the case that no censored observations exist, the KM estimator is the same as the empirical estimator of the survival function.

An alternative estimator of the survival function is defined as:

$$\hat{S}_{NA}(t) = \exp(-\hat{H}_{NA}(t)) \tag{3.32}$$

where $\hat{H}_{NA}(t) = \sum_{t_i \leq t} \frac{e_i}{r_i}$ is the estimate of the cumulative hazard function known as the Nelson-Aalen (NA) estimate which was introduced by Nelson [105, 106] and generalised by Aalen [1, 2]. The NA estimator is an increasing right-continuous step function with increments at the event times.

The standard approach to estimate the survival function and graphically display the survival data is by using of the Kaplan-Meier (KM) estimator. This non-parametric estimate of the survival function does not make any assumptions about the distribution of the survival times. The Kaplan-Meier estimator is based on the following premises [60]:

- at any time patients who are censored have the same survival likelihoods as those who continue to be followed (i.e. censored and observed times are independent);
- the survival probabilities are the same for subjects recruited early and late in the study;
- the event happens at the time specified (i.e. there is no measurement error in the event times).

The KM estimates of the survival function are a step function with jumps at the observed event times. In other words, the survival probability remains constant between the events, even if there are some intermediate censored observations [60]. The median is easy to obtain from a plotted KM estimate by locating the time where the survival probability is equal to 0.5. Figure 3.2 shows the KM estimate of the Lung cancer data in Section 1.2.3 which is used to compare the survival probabilities of males vs females.

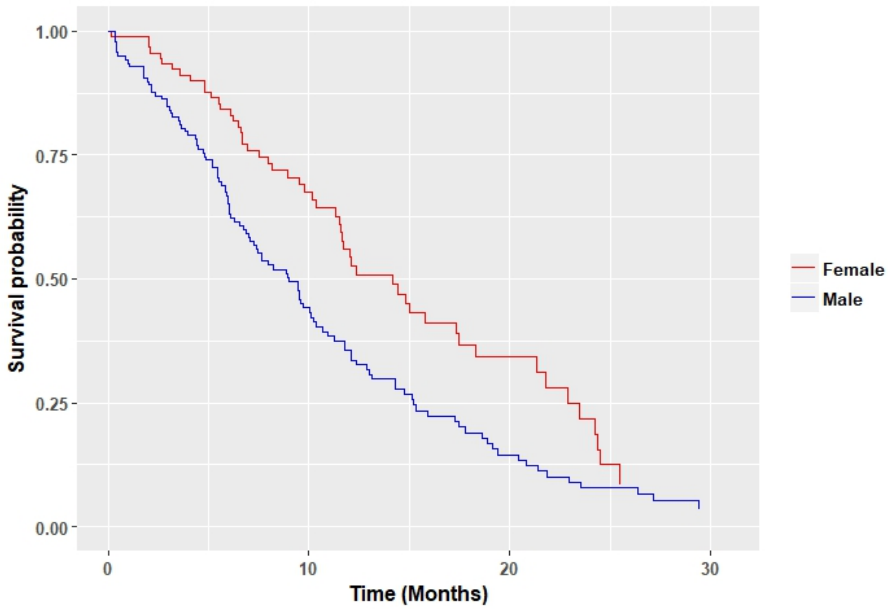

Figure 3.2: KM estimate of the survival function for patients with advanced lung cancer grouped by sex (Lung cancer data)

Figure 3.2 indicates that females with lung cancer are expected to have a higher survival time with a median survival of 14.2 months compared to males with a median survival of 9 months. A different example of the Kaplan-Meier estimate is given in Figure 3.3 where the time on treatment until healing is estimated for low birth weight infants with bronchopulmonary dysplasia, in order to compare the therapy users with the non-therapy group in the BPD data in Section 1.2.5. The variable of interest is time-to-cure which represents a positive event, unlike the death in the Lung cancer dataset. Therefore the lower survival function in the therapy group (in blue) indicates a better situation compared to the control group (in red), although they was similar until about 60 days. Hence, the time-to-cure is shorter for those infants on the oxygen therapy as evidenced by the more rapidly decreasing survival function.

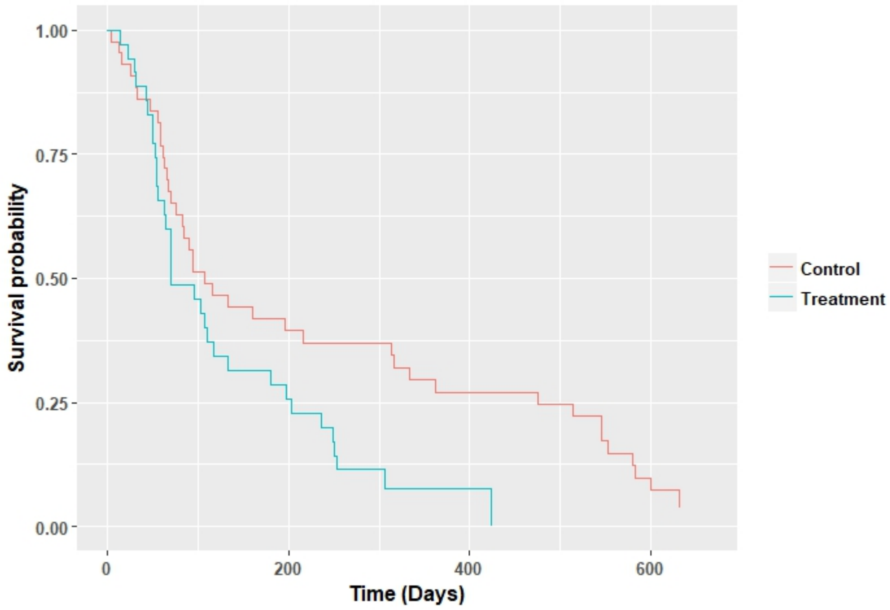

Figure 3.3: The estimated survival function of time-to-cure grouped by therapy in BPD data

Variance of the estimated survival function

Peto [114] and Greenwood [63] introduced formulas to compute the standard errors of the KM estimate at any given time. The Greenwood variance estimator formula for the KM estimator in (3.31) is given by:

$$\hat{V}(\hat{S}(t)) = \hat{S}(t)^2 \sum_{t_i \leq t} \frac{e_i}{r_i(r_i - e_i)} \tag{3.33}$$

where $\hat{S}(t)$ is the survival estimate which could be obtained by any method such as the KM estimator.

The variance of the Nelson-Aalen estimator [19] will be computed as:

$$\hat{V}(\hat{H}(t)) = \sum_{t_i \leq t} \frac{(r_i - e_i)e_i}{(r_i - 1)r_i^2} \tag{3.34}$$

If we can assume that events are not happening at the same time (i.e. there are no tied observations), the variance in (3.34) is:

$$\hat{V}(\hat{H}(t)) = \sum_{t_i \leq t} \frac{1}{r_i^2} \tag{3.35}$$

The variance of the survival function in either (3.33) or (3.34) would give an insight into the accuracy of the estimation; however, a confidence interval for the survival estimate provides better intuition about the precision of the estimation.

Confidence Intervals for the Survival Function

The standard error in (3.33) can be used to create confidence intervals for the survival function at any fixed time t_0. A $(1-\alpha) * 100\%$ confidence interval for the survival function at a given time t_0 can be obtained using the KM estimator $\hat{S}(t)$ as:

$$\hat{S}(t_0) \pm z_{1-\frac{\alpha}{2}} \sigma_n(t_0) \hat{S}(t_0) \tag{3.36}$$

where $z_{1-\frac{\alpha}{2}}$ is the $1-\frac{\alpha}{2}$ quantile of the standard normal distribution, $\sigma_n(t) = \frac{\hat{V}(\hat{S}(t))}{\hat{S}(t)^2}$ and $\hat{V}(\hat{S}(t))$ is the Greenwood's variance estimator in (3.33) [86].
Borgan [20] proposed improved confidence intervals constructed using the log-minus-log transformation of the survival estimation $(\hat{S}(t))$ to map onto $\mathbb{R}$. The $(1-\alpha) *$ 100% log-minus-log transformation confidence interval is defined as:

$$\hat{S}(t_0)^{\exp\left\{\pm \frac{z_{\frac{\alpha}{2}} \sigma_n(t_0)}{ln(\hat{S}(t_0))}\right\}} \tag{3.37}$$

These pointwise confidence intervals for the survival function are acceptable when making a related inference in a single time-point. One common mistake is to try to make inference based on these pointwise intervals which is just a collection of confidence intervals for all time-points, and they do not maintain the coverage level overall.
Two methods by Nair [103] and Hall and Wellner [68] are available to construct the confidence bands over the range $[a_L, a_U]$, which contain the actual survival function

for all t with a given confidence level. They search for the two random functions $L(t)$ and $U(t)$ such that $P\left(L(t) \leq S(t) \leq U(t); t \in (a_L, a_U)\right) = 1 - \alpha$.
The first method by Nair is known as the equal probability (EP) bands, and the log-transformed form of it expressed as:

$$\left(\hat{S}(t)^{\frac{1}{\theta}}, \hat{S}(t)^{\theta}\right), \quad \text{where } \theta = \exp\left\{\frac{c_\alpha(a_L, a_U)\sigma_n(t)}{ln(\hat{S}(t))}\right\} \tag{3.38}$$

The $c_\alpha(a_L, a_U)$ is the confidence coefficient of EP confidence band presented in Table C.3 in Appendix C of the Klein book [86]. The alternative log-transformed form of the confidence bands by Hall and Wellner is expressed as:

$$\left(\hat{S}(t)^{\frac{1}{\theta}}, \hat{S}(t)^{\theta}\right), \quad \text{where } \theta = \exp\left\{\frac{k_\alpha(a_L, a_U)[1 + n\sigma_n(t)]}{n^{\frac{1}{2}ln(\hat{S}(t))}}\right\} \tag{3.39}$$

The $k_\alpha(a_L, a_U)$ is the confidence coefficient of Hall-Wellner confidence bands presented in Table C.4 of Appendix C in the Klein book [86].
An illustrative example of the confidence interval and confidence band of the survival curve estimated for time to death in the Lung cancer data is demonstrated in Figure 3.4.

Methods to compute confidence intervals for survival functions are available in almost all statistical software packages; however, the confidence bands usually require a little more effort. For example in **R**, the '`survival`' package, which is the standard package for survival analysis, estimates and plots only confidence intervals. The confidence bands are accessible through other packages. For example, the '`OIsurv`' package includes a function ('`confBands`') to estimate the confidence bands for a survival curve using the both 'Hall-Wellner' and 'EP' methods.

Comparison of different survival populations

In applications, it would be of interest to compare the survival prospects of two or more groups, for example, to compare the survival probabilities of patients on a standard therapy with a new therapy. One simple comparison mentioned by Goel [60] is by plotting the survival curve and looking for the vertical and horizontal

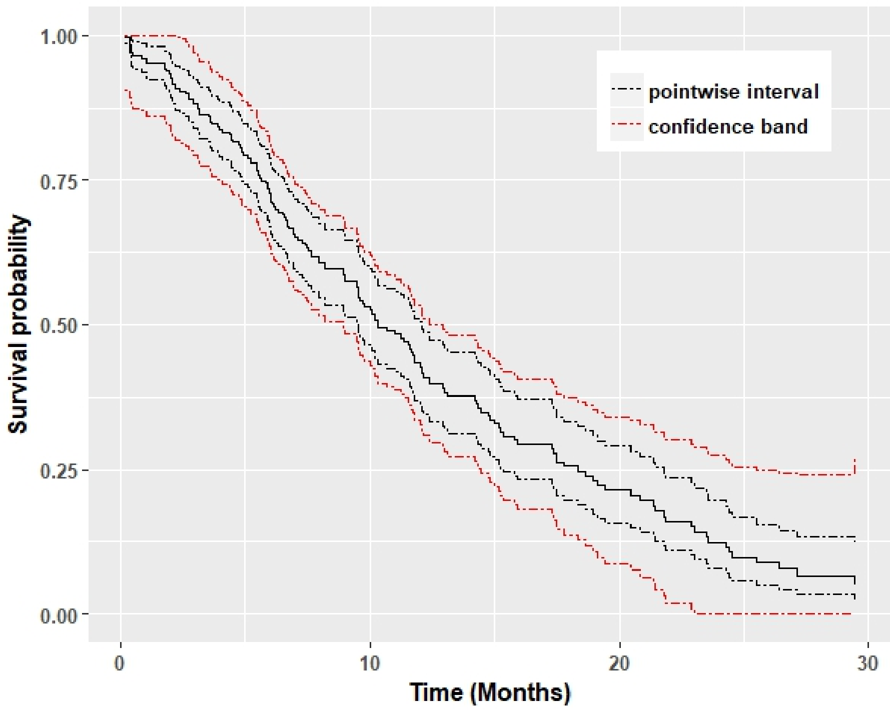

Figure 3.4: The pointwise confidence interval and the Hall-Wellner confidence band estimated for the Lung cancer data

gaps. A vertical gap shows the difference in the survival probability of two groups for any specified time-point, while, a horizontal gap suggests that it took longer for one group to experience the event.

Including confidence bands to explore if they overlap is useful and common; however, they would be difficult to interpret when there are several intervals. In order to represent the uncertainty of $\hat{S}(t)$, the 'number of subjects at risk' could be incorporated to the plot since this indicates the relative sample size used to make the estimation. This measure is obviously decreasing over time as subjects either experience the event or are censored. This information can be incorporated to the KM graph by attaching a table containing the number at risk at each time-point below the x-axis. An example for the BPD data is given in Figure 3.5.

The alternative way to incorporate the number of subject at risk is to use transparency (also known as alpha-blending) in the lightness of the curve as a function of the number at risk at each time-point. Thus the survival curve becomes lighter over time as the risk set get smaller. An example of the use of alpha-blending for the BPD data is given in Figure 3.6.

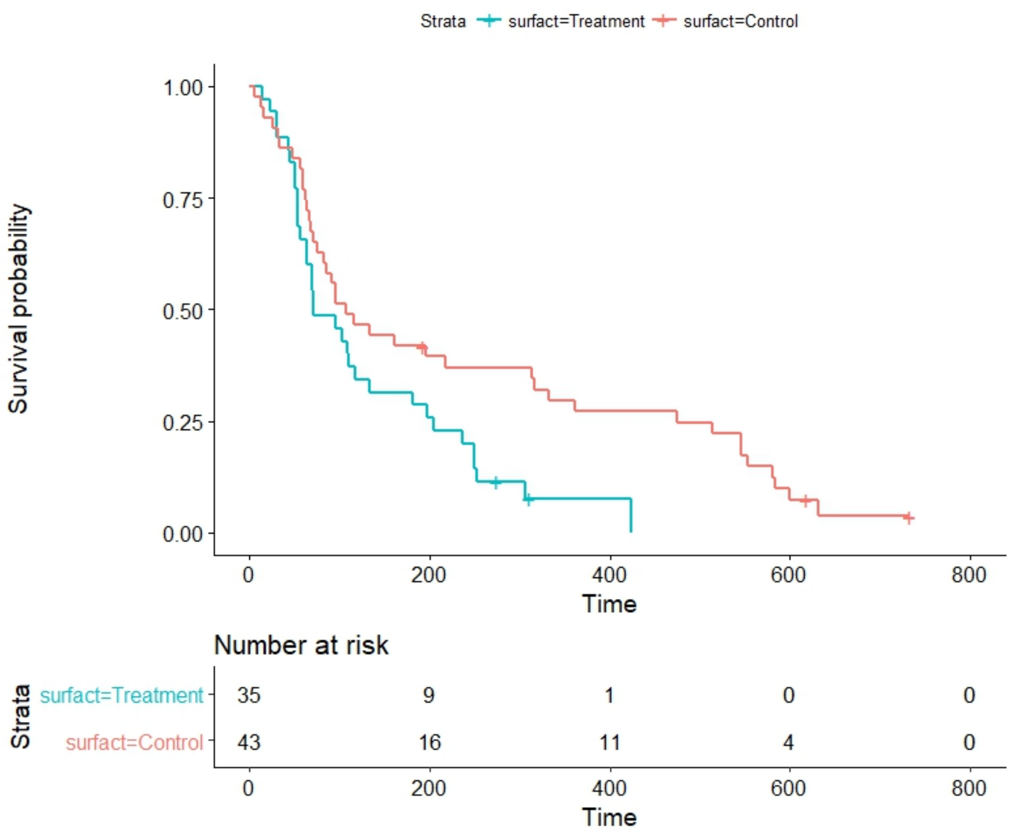

Figure 3.5: The KM plot with at risk table for the BPD data

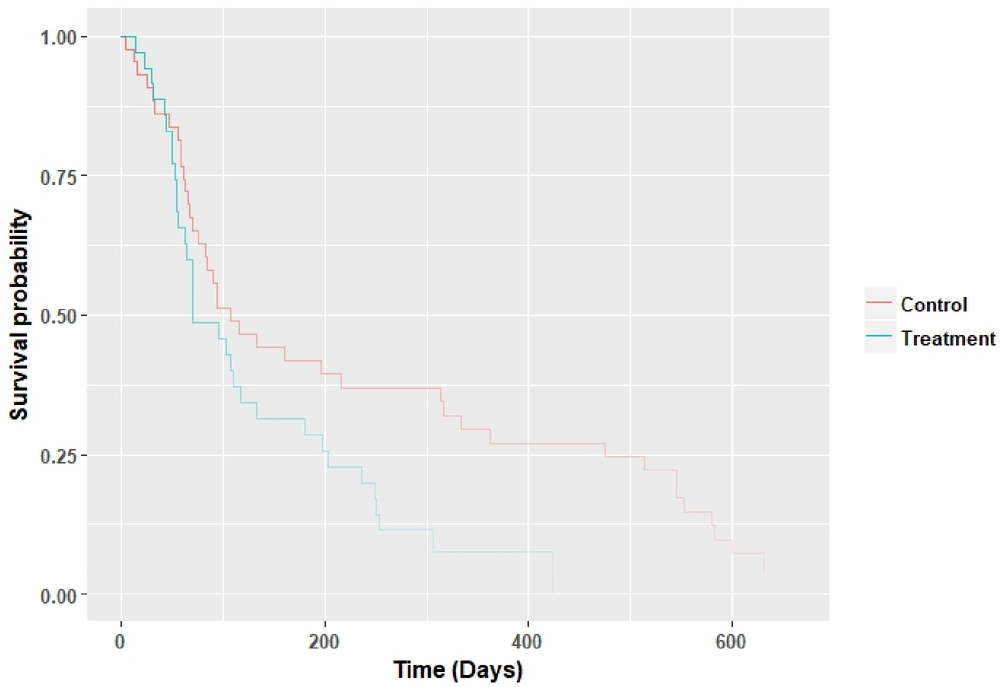

Figure 3.6: The KM plot with alpha-blending for the BPD data

The use of alpha-blending also provides a notion of how accurate an estimation is by not adding more lines. Although confidence intervals provide more informative measurements of the precision, the alpha-blending is beneficial when having more than one survival curve. In these cases, plotting several confidence intervals may clutter the graph and prevent a clear understanding. The use of coloured shaded regions to represent the uncertainty could be beneficial when only two groups are present but may be difficult for three or more groups.

Alternative ways proposed to avoid this confusion for the case of a two-level comparison is by collapsing the dimension, and ploting a new estimated measure along with its confidence intervals. Parzen et al. [112] have suggested the difference in survival functions with its approximate confidence interval and to use the line at 0 as the reference line. The '`survdiffplot`' function in `rms` package [73] in **R** computes and plots the survival difference with an approximate confidence interval. An example of it is given in Figure 3.7.

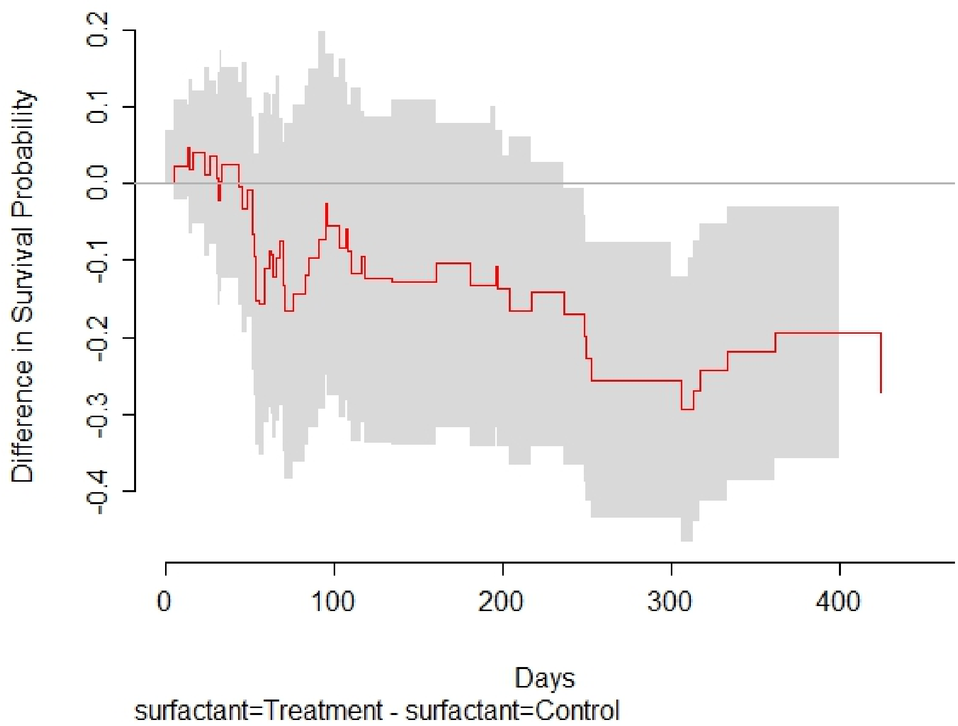

Figure 3.7: The survival difference plot with the confidence interval for the BPD data

Newell et al. [108] suggested the ratio of survival functions as the measurement of interest and proposed a method based on resampling techniques to produce uncertainty band.

The log-rank test

All the previous methods provide comparisons of different survival curves at specific time-points; however, we might be interested in the comparison of the total survival experience of two or more survival distributions. Hypothesis tests are available using the log-rank test or the z-test for the coefficients in the Cox proportional hazards model. The log-rank test is the most commonly-used statistical test used for such a comparison. The null hypothesis is that the survival probabilities are identical at any time-point in all the populations; in the case of two levels, the hypotheses can be written as:

$$\begin{aligned} H_0 &: S_1(t) = S_2(t) \\ H_1 &: S_1(t) \neq S_2(t) \end{aligned}$$

The log-rank test is a type of chi-squared test. It is a non-parametric test which is approximated as:

$$\text{Log-rank statistics} \approx \sum_i \frac{(O_i - E_i)^2}{E_i} \tag{3.40}$$

where O_i is the observed number of events and E_i is the expected number of events in group i if there were no difference between groups [87]. Alternative tests for comparison of survival distributions are also available in the literature; for example, Wilcoxcon [146], Tarone-Ware [130] or Peto [113] tests which have similar test statistics to the logrank test but using different weighting methods.

An example of the result of a logrank test performed for the BPD data in Section 1.2.5 is illustrated in Table 3.4.

Table 3.4: The log-rank summary for the BPD data

	Chi-Square	df	Sig.
Log-rank test	5.6	1	0.0178

The small p-value (0.0178) in Table 3.4 provides evidence that the difference in time to get off the oxygen treatment of infants with bronchopulmonary dysplasia is statistically significant for those who used the surfactant therapy compared to those not on the treatment. Therefore, we can conclude that overall, time-to-cure is shorter for the surfactant therapy users as illustrated in the plot of their estimated survival function (Figure 3.3).

Bland [18] suggests to always plot the survival curve when using the log-rank test as it does not distinguish the difference when survival curves cross. One practical idea is to add the log-rank test p-value to the KM plot when it is applicable.

The log-rank test provides the total comparison of survival experience but not an estimate of the magnitude of the difference between populations or a confidence interval. The Cox proportional hazards model, which is discussed in the next section can be used here.

3.3.2 Cox proportional hazards model

The Cox proportional hazards model is the most popular statistical technique to model time-to-event data [37]. This model investigates the potential effect of prognostic factors on the survival time by expressing the hazard function as a function of the explanatory variables and an unknown baseline hazard function [37]. The Cox model structure allows us to estimate the treatment effect after adjusting by other predictors. The proportional hazards models for a set of variables (X) is formulated by:

$$h(t) = h_0(t)e^{X^T\beta} \tag{3.41}$$

where $h_0(t)$ is the baseline hazard function which is an unknown function of time.

The Cox proportional hazards model is a semi-parametric model that uses no particular parametric assumption for the survival times. In other words, it assumes that the hazard at time t is the product of a baseline hazard function that depends on t and an exponential form of a linear function of all explanatory variables.

The main assumption of this model is the proportionality of hazards. It means that the hazard functions of different individuals are assumed to be proportional, for example, if a covariate doubles the risk of the event at a given time, it always doubles the risk of the event on any other time. This can be demonstrated when dividing the hazards models for two distinct individuals (X_1 and X_2) as follows:

$$\frac{h(t|X_1)}{h(t|X_2)} = \frac{h_0(t)e^{X_1^T\beta}}{h_0(t)e^{X_2^T\beta}} = \frac{e^{X_1^T\beta}}{e^{X_2^T\beta}} = e^{(X_1-X_2)^T\beta}$$

One of the key advantages of this model is that it does not require specification of the underlying distribution. The model parameters can be estimated by the partial likelihood method while the baseline hazard is undefined [37].
A positive coefficient β_i indicates that, as the covariate increases, the event risk increases so that the survival probability decreases; in contrast, a negative coefficient $\hat{\beta}_i$ negatively associates with the event risk. This is also visible in the hazard ratio (HR) which is defined by e^{β_i}. To sum up, the covariate increases if:

- **HR $>$ 1:** hazard increases, survival decreases;
- **HR $<$ 1:** hazard decreases, survival increases;
- **HR $=$ 1:** hazard and survival remain constant.

A Cox proportional hazards model is fitted to the Lung cancer data where age ('Age'), gender ('Sex') of patients, the ECOG performance score (varying from 0 = *good* to 5 = *dead*) and the amount of weight loss in last six months ('wt.loss') are considered as explanatory variables. Table 3.5 presents the parameter estimates and 95% confidence intervals and the p-values of the Cox proportional hazards model fitted to the Lung cancer data on the log scale and as well as the hazard ratios.

	Estimate	Std. Error	HR	HR CI	z-value	p-value
age	0.0134	0.0096	1.01	c(0.99, 1.03)	1.39	0.1650
ph.ecog	0.5151	0.1260	1.67	c(1.31, 2.14)	4.09	<0.0001
wt.loss	-0.0090	0.0067	0.99	c(0.98, 1.00)	-1.35	0.1761
sex(female)	-0.5908	0.1753	0.55	c(0.39, 0.78)	-3.37	0.0008

Table 3.5: The Cox proportional hazards model summary for the Lung cancer data

In general, there is a significant positive effect of the ECOG performance score and significant effect of gender. The risk of death in female patients is about half of male patients (HR= 0.55). Also, a one unit increase in the ECOG score is associated with approximately 0.52 increase in the log hazard rate. In other words, there is a 67% risk increase as the ECOG score increases by one unit.

3.3.3 Dynamic prediction of survival (DPS) function

Predicting the response is of crucial importance in clinical research involving patients. There is an enormous amount of studies for the prediction of time to events, typically using a modeling approach. There is also an enormous amount of studies for prediction of survival after diagnosis based on statistical models [109]. They mostly use static models, in particular, the Cox proportional hazards model for remaining lifetime after diagnosis or treatment. In contrast, dynamic models by using landmark models focus on later points in time [139].

The word landmarking [13] is a term used in the medical literature to refer to the process that dynamically adjusts predictive models for survival data during the follow up [140]. The multi-state models [117, 100] use landmarking to compute such a predictive probability. The dynamic prediction for survival analysis was proposed by van Houwelingen and Putter [140, 141, 139] as an alternative approach to the multi-state model. This concept has also appeared in the former literature; see Madsen (1983) [94], Christensen (1986) [34] and Klein (1993) [85], and more recently by Proust-Lima and Taylor, (2009) [115] and Parast et al., (2011) [111].

In this context, the 'conditional failure function' is defined by van Houwelingen and Putter [139] as:

$$F_w(t) = P(T \leq t + w | T \geq t) \tag{3.42}$$

where w represent a fixed window width ($w > 0$). The window width is a fixed arbitrary value which is chosen by considering the overall prognosis time and based on the period of follow-up. The function in (3.42) represents the probability of occurrence of the event within a fixed window given survival up to that point, which supply useful information for the patient at any time during the follow-up. It can be written as:

$$F_w(t) = 1 - S(t + w | T \geq t) = 1 - \frac{S(t + w)}{S(t)} \tag{3.43}$$

where $S(.)$ is the survival function. We call the expression $F_w(t)$ the dynamic prediction of survival (DPS) function. The DPS function is estimated on the negative exponential of the Nelson-Aalen (NA) estimate in (3.32), it then can be transformed to the probability scale using the following formula:

$$F_w = 1 - e^{-H_w} \tag{3.44}$$

where $H_w = H(t + w|T \geq t)$ is the conditional cumulative hazard.

The endpoint in the Lung cancer data in Section 1.2.3 is time to death and the maximum survival time is 29.4 months. A window width of 6 months ($w = 6$) is chosen to represent the DPS estimate for this data in Figure 3.8.

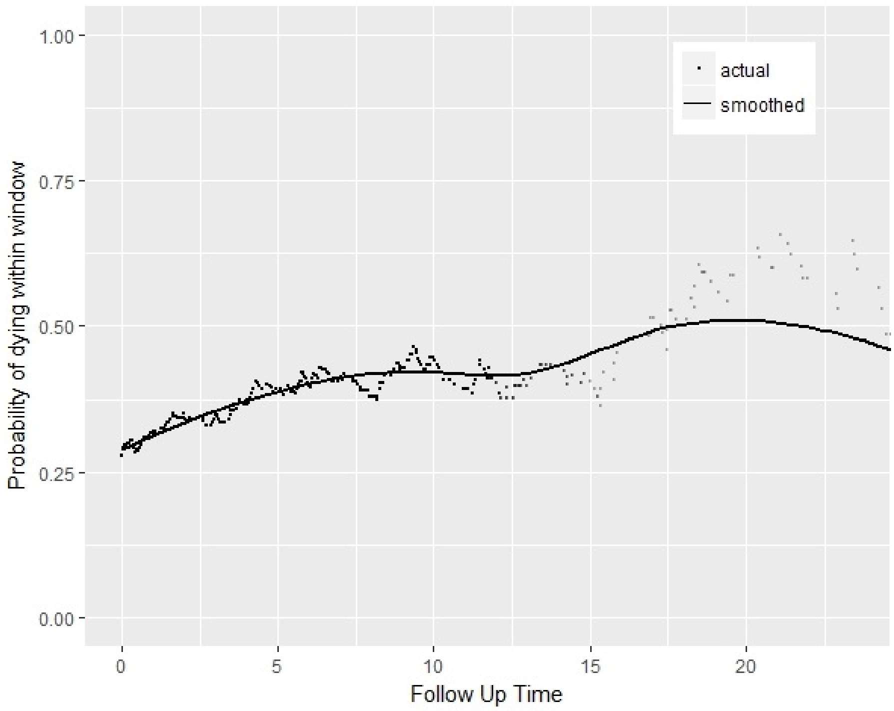

Figure 3.8: Probability of dying within the next 6 months for Lung cancer dataset; points represent the actual estimations, and the line represents a smoothed estimation line

The probability of dying within the next six months in Figure 3.8 is rising gradually from 0.3 to 0.6. The DPS estimate in Figure 3.8 shows that, for example, the probability of dying within the next 6 months for a patient who is alive after 18 months is 0.5.

The estimate of DPS function for this dataset grouped by gender is also given in

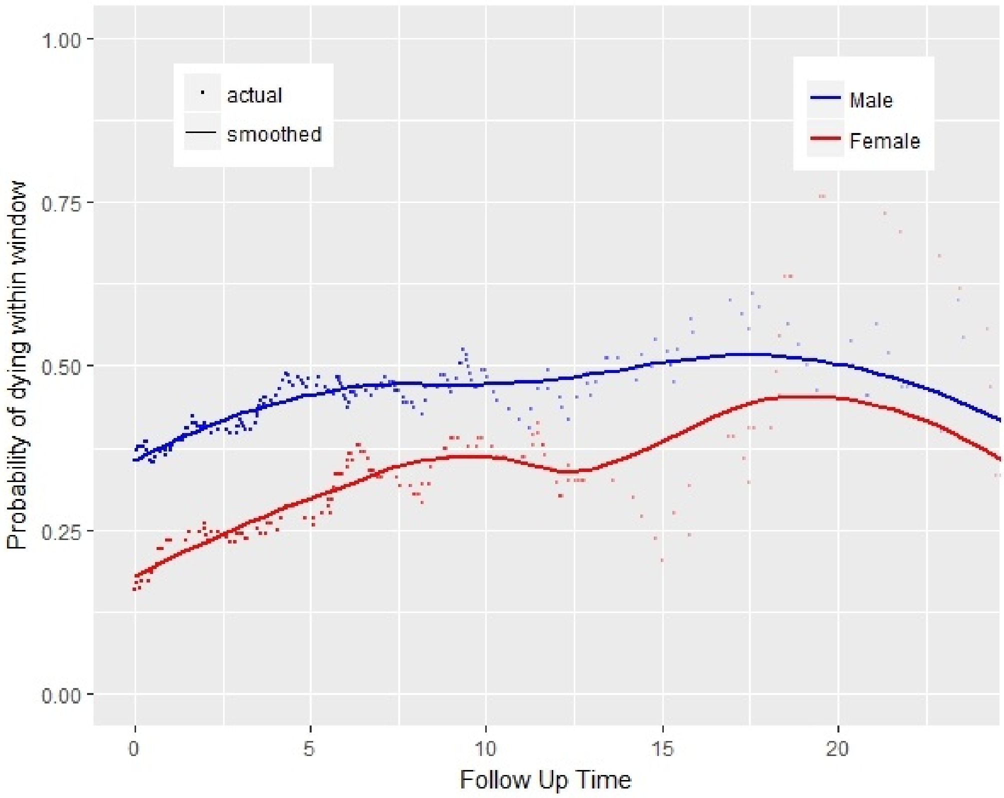

Figure 3.9: Probability of dying within the next 6 months for males and females in the Lung cancer dataset; points represent the actual estimations, and lines represent smoothed estimation lines

Figure 3.9 where lines are smoothed to help the interpretation of the plot. It indicates a similar pattern for both groups where the probability of dying within the next 6 months for females are about 0.2 lower than for males at the start; however there is only a slight difference of about 0.05 after 20 months.

The variance of DPS estimate can also be estimated in the same way as its estimation using the Nelson-Aalen estimate in (3.34). Therefore, the pointwise confidence intervals are fairly simple to compute in the cumulative hazard scale and transformed to the probability scale using in (3.44) afterwards. Putter has made available an **R** package called '`dynpred`' [116] which contains functions for estimating DPS functions with pointwise confidence intervals.

3.3.4 Mean residual life (MRL) function

The mean residual life (MRL) function has been used as far back as the third century A.D. [42, 51]. The MRL function at any given time provides the expected remaining lifetime given survival up to that time. Its use in engineering and reliability studies was introduced by several authors (see Guess (1988) [64], Hollander (1975) [75] and Bryson (1969) [25]). The use of the MRL function in the biomedical context as a summary of the patient's survival experience has several advantages.

In practice, the summary given to the patient is typically the median survival time which is open to misinterpretation. Although the median is given in units of time, it has a probabilistic interpretation that is unlikely to be relevant at the patient level. The typical summary based on median might be "the treatment option gives the patient, the median survival time of t_1 months compared to t_2 months without the medical care".

In contrast, the MRL is defined in units of time, and the message, for example, could be "The expected remaining time of a patient using treatment (given survival up to a particular time t) is n extra months compared to a patient with the same background but not using the treatment". This supplies useful information for the patient at any given time during the follow-up. The MRL function at time $t \geq 0$ is defined as:

$$m(t) = E(T - t|T > t) = \frac{1}{S(t)} \int_t^{\infty} S(u)du \qquad (3.45)$$

where $S(t)$ is the survival probability at time t.

Oakes and Dasu in 2003 [110] show that $m(t)$ exists for all t if and only if $m(0)$ is finite, this is equalvalent to $E(T)$ being finite. The characterization theorem of Hall and Wellner [69] provides necessary and adequate assumptions for the MRL function of a continuous non-negative random variable. The concisely summarised version of it described by McLain and Ghosh (2011) [99] is:

1. $m(t) \geq 0$ for all $t \geq 0$,
2. $m(t) + t$ is nondecreasing in t,
3. if there exists a w such that $m(w) = 0$, then $m(t) = 0$ for all $t \geq w$, otherwise,

$$\int_{0}^{\infty} m(t)^{-1}dt = \infty.$$

Not only is the MRL easier to interpret, but also completely characterises the survival distribution (with a finite mean) and in that sense, it is equivalent to the density function or the moment generating function [64]. For a better understanding, assume four typical survival scenarios; Stable, Recovery, Terminal and Infectious. They relate to different disease mechanism and could be derived from the following distributions:

- **Stable:** exponential distribution where hazard and MRL functions are constant over time,
- **Recovery:** gamma distribution (shape parameter < 1) where hazard function is decreasing and MRL is increasing,
- **Terminal:** gamma distribution (shape parameter > 1) where hazard function is increasing and MRL is decreasing,
- **Infectious:** log-normal distribution where hazard function is first increasing at then decreasing over time and an opposite pattern for the MRL.

Figure 3.10 represent survival, hazard and MRL functions of intervention and control groups for the four assumed disease scenarios. It shows that the survival function cannot completely represent all the main characteristic of each survival time distribution. For example, the survival functions are quite similar for the Stable and Recovery scenarios. It is because the survival function is merely a fraction of events over time; therefore it would not provide enough information about the mechanism of disease. The hazard function can be used to distinguish this difference; however, it could be confusing and difficult to communicate. Spruance [127] presents some examples of this confusion when clinicians and clinical investigators interpret and communicate the results of clinical trials to patients.

The hazard function focuses on what is happening in the short period of time, while, the MRL function concentrates on the long-term survival experience and provides the expected remaining lifetime.

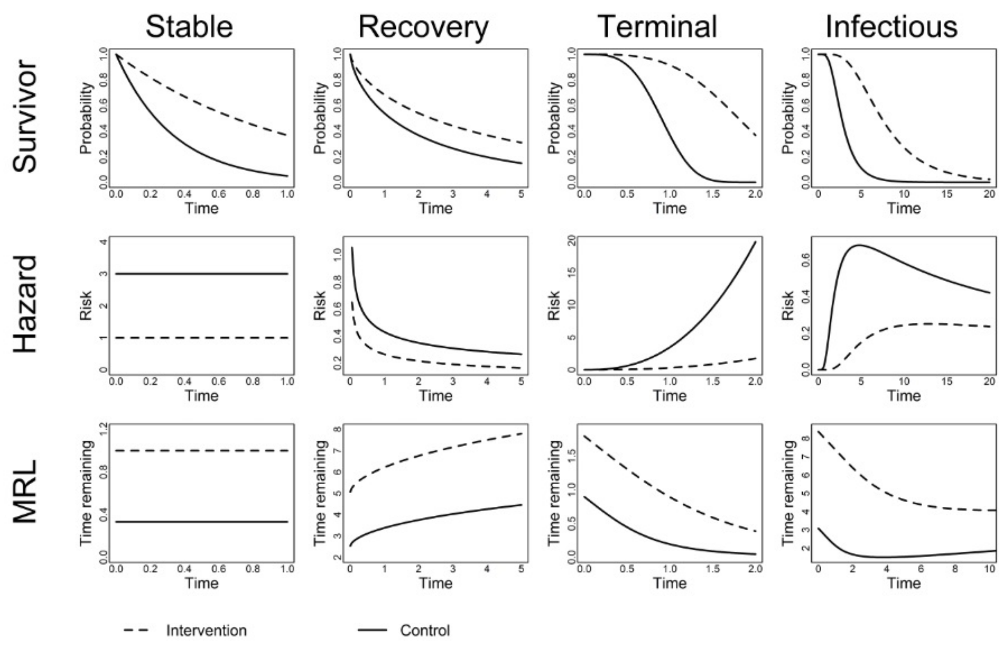

Figure 3.10: Graphical summaries of different standard time distributions

This characteristic of the MRL function is particularly interesting when one tries to convey the results involving time-to-event data. Rather than using a probability or hazard ratio, it will be more understandable for patients if describing the effectiveness of therapy by the expected remaining lifetime (using the MRL). It is a useful alternative summary of survival times as the inference based on the expected remaining time is more intuitive to communicate compared to summaries on a probabilistic scale or as ratios of hazards. In the following section, different methods for estimating the MRL function will be discussed.

Mean residual life estimation

Based on the definition of the MRL function in (3.45), the estimation of MRL would be straightforward if the complete survival function could be specified ($\hat{S}(t)$ for all $t > 0$). However, it is not always the case due to the Type I censoring (study termination). Yang (1978) [150] assumed no censoring cases and proposes the estimate of the MRL function using the empirical survival function. Another suggested approach includes using a kernel smoothing based estimator for the MRL function [65], and computing smooth estimators for S(t) and f(t) [31]. All these methods assume the complete case data which is not feasible in almost all the biomedical studies.

The estimation of the MRL in the presence of the censored observations has been proposed by Gill [59], Yang [149], Ruiz and Gulliamon [121] and Chaubey and Sen [31, 30]. Zhou (2006) [153] proposed an empirical likelihood inference method to estimate the MRL function using the non-parametric estimate (KM estimator) of $S(t)$. However, in the case of Type I censoring, they do not provide appropriate estimates of the MRL function [11].

The MRL formula in (3.45) can also be represented in the following form:

$$\begin{aligned} m(t) &= \frac{\int_t^\infty S(s)ds}{S(t)} \\ &= \frac{\int_0^\infty S(s)ds - \int_0^t S(s)ds}{S(t)} \\ &= \frac{\mu - \int_0^t S(s)ds}{S(t)} \end{aligned} \tag{3.46}$$

In practice in the presence of Type I censoring, the follow-up time can be divided into two separate parts; before the study termination and after it as indicated in Figure 3.11.

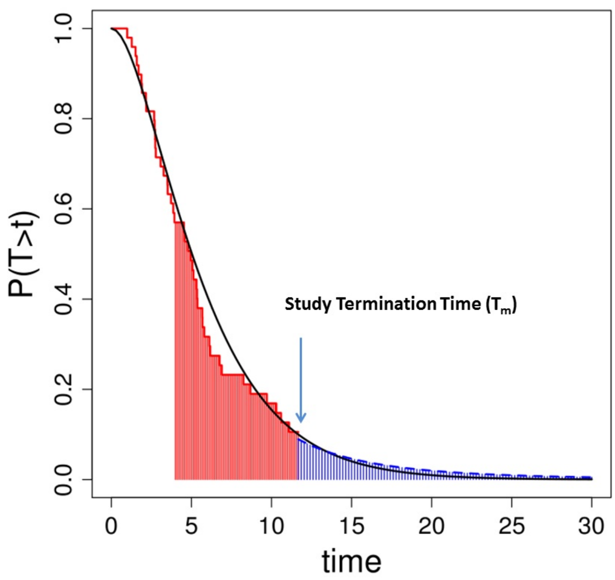

Figure 3.11: Hypothetical survival distribution in black and the estimation of survival time in red

The estimation of the red part in Figure 3.11 ($S(t)$ and consequently $\int_0^t S(s)ds$) is well studied using non-parametric or parametric estimators, thus the challenge is adopting a method to evaluate the missing tail of the distribution in order to estimate μ in (3.46). In other words, the critical issue with estimation of the MRL under Type I censoring depends on reliable tail estimation.

Alvarez et al. (2015) [11] proposed a method to estimate the MRL function for non-informative right-censored data in the presence of Type I censoring. This method uses a semi-parametric hybrid estimator which combines an extrapolated tail estimation with a non-parametric estimate of the survival function. The upper tail of the survival distribution is approximated by an extreme value model (e.g., the generalised Pareto distribution). The distribution parameters can be computed applying maximum likelihood approach to the limited sample observations after a fixed time-point u (called threshold).

The 0.8 quantile of the uncensored observed times is suggested as a reasonable threshold choice according to Alvarez et al. [11] simulations and applications. This provides 20% of uncensored observed times. However, the optimal value of the threshold needs careful investigation. The selection of the threshold depends on a trade-off between the higher bias due to the weaker asymptotic approximation (lower value of u) or the larger parameter estimation uncertainty due to the limited sample of excesses (higher value of u) [124]. A review of the advances and traditional threshold estimation approaches is presented by Scarrott (2012) [124].

This approach (with the 0.8 uncensored quantile) is used to estimate the MRL through this thesis. The **R** function written to estimate the MRL using this approach is given in A.2. Note that, the MRL will be estimated and presented up to the threshold as it would be unreliable afterwards due to the small number of events.

An example of the MRL estimate for the Lung cancer data is given in Figure 3.12. It reveals that the average time remaining at the start of follow-up is about 12.7 months i.e. about one year of surviving. However, this time remaining decreases over time and reaches 9 months, conditional on surviving for one year. This reveals a bad prognosis for cure in patients with advanced lung cancer.

The same approach can be employed in order to study the survival experience of two or more populations. A simple graphical approach could make the comparison; by plotting the MRL estimates for all population of interest. A different approach

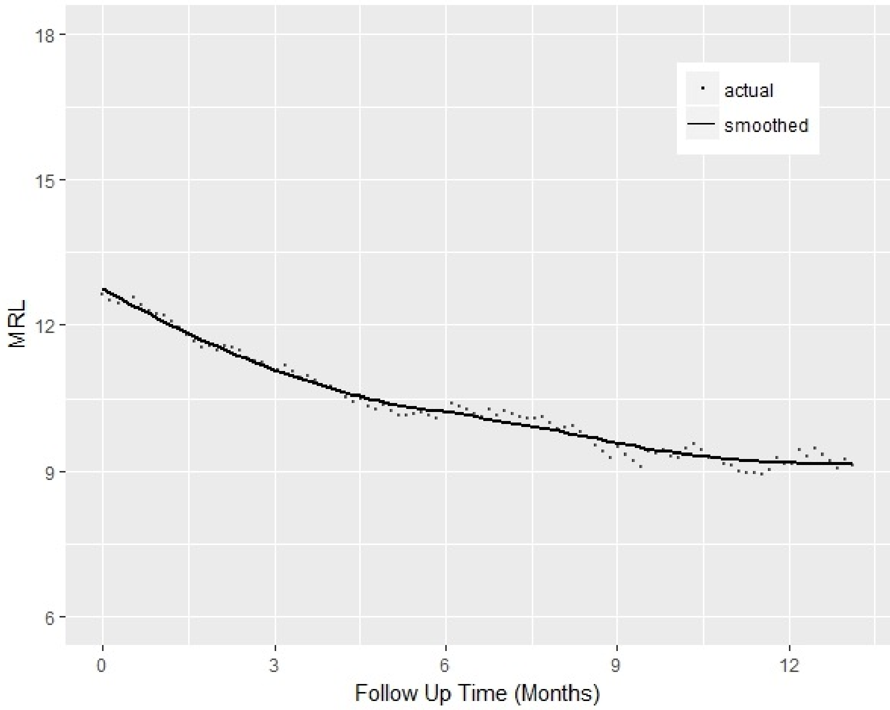

Figure 3.12: The MRL estimates (points) and the fitted smoothing spline (line) for the Lung cancer data; points represent the actual estimations, and the line represents a smoothed estimation line

would be to calculate the difference in MRL estimates (i.e. $\hat{m}_1(t) - \hat{m}_2(t)$) at each follow-up time-point. The MRL difference is especially beneficial when comparing one or more populations with a control population, for example, studying the effects of a new treatment and the standard therapy.

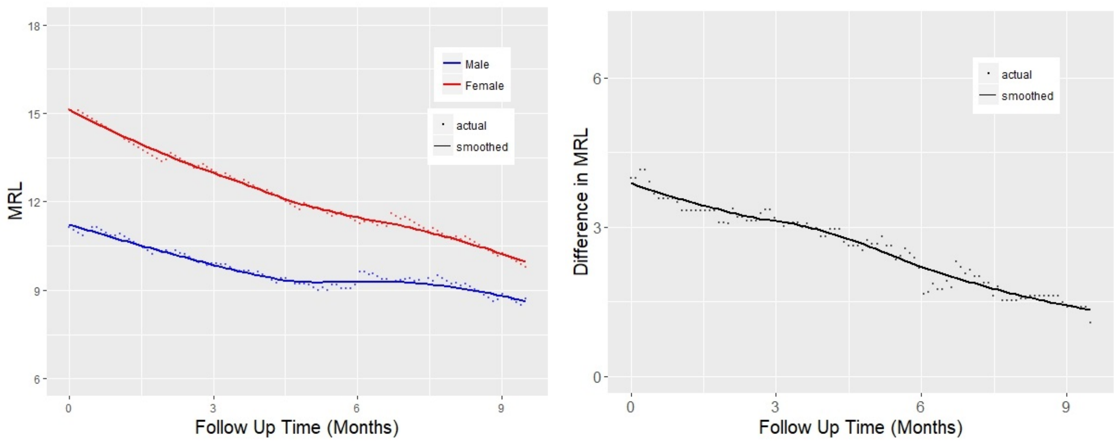

Figure 3.13: The MRL estimates of the Lung cancer data grouped by gender (left), and the estimate of MRL difference (right); points represent the actual estimations, and lines represent smoothed estimation lines

Figure 3.13 presents graphical summaries for the average remaining lifetime of the Lung cancer data grouped by gender or the difference plot $(\hat{m}_f(t) - \hat{m}_m(t))$. Both graphs display similar disease scenario where the average time remaining for both males and females are decreasing over time (this decrement is sharper for the first four months). However, a longer remaining lifetime is expected for female than male. This difference is approximately four months at the start which will decrease over time and finally reduce to about one month for patients who are still alive after about nine months.

A careful analysis needs to be done in order to make a reliable inference about the MRL difference in two group populations. For example, this will be addressed using the following hypothesis test:

$$\begin{aligned} H_0 &: \quad m_1(t) \leq m_2(t) \quad \text{for some } t \\ H_1 &: \quad m_1(t) > m_2(t) \quad \text{for all } t \end{aligned} \tag{3.47}$$

Berger et al. (1988) [16] proposed a non-parametric test statistic to compare two independent MRL functions. A permutation test can also be used to investigate the difference of two MRL functions. The test p-value is calculated as the proportion of all sampled permutations which have the absolute difference greater than or equal to the absolute observed value. In Chapter 4, methods to describe the generation of variability bands for the MRL will be introduced. The inference about the difference in MRL of various populations can also be assessed by investigating the MRL variability bands calculated for each group.

3.4 Summary

In this chapter, various topics related to the statistical modelling have been reviewed. The Generalised Linear Models framework was revisited where different confidence intervals are discussed. The logistic and Poisson regression models were reviewed as the two popular special cases of GLM. The Cox PH model was introduced as a popular model for time-to-event data. The survival and hazard function were presented as two standard survival summaries where their low ability to convey the survival information efficiently was discussed. The dynamic prediction of survival and the mean residual life functions were introduced as two alternative summaries for time-to-event data. The DPS function provides additional survival probability dynamic during follow-up time. In contrast, the MRL function provides an informative summary in units of time which can easily translate the survival information to the clinicians and patients. The uncertainty of the DPS estimate is studied and provided in the **R** package written by Putter [116]. However, calculating the variance of the MRL estimate was always a challenge due to the complexity that arise from the nature of hybrid estimators. We will be considered this issue in the next chapter.

Chapter 4

Simulate

4.1 Introduction

The natural extension to the MRL estimation is determining the uncertainty of this estimate and generating the confidence region. The variance formula for the mean residual life function can be derived as [69]:

$$\sigma(t) = Var(T - t|T > t) = \frac{\int_t^\infty m^2(\nu) f(\nu) d\nu}{S(t)} \tag{4.1}$$

Zhou (2006) [153] developed a confidence interval for the MRL estimate based on empirical likelihood method using the non-parametric estimate (KM estimator) of $S(t)$. Generating confidence bands for MRL estimates that incorporate the censored information, in particular, Type I censoring is challenging. For example, the MRL estimate introduced in Section 3.3.4 is a hybrid estimator, which combines a non-parametric Kaplan-Meier part with a parametric part using the extreme value theory and the Generalised Pareto distribution.

Gong and Fang (2012) [61] proposed a closed formula for the variance of such an MRL estimation when $t = 0$. They considered the hybrid mean survival estimator of $\hat{\mu}_H$ as:

$$\hat{\mu}_H = \hat{\mu}_{KM} + \hat{\mu}_P = \int_0^\tau \hat{S}_{KM}(t) dt + \int_\tau^\infty \hat{S}_p(t) dt \tag{4.2}$$

where τ represent the threshold, $\hat{S}_{KM}(t)$ is the Kaplan-Meier estimate of the survival function defined up to the threshold and $\hat{S}_p(t)$ is a parametric estimate of survival function defined from threshold to infinity. The asymptotic distribution $\hat{\mu}_H - \mu \sim N(0, \sigma_H)$ derived using the delta method, where σ_H is a linear function of the variance and covariance of $\hat{\mu}_{KM}$ and the parameter vector of the distribution fitted to the tail ($\hat{\beta}$). This method computes the estimate of the variance of the MRL estimator at time $t = 0$.

Calculating the variance of hybrid estimators was always a challenge in the literature, and the uncertainty of such estimators is usually obtained using a resampling technique [58]. In this chapter, we are seeking to find a quick but effective answer to the problem of generating a variability band for the MRL estimate. We begin with an introduction to the bootstrap method as a useful resampling technique which can be used to estimate the variability bands of any complex estimators such as the hybrid MRL estimator introduced in Section 3.3.4. The common bootstrap methods to construct confidence bands will be introduced. A simulation study is performed to access the adequate bootstrap approach and to investigate effects of various factors (the censoring percentage, sample size or the underlying distribution) on the bootstrap performance in terms of the bandwidth and coverage probability. Finally, a Shiny application will be presented to provide the estimate MRL with its variability bands for any uploaded time-to-event data.

4.1.1 Bootstrap method

Bootstrapping, jackknifing and permutation tests are common types of a larger class of methods called resampling procedures [33]. They allow us to assess the bias, standard deviation and confidence interval of estimates and also perform significance tests.

The bootstrap method is a simple and straightforward resampling technique where data is resampled with replacement, and the resampled data is used to calculate the quantities of interest. It is a useful method to estimate the sampling distribution of almost every statistic using the random sampling methods. They are beneficial when inference is based on a complicated method, and the theoretical basis is not available or even not helpful for the sample size taken in practice. Also, they are used to test the performance of standard approximation of parametric approaches and to improve them if they show deficient inferences [40]. However, one of the main uses of bootstrapping is to measure the uncertainty which could be defined regarding the estimation bias, standard error, confidence interval or prediction error. Briefly, the aim of the bootstrap is to provide an empirical estimates of a sampling distribution. It can be estimated using parametric or non-parametric bootstraps. In the parametric approach, the distribution F is assumed to be a member of a class of distribution functions depends on a parameter vector ψ, so the estimate of F can be achieved by estimating the vector parameter (i.e. $\hat{F} = F_{\hat{\psi}}$).

However, in the non-parametric approach, F will be estimated using the empirical distribution function $\hat{F}$. In this case, samples of $\hat{F}$ will be taken using resampling from the observed dataset. Let $x_1^*, x_2^*, ...x_n^*$ be a set of n random sample taken with replacement from the observed dataset $(x_1, x_2, ...x_n)$. Suppose the interest is a function of distribution $(\theta = t(F))$ which is estimated by $t = t_{(\hat{F})}$. Then, we can approximate the distribution of $t - \theta$ by the empirical distribution of $t^* - t$ where $t^* = t_{(\hat{F^*})}$ and $\hat{F_r^*}$ is the empirical distribution correspond to the r^{th} set of the bootstrap sample. Therefore, the estimates of the bias (b) and variance (v) of the estimator t can be computed as:

$$b = \bar{t}^* - t \tag{4.3}$$

$$v = \frac{1}{R-1}\sum_{r=1}^{R}(t_r^* - \bar{t}^*)^2 \tag{4.4}$$

where $\bar{t}^*$ is the parameter estimate using the average of t_r^* over all replicates (i.e. $t_r^* = \frac{1}{R}\sum_{r=1}^{R} t^*$) and R is the number of replicates; for more detail see Canty (2002) [26]. The R needs to be chosen large enough to make the simulation error minimal [40]. A 100-500 number of replicates is usually suggested as a good choice for most cases, but a more conservative R might be about several thousand replicates.

The idea of the bootstrap confidence intervals were introduced by Efron (1979) [46], this method is then extended by him in 1981 for censored data [47], and further developed by other authors such as Efron [48] , Hall [67], Efron and Tibshirani [49] and DiCiccio [43]. In general, a confidence interval with $1-2\alpha$ coverage probability will be defined when the limits $\hat{\theta}_{\gamma_1}$ and $\hat{\theta}_{1-\gamma_2}$ satisfy in the following equation:

$$P(\hat{\theta}_{\gamma_1} < \theta < \hat{\theta}_{1-\gamma_2}) = 2\alpha \tag{4.5}$$

where γ_1 and γ_2 are respectively the left-tail and right-tail error probabilities. In most cases, the left-tail and right-tail errors are considered as equal (i.e. $\gamma_1 = \gamma_2 = \alpha$); otherwise, the likelihood property [41] can be used to choose γ_1 and γ_2 which create the shortest approximate interval.

The simplest technique to compute the confidence interval is by the normal bootstrap which is based on using an asymptotic normal distribution for $t-\theta$. In the case of large samples, the following distribution may be appropriate [40]:

$$Z = \frac{t-\theta-b}{v^{\frac{1}{2}}} \sim N(0,1)$$

where b and v are respectively the bias and variance of the estimator t. Then, the confidence interval for the parameter of interest will be simply computed by using the bootstrap approach and the equations (4.3) and (4.4).

In some cases, the normal assumption would not be valid. Most of the alternative approaches are based on choosing the best error probabilities (γ_1 and γ_2), then using the corresponding sorted bootstrap estimates ($t^*_{(r)}$) for the confidence interval limits. The basic, percentile and bias-corrected adjusted (BCa) bootstraps are three well-known alternatives [40].

The basic bootstrap aims to approximate the distribution of $t-\theta$ by the empirical distribution of t^*-t. The $1-2\alpha$ basic bootstrap confidence limits are constructed using α-quantile and $(1-\alpha)$-quantile of t^* as:

$$\left[t-(t^*_{((R+1)(1-\alpha))}-t),\ t-(t^*_{((R+1)\alpha)}-t)\right] \tag{4.6}$$

where $t^*_{(1)} < t^*_{(2)} < ... < t^*_{(R)}$ are the sorted bootstrap estimates. Similarly, the percentile bootstrap confidence limits are:

$$\left[t^*_{((R+1)\alpha)},\ t^*_{((R+1)(1-\alpha))}\right] \tag{4.7}$$

The basic and percentile bootstraps are asymptotically equivalent to the normal bootstrap but they have less limiting assumption [26].They have the equal left-tail and right-tail error probabilities (i.e. $\gamma_1 = \gamma_2 = \alpha$).

In contrast, the BCa bootstrap method corrects for bias made when the assumption of normality is not met. It substantially improved the properties of the interval in both theory and practice by choosing the error probabilities cleverly. The BCa bootstrap confidence limits are:

$$\left[t^*_{((R+1)\alpha')},\ t^*_{((R+1)(1-\alpha''))}\right] \tag{4.8}$$

More detail of the BCa method and the calculation of the error probabilities (α' and α'') can be found in Efron and Tibshirani (1992) [49].

4.2 Simulation

One of the ultimate goals of this study is to construct variability bands for the MRL estimate introduced in Section 3.3.4 using the bootstrap technique. For this purpose, a simulation study is carried out to assess the performance of different bootstrap techniques for the estimation of the variability bands in terms of their coverage and the width of the intervals. The results of the simulations will be used to inform the adequate selection of the bootstrap method to generate confidence bands for the MRL estimate and investigating the factors affecting the bootstrap performance.

4.2.1 Proposed Scenarios

In this simulation study, different scenarios are assumed to investigate the bootstrap sensitivity depending on the sample size and censoring percentage. The coverage probability is fixed to 0.95 as the most commonly desired coverage level. The four survival families explained in Section 3.3.4 are considered as the simulation disease scenarios. They include constant hazard ('Stable'), decreasing hazard ('Recovery'), increasing hazard ('Terminal') and the survival experience with first increasing hazard and decreasing then ('Infectious') which cover the most common survival experiences. The survival distributions used for the simulation study are:

1 **Stable:** Exponential ($\lambda = 3$) with constant MRL;

2 **Recovery:** Gamma ($shape = 0.7, scale = 3$) with increasing MRL;

3 **Terminal:** Gamma ($shape = 2, scale = 3$) with decreasing MRL;

4 **Infectious:** Lognormal ($\mu = 1, \sigma = 0.5$) with decreasing MRL first and increasing after.

For each survival distribution, we consider different cases to include various right censoring percentages (in two mechanisms of Type-I censoring and random censoring) and in three different sample sizes. The total of 27 cases for each survival distribution corresponds to:

- Sample size: $n \in 30, 100, 500$;
- Type-I censoring percentages: $cp1 \in 0, 0.1, 0.2$
- Random censoring percentages: $cp2 \in 0, 0.1, 0.2$

A total of 500 datasets are generated for each case, then four different bootstrap approaches discussed in Section 4.1.1 with 1000 replicates (basic, percentile, normal and BCa bootstrap) are used to construct a pointwise confidence band for the estimated MRL for each of the generated data. We consider three different fixed time points for each confidence band. They include $T0 = 0$, $T1 = \frac{1}{3}t_u$ and $T2 = \frac{2}{3}t_u$ where t_u is the fixed threshold calculated for each case.

4.3 Simulation results

It is reasonable to think of the normal bootstrap as the most suitable bootstrap option since the MRL nature is a conditional mean. In order to gain the basic understanding of the simulation result and set up further study goals, we first consider the cases from the 'Stable' scenario where the normal bootstrap is used to make pointwise confidence bands for the MRL. Each box-plot in Figure 4.1 describes the distribution of the confidence bandwidth for all different censoring percentages and sample sizes, where the overall coverage probability is assigned at the top of each of them.

The distribution of coverage probability in Figure 4.1 suggests that the coverage may be low compared to the desired (95%) in some cases, but they are almost comparable to each other and close to the desired level (95%). This becomes more evident when $n = 500$ and there is Type-I censoring. Similar trends are observed for each of the censoring percentages where the bandwidth is decreasing as n is increasing, and the width is rising as time raises in each case. These result were not unexpected as larger sample sizes provide more accurate estimates. Also, the MRL estimation accuracy is expected to decrease for the further time point due to the larger proportion of this hybrid estimate that is extrapolated rather than using the non-parametric KM estimate.
The primary simulation study purpose was to select an adequate bootstrap technique to generate confidence bands for the MRL estimate. We fixed the selected time point to $\frac{1}{3}T_u$ and considered moderate and extreme situations. In the moderate, Type-I and random censoring percentages are set to 0.1 (a total of 20% right censoring); while in the extreme, both are set to 0.2 (a total of 40% right censoring).

Table 4.1 shows the coverage probabilities of the four bootstrap approaches in the moderate situation for different scenarios and sample sizes. The basic bootstrap has obviously the worst coverage probabilities as the most of the coverages were in the range of about 0.8 to 0.9. The normal bootstrap performs better than the basic method, but coverages are still not as good as the percentile and BCa bootstrap methods. The percentile approach seems to be dependent on the sample size since coverages increase as sample size increases. Overall, the percentile approach seems to represent the best performance in terms of maintaining the desirable coverage since the basic, normal and BCa bootstrap methods always under-covered the confidence level.

Moreover, Figure 4.2 displays roughly the distributions of confidence bandwidths

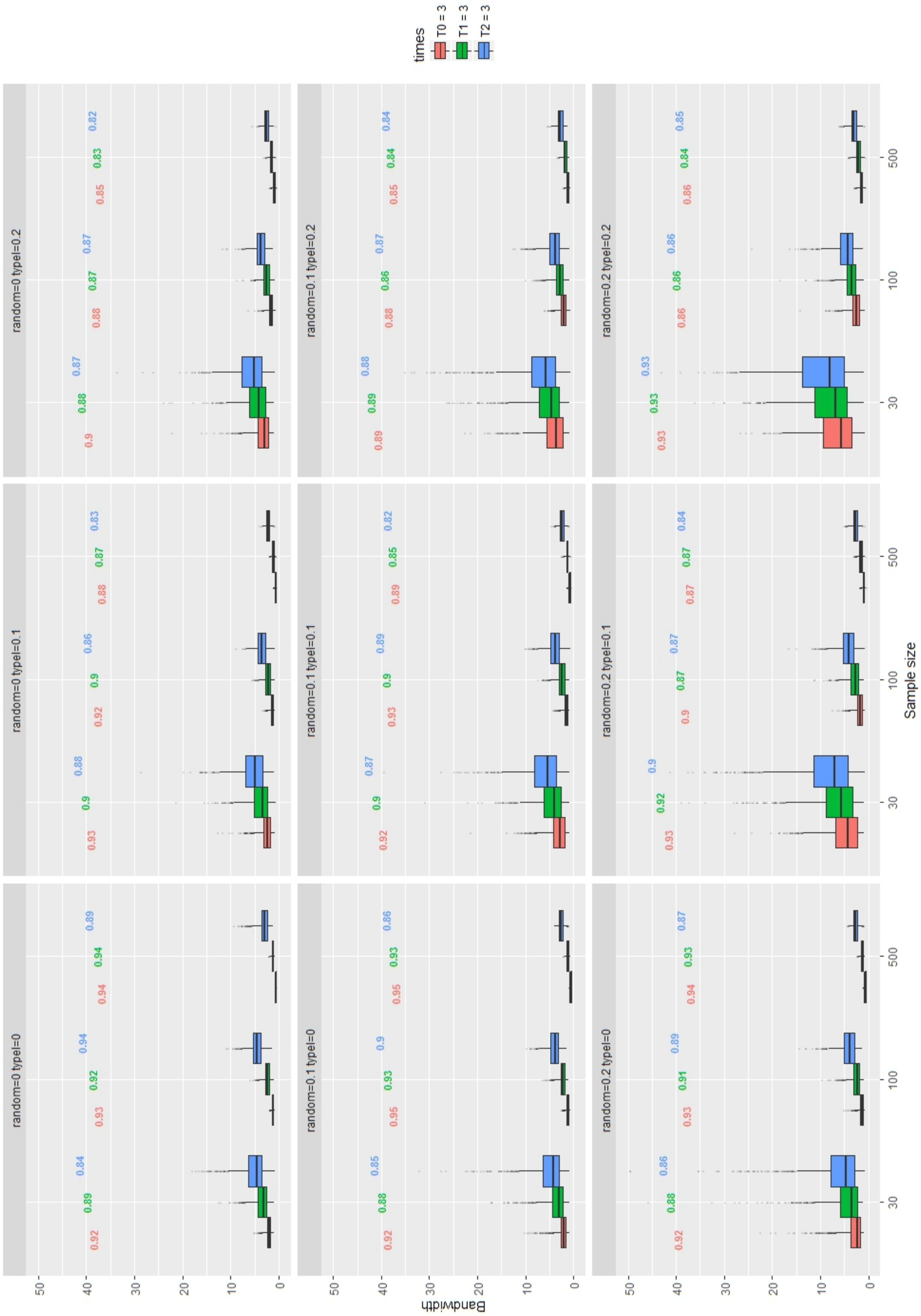

Figure 4.1: The confidence bandwidths distributions of the ‘Stable’ scenario using the normal bootstrap by different sample sizes and time points divided by censoring percentage

		Bootstrap approaches			
		Basic	Percentile	Normal	BCa
Stable	n=30	0.84	0.92	0.90	0.92
	n=100	0.83	0.95	0.90	0.92
	n=500	0.80	0.97	0.85	0.91
Recovery	n=30	0.78	0.90	0.88	0.92
	n=100	0.79	0.94	0.87	0.92
	n=500	0.80	0.97	0.88	0.92
Terminal	n=30	0.88	0.93	0.92	0.92
	n=100	0.88	0.96	0.91	0.92
	n=500	0.82	0.96	0.87	0.90
Infectious	n=30	0.90	0.94	0.93	0.92
	n=100	0.90	0.96	0.92	0.93
	n=500	0.88	0.97	0.91	0.93

Table 4.1: The coverage probabilities of the four bootstrap approaches in the moderate situation

for four bootstrap approaches in the moderate situation. In the large sample size cases, there is no noticeable difference between methods. However, the BCa bootstrap method seems to provide slightly wider bands when the sample size is small. Regarding different disease scenarios also, there is no special difference between the 'Stable', 'Recovery' and 'Terminal' scenarios; however, the infectious diseases constantly had narrower confidence bands which could be due to its nature which has smaller tails (lower values) and consequently smaller variance.

Table 4.2 shows the coverage probabilities of the four bootstrap approaches in the extreme situation for different scenarios and sample sizes which provide similar results, where the percentile, normal and BCa approaches perform nearly as well as their performance in the moderate situation. However, the Basic approach coverage is extremely affected by the high censoring proportion, where it gave the worst coverages range from 0.62 to 0.77. The percentile approach repeatedly is the only approach which did not under-cover. To sum up, the percentage bootstrap had the best performance regarding the coverage probability. It returns a reasonable coverage even in the extreme situation where there is a total of 40% right-censored observations.

Furthermore, Figure 4.3 displays the approximate distribution of confidence band-

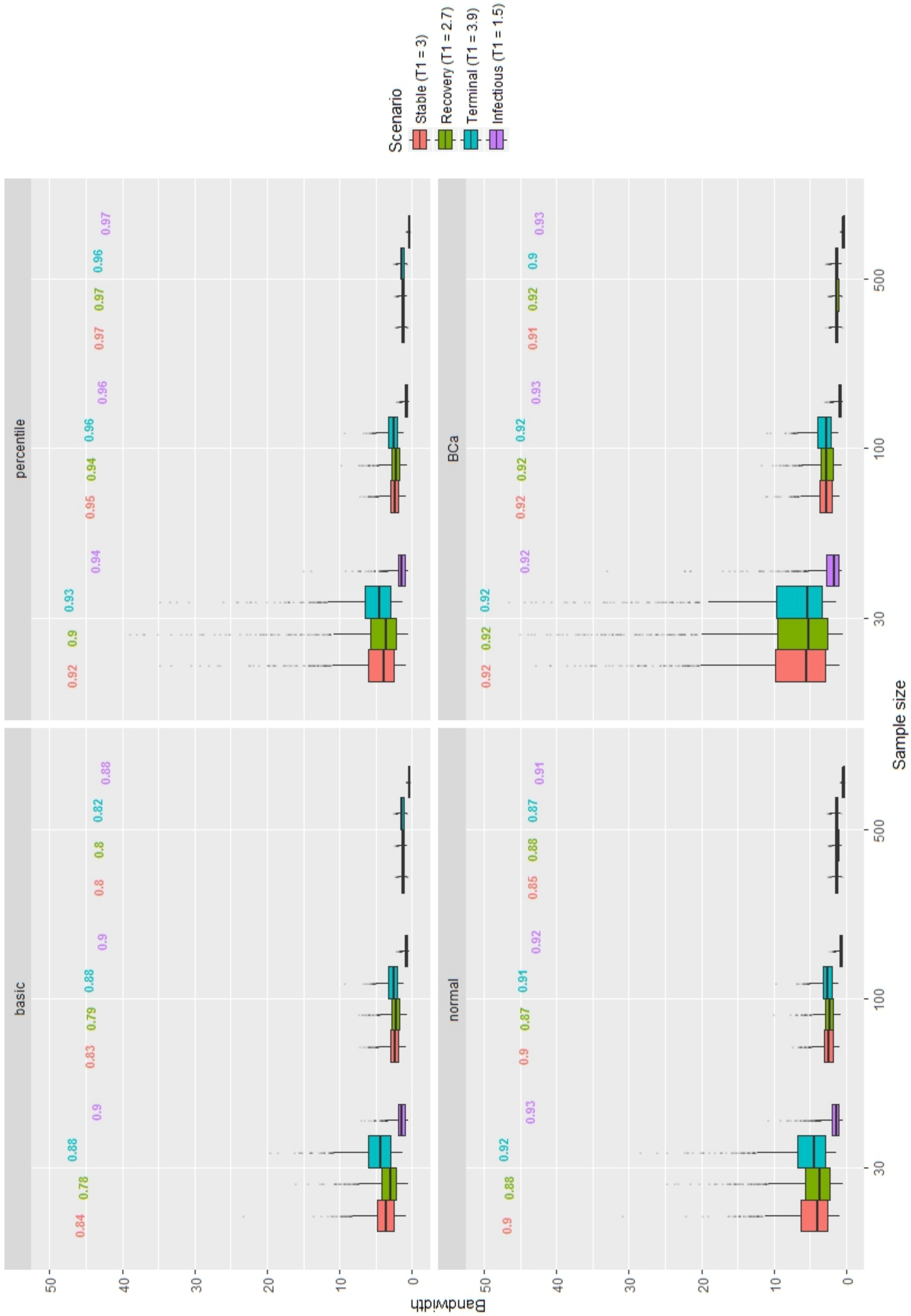

Figure 4.2: The confidence bandwidths of the four bootstrap approaches in the moderate situation

		Bootstrap approaches			
		Basic	Percentile	Normal	BCa
Stable	n=30	0.69	0.95	0.93	0.90
	n=100	0.67	0.96	0.86	0.90
	n=500	0.71	0.97	0.85	0.95
Recovery	n=30	0.62	0.94	0.92	0.87
	n=100	0.64	0.95	0.83	0.88
	n=500	0.72	0.97	0.82	0.93
Terminal	n=30	0.73	0.94	0.91	0.90
	n=100	0.72	0.97	0.90	0.92
	n=500	0.62	0.98	0.81	0.92
Infectious	n=30	0.77	0.93	0.90	0.90
	n=100	0.73	0.96	0.89	0.91
	n=500	0.68	0.97	0.83	0.93

Table 4.2: The coverage probabilities of the four bootstrap approaches in the extreme situation

widths for four bootstrap methods in the extreme situation. Similar to the moderate, there is no distinct difference between methods; however, the BCa and percentile bootstraps returns wider bands in the cases with small sample sizes. The extreme situation is the worst in terms of the censoring percentages which returns remarkably wider bands for all scenarios, especially when the sample size is small. The 'infectious' scenario again had consistently narrower bands; however, the recovery scenario has the widest band as a result of a small sample size in large censoring percentages situation for the gamma distribution with decreasing hazard which make commonly long missing tails.

Both the results of moderate and extreme situations lead to similar conclusions where the percentile bootstrap method gives a more practical confidence bands in general as there is always a chance of reaching the desired coverage when using this approach. Davison mentioned that the BCa bootstrap is expected to slightly undercover the exact coverage probability [40]; however, the results presented here suggest that this also applies to the basic and normal bootstraps.

To sum up, the percentile method is recommended as the best technique to access the most reliable confidence band for the MRL estimate. However, in the case of small sample size (e.g. $n = 30$) and the large right censoring (e.g. 40% censoring), using the normal bootstrapping is suggested as an attractive alternative as it ex-

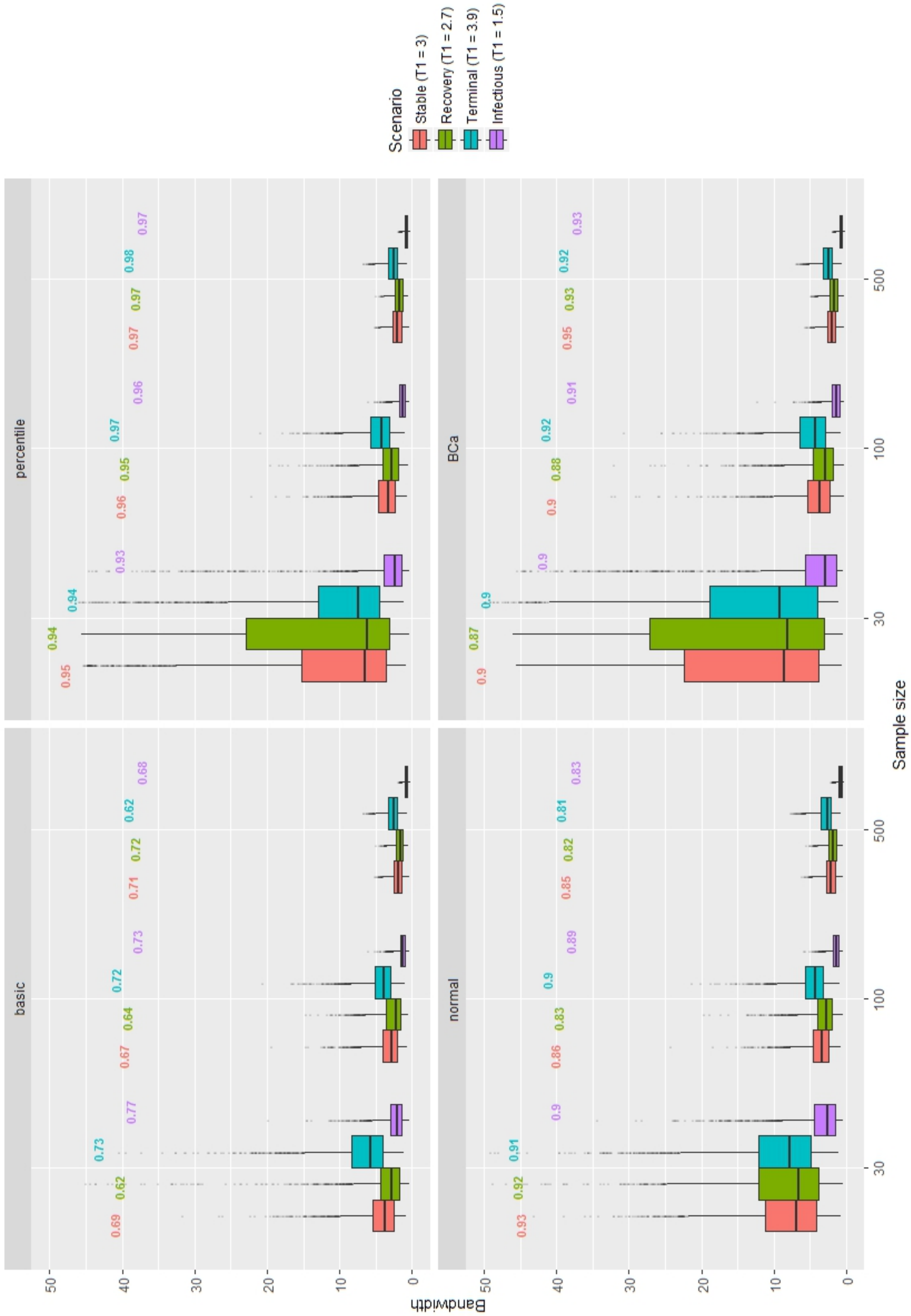

Figure 4.3: The confidence bandwidths of the four bootstrap approaches in the extreme situation

pected to provide close enough coverage as well as narrower bandwidths.
In the last stage of the simulation study, we applied the percentile bootstrap approach to investigate how the bootstrap performance is influenced by the censoring proportion and sample size in different scenarios.

Figures 4.4, 4.5, 4.6 and 4.7 show the simulation results of the confidence bandwidth distributions using the percentile bootstrap under the four different scenarios. Similar trends for all scenarios are observed where the bandwidths always increase over time due to the higher variance arising from the more weight of the extrapolated part of MRL estimator. They also rise as the censoring percentage (Type-I or random censoring) rises or the sample size drops. This once again confirms that the MRL estimation precision is decreasing by time, and the more sample size and less censored observation provide a more accurate estimate of the MRL.

The confidence band coverage and the bandwidth for each case in these four figures, revealed that the percentile method provides reliable bands for all combinations of diverse sample sizes and censoring percentages in different time points. All of these confidence bands have coverages close to the coverage of interest (95%). The boxplots of the coverages by the scenario in Figure 4.8 represents satisfactory situations where the coverages probabilities of all the scenarios are distributed around 0.95. The 'Stable', 'Terminal' and 'Infectious' medians are directly on the desired coverage. The coverage median of the 'Recovery' was 0.941. Overall, the coverage probabilities are reasonably well distributed around 0.95; however, the recovery mechanism due to their long tails had the worst bootstrap performance in terms of the coverage in Figure 4.8 as well as bandwidth in Figure 4.3.

The Lung cancer data in Section 1.2.3, which is used as an example of time-to-event data through this thesis, includes about 28% of censored observations (5% due to the study termination) and can be classified as terminal type due to the severity of the disease. The MRL estimate for this data was presented before in Figure 3.12 which was decreasing over time. A 95% pointwise confidence band for this estimate based on the percentile bootstrap approach is given in Figure 4.9. The confidence limits at the start of follow-up are from about 11.5 months to about 15 months which increases by time from 7 months to about 13 months if the patient survives

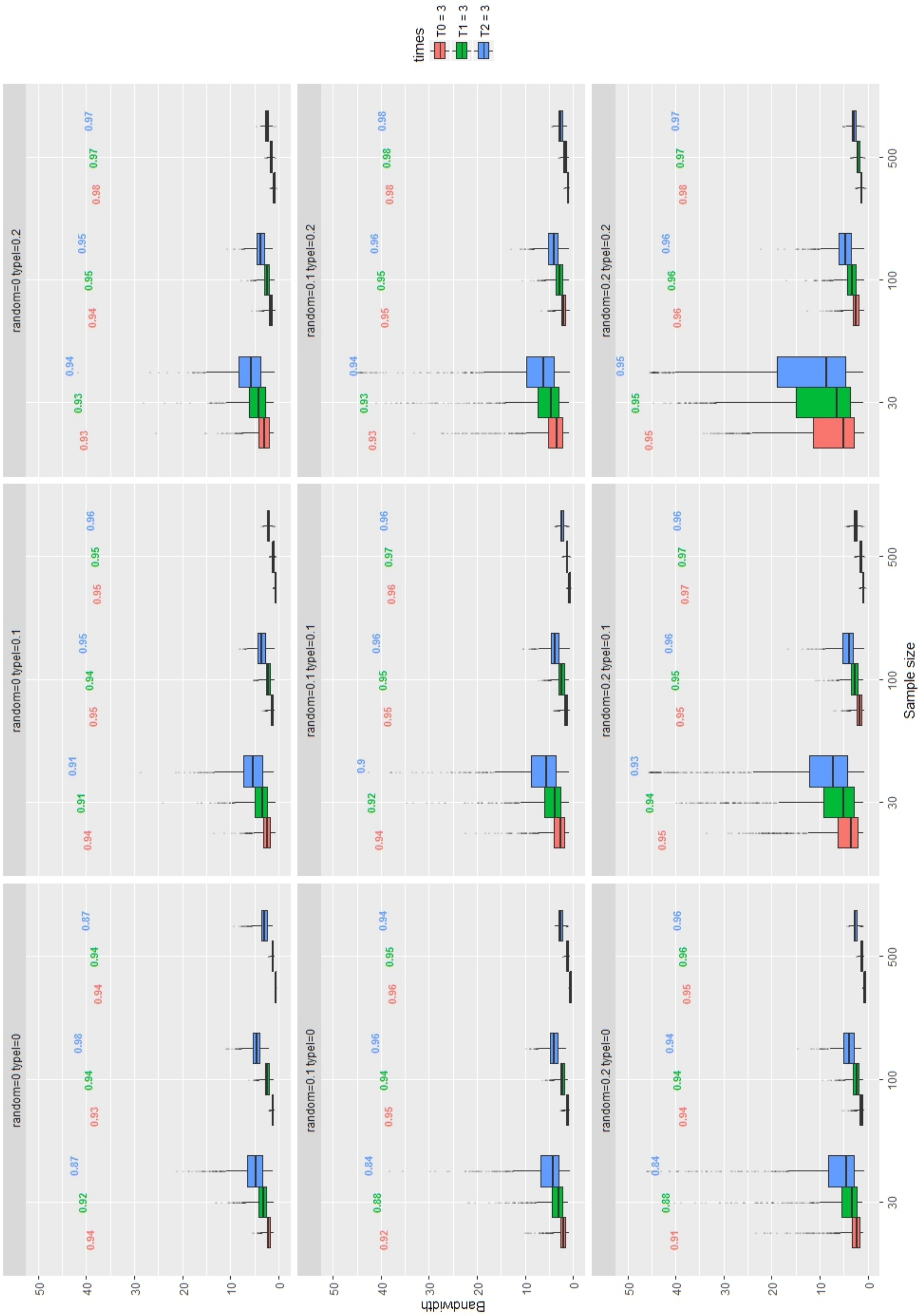

Figure 4.4: The confidence bandwidths distributions of the 'Stable' scenario using the percentile bootstrap by different sample sizes and time points divided by censoring percentage

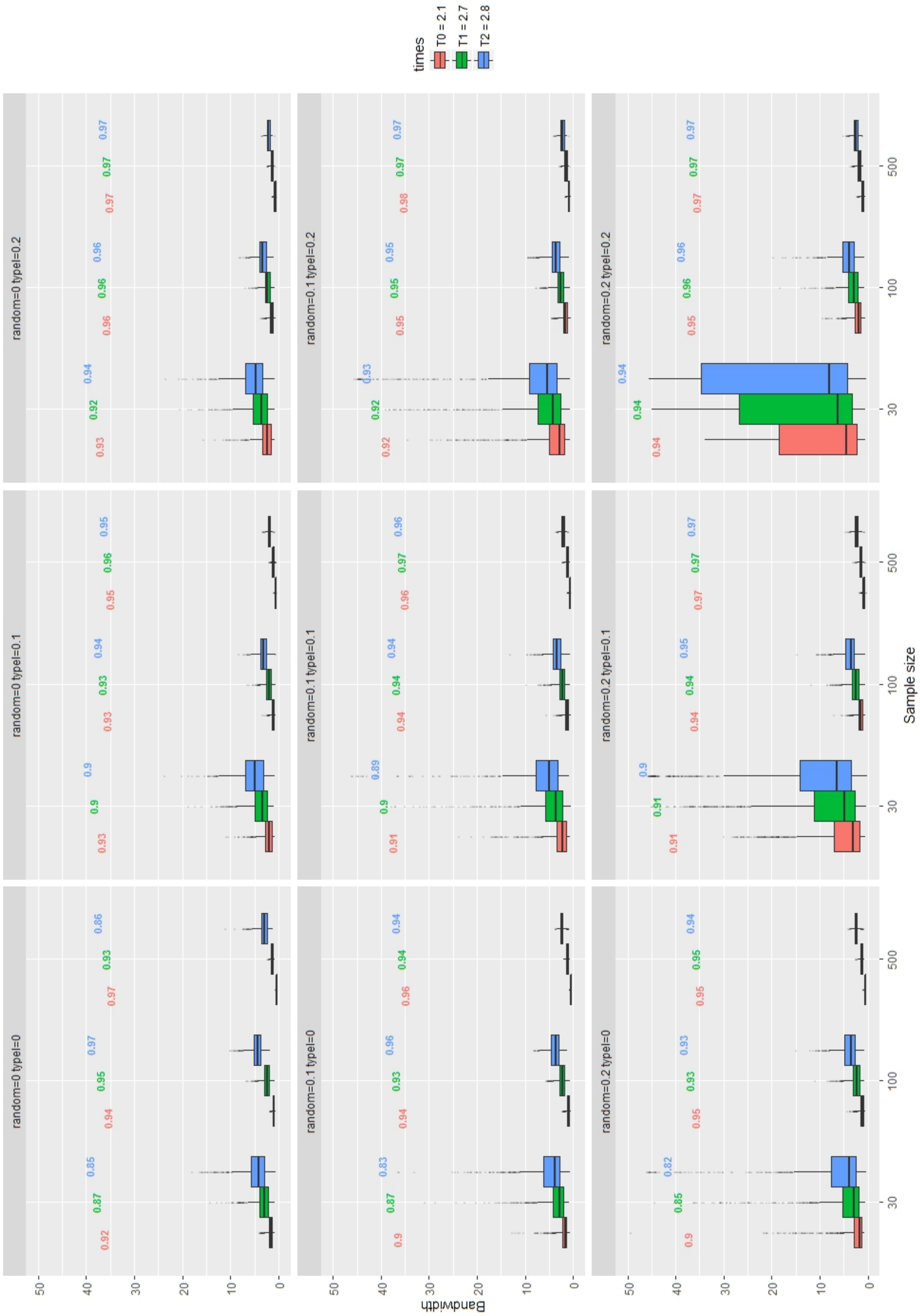

Figure 4.5: The confidence bandwidths distributions of the 'Recovery' scenario using the percentile bootstrap by different sample sizes and time points divided by censoring percentage

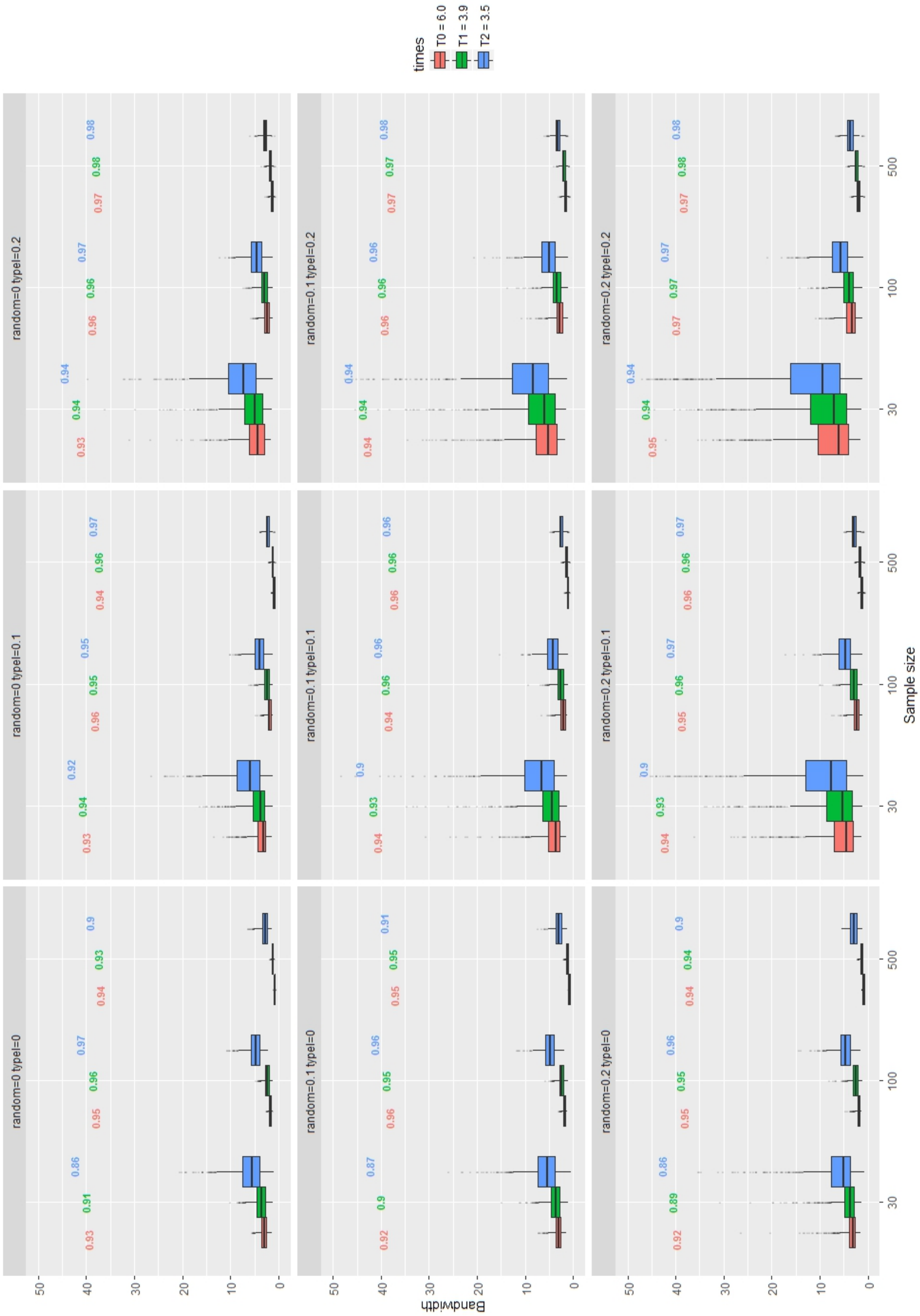

Figure 4.6: The confidence bandwidths distributions of the ‘Terminal’ scenario using the percentile bootstrap by different sample sizes and time points divided by censoring percentage

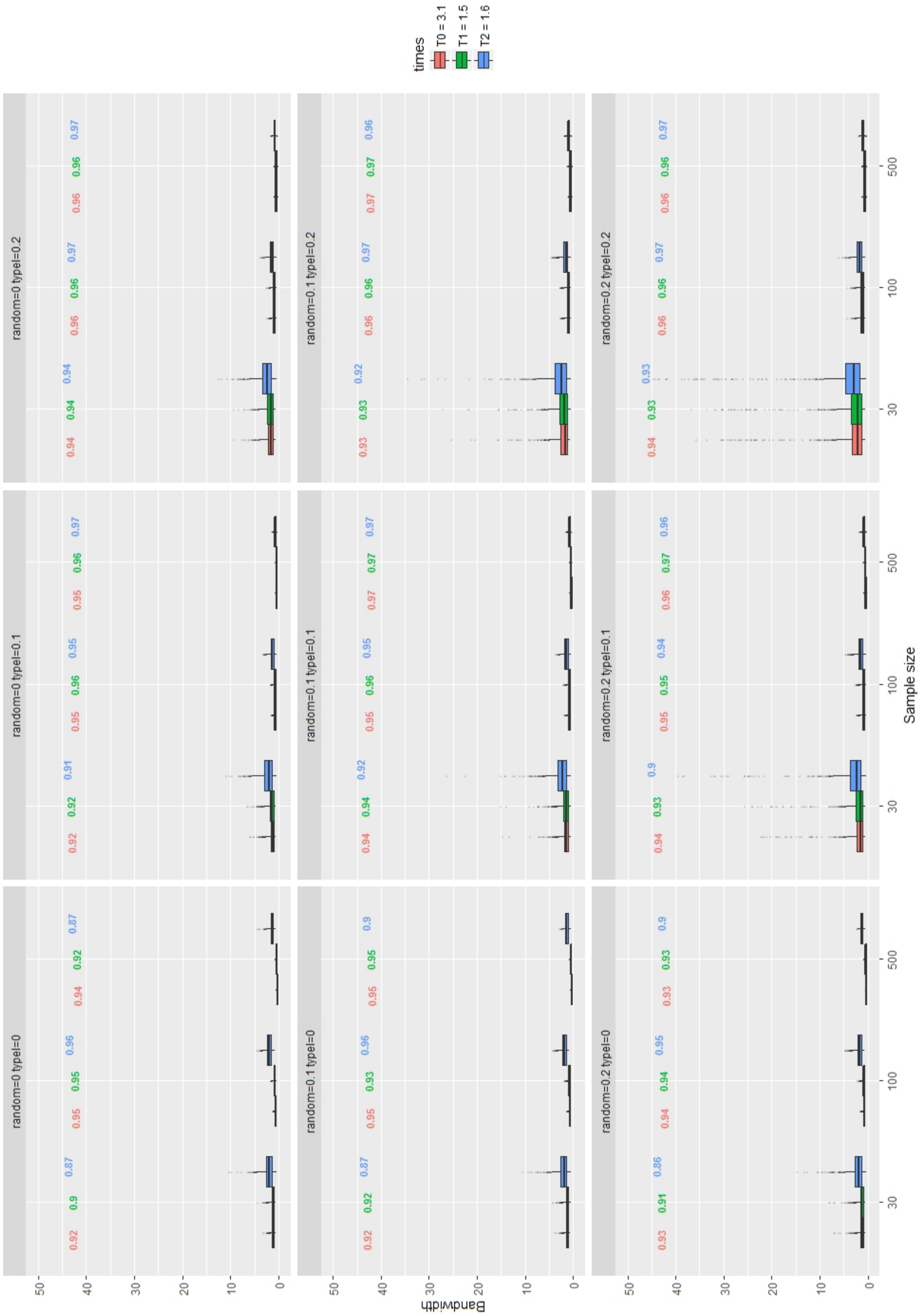

Figure 4.7: The confidence bandwidths distributions of the 'Infectious' scenario using the percentile bootstrap by different sample sizes and time points divided by censoring percentage

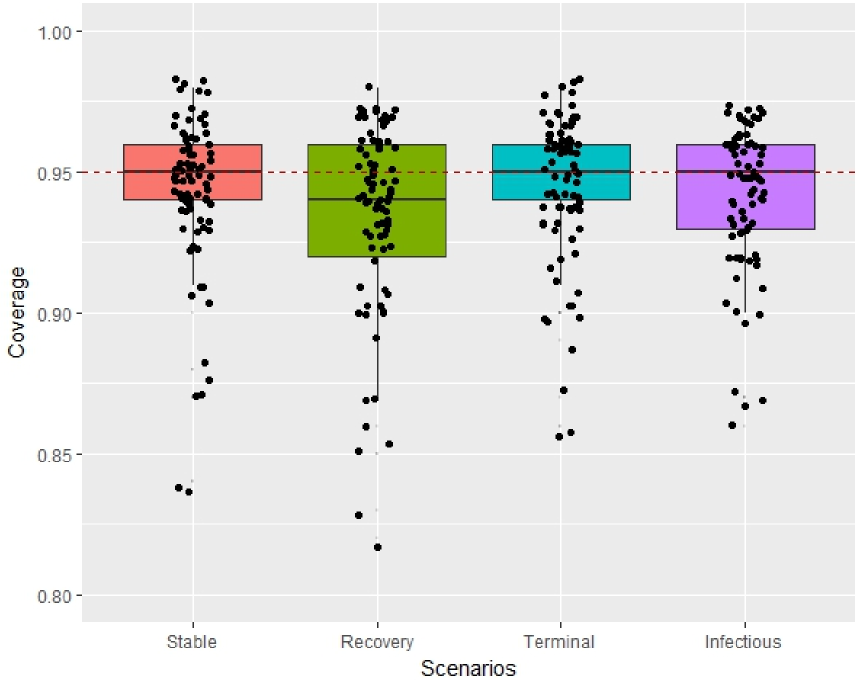

Figure 4.8: The coverage probability distributions of different scenario using the percentile bootstrap

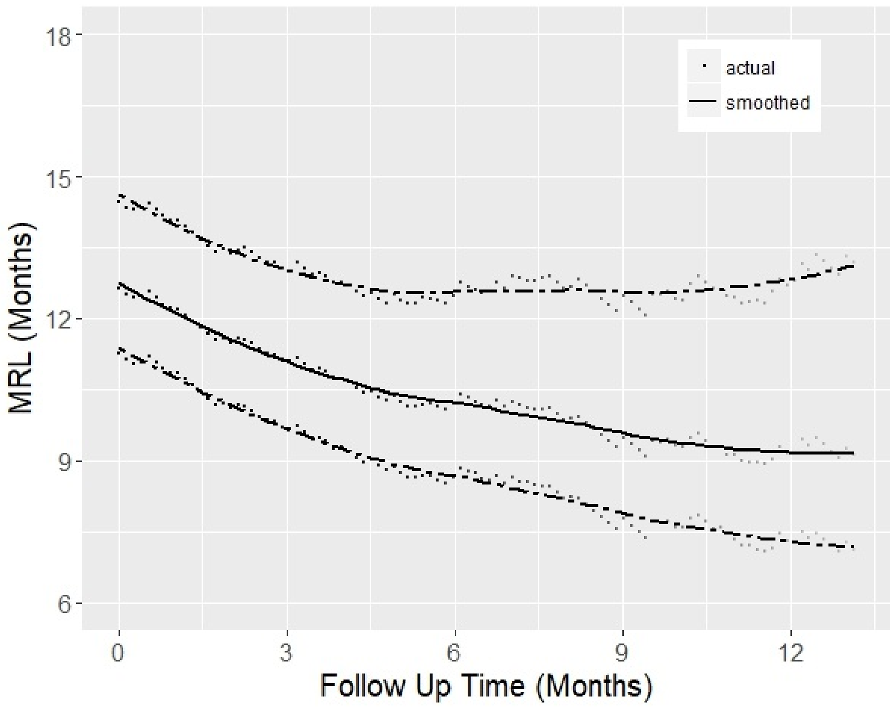

Figure 4.9: The MRL estimates with the 95% pointwise confidence bands for the Lung cancer data; points represent the actual estimations, and lines represent smoothed estimation lines

for one year.

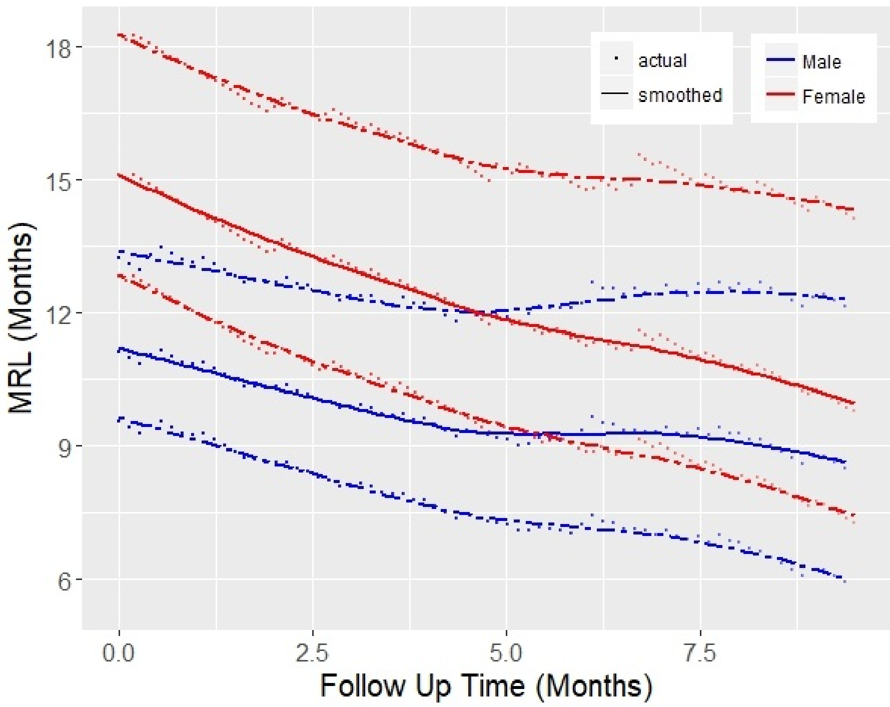

Figure 4.10: The MRL estimates by gender with the 95% pointwise confidence bands for the Lung cancer data; points represent the actual estimations, and lines represent smoothed estimation lines

Figure 4.10 presents an updated version of the graphical summary of the MRL estimate by gender (in Figure 3.13) with 95% confidence bands. As previously mentioned, similar trends are observed where the remaining lifetime is decreasing over time for both groups. But, there is also a noticeable difference suggesting a longer remaining lifetime for female than male. However, these differences are declining over time as patients survive, they also seem to be non significant at any time due to overlapping the confidence bands.

In the final part of this chapter, a Shiny application is created as a tool for estimating the MRL with confidence band. This application receives time-to-event data (including the pair of time and censoring variables) and returns an KM estimate of the survival function along with an estimate of MRL with its variability bands. Once the application receives the time-to-event data, the variables names will be identified and shown in the drop-down menus for the user to select the time and censoring variables. The plots of KM (with alpha-blending) and MRL with confidence band estimates would be shown in separate Tabs. A screenshot of this application applied to the Lung cancer data is given in Figure 4.11.

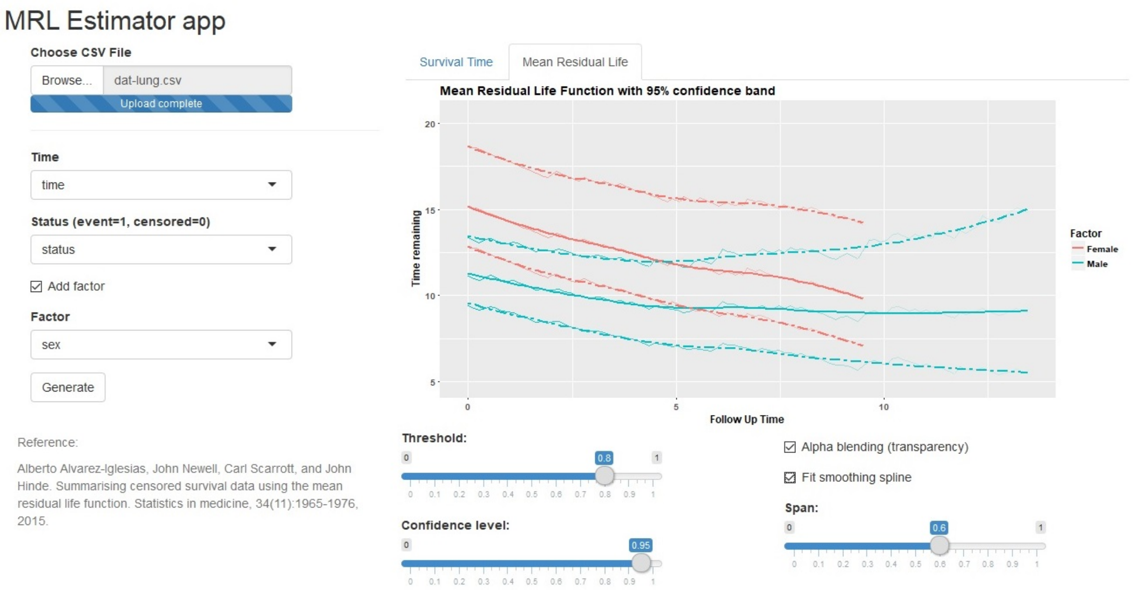

Figure 4.11: Screenshot of the Shiny application providing the MRL estimation with the confidence bands for the Lung cancer data

The MRL function would be estimated using the hybrid estimator with the GPD tail extrapolation introduced in Section 3.3.4 which incorporates the most practical situations. The default choice of threshold is set to 0.8 as suggested by Alvarez et al. [11]; however, the provided slider can be used to modify it to any desired threshold. The MRL confidence bands would be generated using the percentile bootstrapping approach as described in this chapter. More available options include the selection of confidence level and adding the notion of the number of observations for the survival plot using alpha-blending (transparency). This application is developed as a translational tool to facilitate interpretation of time-to-event data through summarising it in the most informative way (i.e in units of time). It is available to use for any time-to-event data at https://amir-jalali.com/apps/mrlapp.

4.4 Summary

In this chapter, the generation of confidence intervals for the estimated MRL function has been studied. The case presented in this chapter, where the tail of the distribution is missing due to Type I censoring, is the most challenging one, due to the lack of information beyond the maximum observed time. There are only a handful of methods available to deal with this issue, and they all propose the use of hybrid estimators, often based on some form of algorithmic approach where a few steps are required to obtain the desired estimate. The missing tail is usually modelled parametrically. Although in general, the MRL estimates are straightforward (following the algorithm), confidence intervals are more difficult to obtain, and people normally make use of resampling techniques.

In this chapter, the uncertainty of the MRL estimate introduced in Section 3.3.4 has been studied using the bootstrap as a natural solution to address this issue. The suitability of basic, percentile, normal and bias-corrected adjusted (BCa) bootstrap methods was investigated by performing a simulation study over a wide variety of survival distributions.

The normal bootstrap method was a natural first step to consider. However, this approach tends not to be valid for MRL estimation at least at all time points when considering the fact that the symmetric assumption for estimation in the tail appeared not to be reasonable. Three other bootstrap methods were also studied as non-parametric alternatives. The basic bootstrap method gives poor coverage probabilities which could be as a result of lacking the property of invariance to transformations mentioned by Davison [40]. The BCa method is second-order accurate with the invariance property, but its coverage was poor.

The percentile bootstrap method is the other approach which maintains the invariance to transformations property. It also gave the best performance in the simulation results providing the most efficient confidence bands regarding maintaining the desired coverage while providing narrow bandwidths. The sensitivity of this approach to the sample size and censoring proportions shows that it gives good performances in the moderate case where there are a large number of uncensored observations. However, it gives wide (well-covered) confidence bands for the extreme cases with small samples and large censoring proportion. In these cases with lack of information available, using a parametric bootstrap approach would be an attractive alternative to access narrower confidence bands. The simulation results confirm that the normal bootstrap in these cases gives narrower but well-covered confidence bands.

Chapter 5

Translate

5.1 Introduction

The need for visualising data before performing any formal analysis is an essential part of the process of any data analysis. This step will give valuable insight into the data, aid in identifying anomalies, data cleaning, providing a framework for generating subjective impressions and in assessing the plausibility of assumptions needed for formal analysis. Visualising continuous variables is often achieved using box-plots, scatterplots and case profile plots for example when considering time-to-event data, a continuous response coupled with a censoring indicator, the challenges a plot of the survival function may pose were discussed in Chapter 1. A translational tool using all the survival summaries presented in Section 3.3 could be of value, and one such tool will be introduced in this chapter. In the sense of visualising the results of a statistical model, the use of nomograms as a graphical tool is popular. Nomograms play the role of a graphical '`predict`' function which facilitates the calculation of a point estimate of the response variable for a particular set of values of the explanatory variables in the corresponding model.

In this chapter, a visualisation tool will be introduced, as a Shiny application, to provide informative graphical summaries for time-to-event datasets. Following this, the use of dynamic nomograms as a visualisation and translational tool will be proposed to further aid the communication of the results of a statistical analysis to a non-statistical audience. The new **R** package `DynNom` for generating dynamic nomograms (as Shiny objects) will be presented for a variety of linear and non-linear models. Examples of their use will be given using the example datasets. A proposal will be made that advocates that models presented in the literature could have an accompanying dynamic nomogram.

5.2 Visualisation tool for time-to-event data

In Section 3.3, various summaries for time-to-event data were introduced and discussed. The survival function was reviewed as the most common (graphical) summary translating the survival information on a probabilistic scale. The use of dynamic prediction of survival and the mean residual life functions were discussed as useful complements or alternatives. It is reasonable to think of using all three summaries simultaneously for time-to-event data as each of them will provide a different and valuable insight into the time-to-event problem of interest.
As part of this research, a web-based interactive Shiny application has been created using the '`shinydashboard`' package introduced in Section 2.3.4. This visualisation application generates the collection of graphical summaries for time-to-event data mentioned here. A screenshot of this application applied to the BPD data is given in Figure 5.1.

The variable of interest in the BPD data is defined as the days that an infant required supplemental oxygen therapy. In Section 3.3.1, the KM estimate of the survival function was presented, where a sharper decreasing function was evident for those infants who were given the Surfactant treatment indicating a shorter time-to-cure compared to those who were not on the treatment. This difference was identified to be statistically significant using the logrank test.
Figure 5.1 shows that the survival times of two groups are fairly similar for about two months, but the effect of using treatment becomes evident then. This effect is also clear from the plot of the difference in survival functions where it is increasing over time for the first ten months, but starts decreasing after about 15 months as the infants in the non-treated group start to 'cure'. However, a careful interpreta-

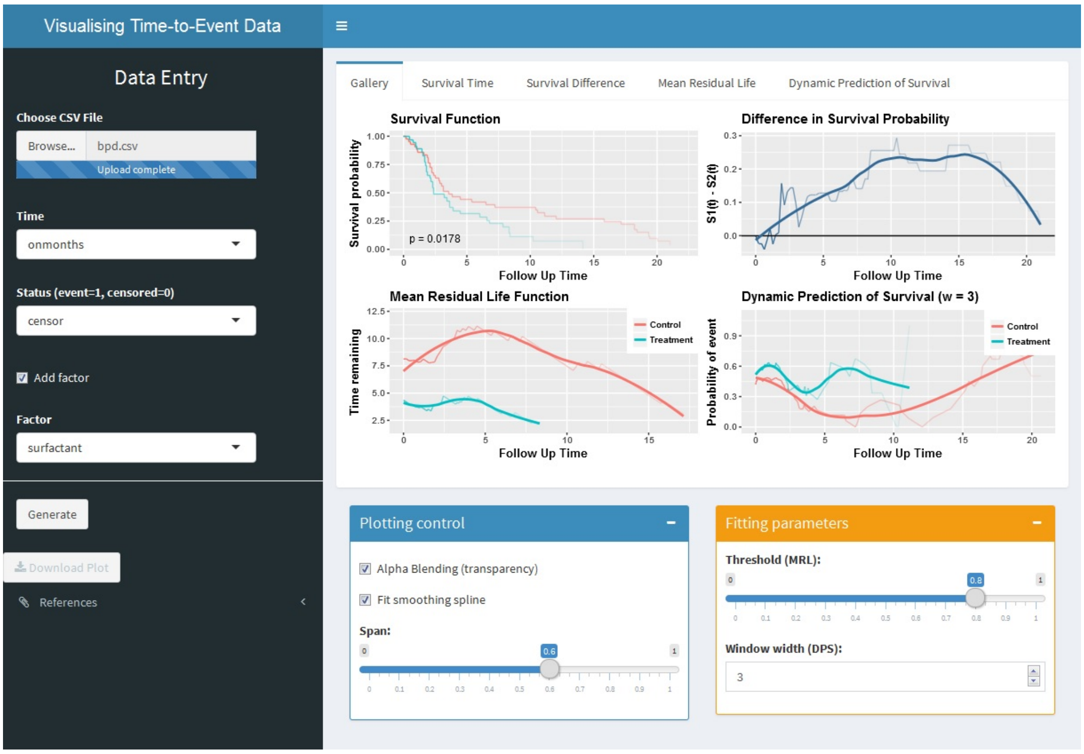

Figure 5.1: Screenshot of the visualisation summaries generated for the BPD data

tion is needed when the time on treatment exceeds one year as there is considerable uncertainty in the survival estimates, highlighted by the use of alpha-blending. The DPS plot is useful in order to project the prognosis of individuals after the start of follow-up. For example, conditional on surviving to a specific time, the probability of being 'off treatment' in the subsequent three months (shown in the bottom right) is always higher for infants in the treatment group. This difference between two groups is negligible at the start of follow-up, but it becomes more prominent as they remain on the therapy longer than about three months.

The MRL plot in Figure 5.1 is the only summary that expresses the difference between the treatments as units of time. At the start, the effect of the treatment is estimated to be about three months, i.e. the mean time-to-cure is four months for the treatment group compared to seven months for the controls. The time-to-cure remains roughly steady for the treatment group across time until five months where it starts to decreases quicker (i.e. those that were not cured up until then are cured more quickly). The equivalent time when this happens for the controls is after 17 months. In conclusion, the children on the Surfactant treatment tend to experience the event of interest (i.e. cure) quicker than those not on the treatment by about four months, on average. If a child on Surfactant has not been cured by eight months, the time-to-cure will be accelerated compared to those not on the treatment. To sum up, the treatment appears to work well, and its effect becomes more evident as infants remain on the therapy.

The survival data, including the pair of time and censoring variables, need to be uploaded in this application. The variables names will be identified and shown in the drop-down menus to identify the time and the censoring variables. A suite of graphical summaries including the estimates of survival function, DPS and MRL function will be created and shown in the dashboard body. Additionally, a plot of the difference in survival will also be included where the user can specify a factor variable (e.g. gender or treatment identifier) if desired. Two useful graphical parameters (smoothing and alpha-blending) are available in the blue box (in the bottom left-hand-side).

Additional display options are provided to allow for more flexible plotting. They include a numerical input menu to enter the window width of interest and a slider input to modify the threshold in the MRL estimation. The default threshold is set as a default to 0.8 of uncensored observations; however, it can be adjusted as needed. This useful translational tool provides an intuitive overview of the results of survival studies. It has been published and is available on https://amir-jalali.com/apps/survapp to use for any time-to-event study.

5.3 Nomograms

The field of nomography was invented by the French mathematician Maurice d'Ocagne in 1880 to provide engineers with fast graphical calculation tools for complicated formulas to a practical level of precision [9, 52]. An early use of nomograms is attributed to Brodetsky 1925 [24], where they were employed in mathematics for calculating elementary arithmetic, quadratic equations and trigonometrical functions. One of the earliest uses of nomograms in statistics was in the calculation of the coefficient of variation, interval estimates for a mean and the comparison of two means and variances [23]. Other examples of nomograms in this domain are the

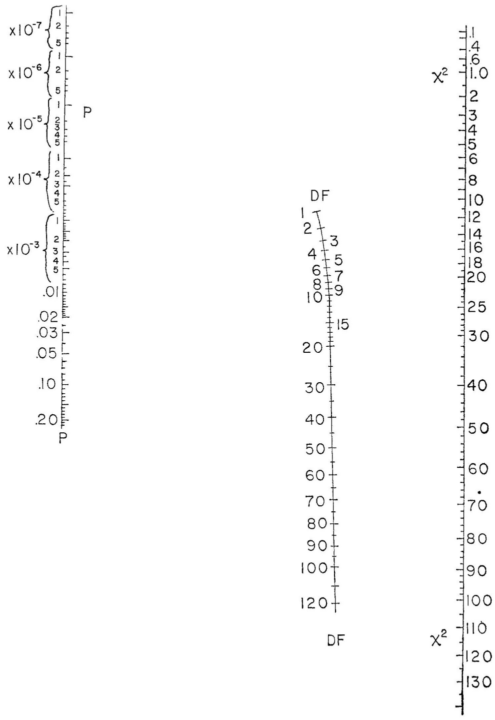

Figure 5.2: A nomogram to calculate the p-value for a chi-square test statistic [22]. To use the nomogram draw a straight line from the value of the chi-square test statistic through the required degrees to read off the corresponding p-value.

chi-square test nomogram (Figure 5.2), Altman's nomogram to calculate the sample size or power [10] and Fagan's nomogram [53] for the applications of Bayes's theorem. Fagan's nomogram, widely used in the context of diagnostic tests, was recently reproduced as the Helds nomogram for the calculation of post-test probabilities [74, 123].

Nomograms have been widely used in applications in science, engineering, the military, clinical research and epidemiology; see [24, 102]. The use of nomograms in clinical decision making includes prediction of prostate cancer [84, 129].
More recently, nomograms have been used to visualise statistical models. A nomogram generated from a model plays the role of a graphical '`predict`' function facilitating the calculation of a point estimate of the response variable for a particular set of values of the explanatory variables [92, 15]. It can also provide a visual representation of the relative effect sizes of each explanatory variable in the model.
Software is available to create nomograms for statistical models in SAS [148], Stata [154], python [45] and as online tools for constructing simple JAVA-based interac-

tive nomograms [81].

The rms package in **R** [73] includes the nomogram function to generate nomograms from a fitted statistical model. For example, the code below creates a nomogram from a logistic regression model used to model the relationships between age, gender and passenger class on surviving the Titanic disaster (data are included in the PASWR package [14]).

```
> library(PASWR)
> library(rms)
> data(titanic3)
> t.data <- datadist(titanic3)
> options(datadist = 't.data')
> fit <- lrm(survived ~ age + pclass + sex, data = titanic3)
> plot(nomogram(fit, fun = function(x)plogis(x)))
```

The resulting nomogram is given in Figure 5.3. The horizontal line at the top labelled 'points' allows the effect size of each variable to be assessed, including direct visualisation of the potential variation in the explanatory variable. The points measure the contribution of the values/levels of the explanatory variable of interest on the 'linear predictor'. The total points are then mapped from the 'linear predictor' to obtain the 'predicted value', which in this case is a probability. It is clear from the graph on top that all three explanatory variables make similar contributions to the linear predictor, over the spread of value/levels that each can take. For example, in the main effects model presented, the effect of gender on the probability of survival is comparable to a change in the age of 70 years. Such an insight is less obvious from the estimated coefficients. The model represented at the bottom in Figure 5.3 incorporates higher order interactions and suggests that effect of age is most prominent in 2^{nd} class males.

When modelling a binary outcome, it has been argued [39] that a summary quoting the underlying probabilities is more informative than one based on ratios of odds or probabilities. Despite this, the model summary is typically a table of odds ratios (and measures of uncertainty) for each explanatory variable. As it was discussed

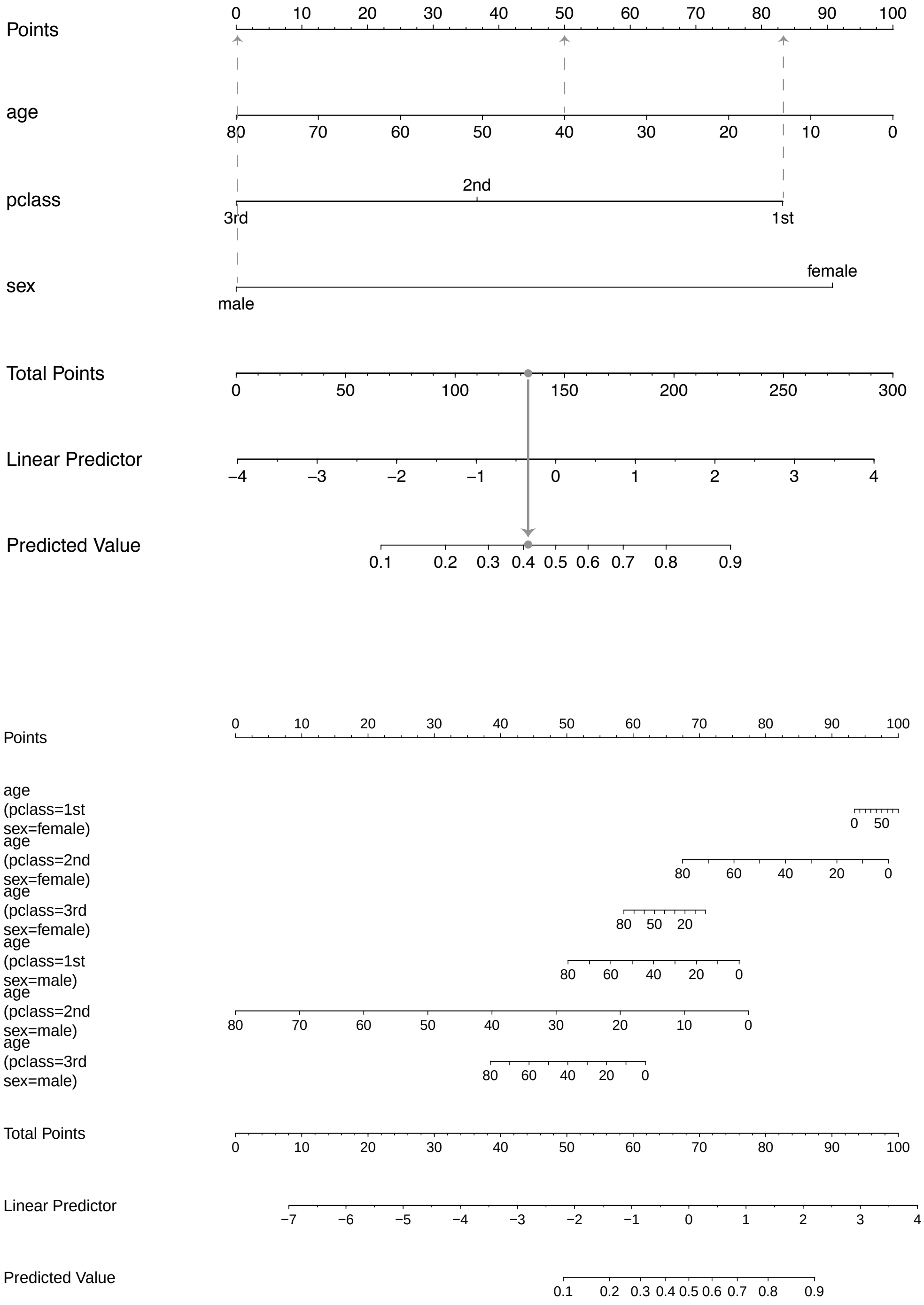

Figure 5.3: Nomograms for calculating the probability of surviving the Titanic disaster using the `rms` package. A main effects logistic regression model containing Age, Sex and passenger class as explanatories is displayed on top while a model including all two-way interactions is shown at the bottom.

in Section 3.2.1, the importance of an explanatory variable is often assessed (incorrectly) by the regression coefficient and the p-value which easily ignore the scale of the variable and the sample size.

Nomograms provide an attractive and useful graphical summary of a proposed model with the ability to make predictions as point estimates. They are a powerful graphical tool to visualise variable importance; however, such (static) nomograms can become cumbersome to generate precise prediction as models become more complex (e.g. higher order interactions) or the inclusion of smoothers [72]. One solution is to create dynamic nomograms as interactive applications to visualise statistical models such that variable effect sizes can be assessed graphically through predicted values with uncertainty estimates.

5.4 Dynamic nomograms

The emergence of the `rpanel` [21] and `Shiny` packages [28] for **R** allowed the creation of interactive, user-friendly graphical interfaces to be displayed either in a standalone window or delivered as a webpage.

The `DynNom` package [77] is built using `Shiny` to generate dynamic nomograms for any generalised linear model or proportional hazards models to allow a reader interact with the model in a user-friendly manner. For example, the **R** code below will create a dynamic nomogram, using a simple function, for a logistic regression model of the Titanic data.

```
> library(PASWR)
> library(DynNom)
> data(titanic3)
> fit <- glm(survived ~ (age + pclass + sex) ^ 3, titanic3,
+    family = "binomial")
> DynNom(fit, titanic3)
```

The dynamic nomogram displays sliders for covariates (bounded by their observed ranges) and drop-down boxes for the factors. The `predict` function maps the appro-

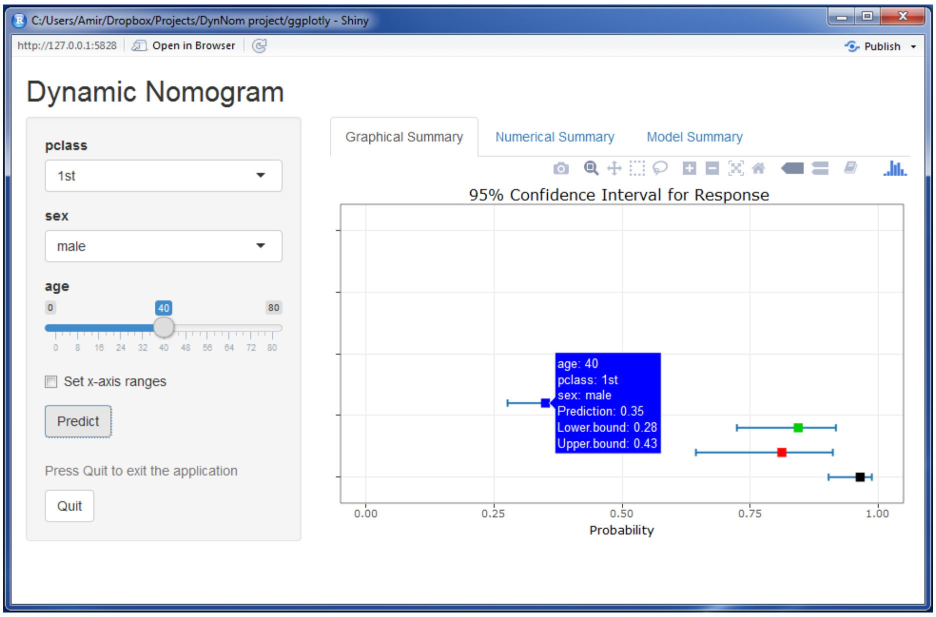

Figure 5.4: Dynamic nomogram created using `DynNom` for the Titanic data using the logistic model with all three-way interactions.

priate inverse link function ($\eta = g(\mu)$) from the generalised linear model object to generate predicted values (on the scale of the response variable) and corresponding 95% confidence intervals (see Figure 5.4). The predicted value and corresponding confidence interval are presented in the tooltip labels and the 'Numerical Summary' tab. Further, the usual model summary or a formatted model output is displayed in the 'Model Summary' tab.

Once the model is fitted, and parameters are estimated, the point estimation for the response variable based on a set of explanatory variables will be accessed using the estimated model equation. However, the predicted quantities need to be accompanied by measures of precision to be useful (e.g. confidence intervals) [97].

The `DynNom` package supports model objects created by the `lm`, `glm` and `coxph` functions in **R**. It also supports `ols`, `Glm`, `lrm` and `cph` models in the `rms` package which allow the inclusion of smoothing splines. When applied to a Cox proportional hazards model object, the estimated survival function will be displayed with either the accompanying confidence bands or alpha-blending (applied to the number of subjects at risk over time) to represent the underlying uncertainty of the estimates. Table 5.1 summarises all the model objects supported by `DynNom` which cover the wide variety of the most common models fitted to almost all typical types of response variables.

Response type		Method	R functions (package)
Discrete			
	Binary	Logistic regression	glm (stats), lrm (rms)
	Count data	Poisson regression	glm (stats), Glm (rms)
Continuous			
	Normal	Linear regression	lm (stats), ols (rms)
	Positive	Gamma regression	glm (stats), Glm (rms)
	Failure time	Cox proportional hazards	coxph (survival), cph (rms)

Table 5.1: **R** model objects (package) supported in `DynNom`

5.5 Examples

For illustration, dynamic nomograms for models arising from the sample datasets will be given. They include a Poisson response, a linear regression incorporating a smoothing spline for an explanatory and a Cox proportional hazards model for time-to-event data.

5.5.1 Count response

The 'Crabs' dataset [6] is used here as an example of visualising a Poisson regression model. A Poisson regression model has been used previously [6] to model the relationship between the width and colour on the number of male partners. The summary of the fitted Poisson model is given previously in Table 3.3, where the effect of Width and Dark are identified as significant (at the 5% level).
The accompanying dynamic nomogram (Figure 5.5) can be created as follows:

```
> library(glm2)
> library(DynNom)
> data(crabs)
> fit2 <- glm(Satellites ~ Width + Dark , family = poisson(log),
+     data = crabs)
> DynNom(fit2, crabs)
```

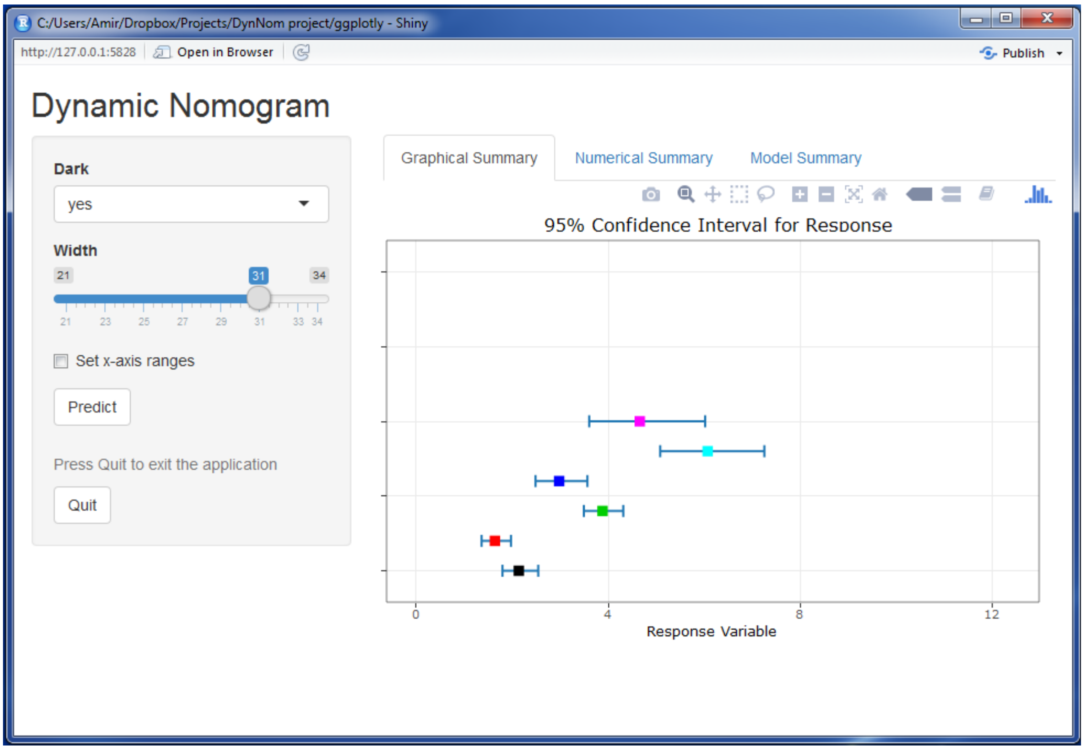

Figure 5.5: Screenshot of the 'Graphical Summary' tab of the dynamic nomogram for the Poisson regression model fitted to the Crabs data.

The estimates displayed in Figure 5.5 are for the predicted (mean) response for the two levels of crab colour and three values of width. The multiplicative effect of increasing age (and corresponding uncertainty estimates) on the response is evident which would be more difficult to glean from the model summary.

5.5.2 Continuous response with smoothing spline

Interpreting the role explanatory variables play can be difficult in a model when the functional form of an explanatory is modelled as a smooth function. For example, the 'Ragweed' dataset in Section 1.2.4 is an example where a regression model can be used for a continuous response incorporating a smoothing spline for the pollen season day ('day.in.seas') as it has a nonlinear relationship with the square root of ragweed level. The data are visualised and discussed in Section 2.2. The scatter-plot in Figure 5.6 shows the marginal relationship between the day in season and the square root of ragweed.

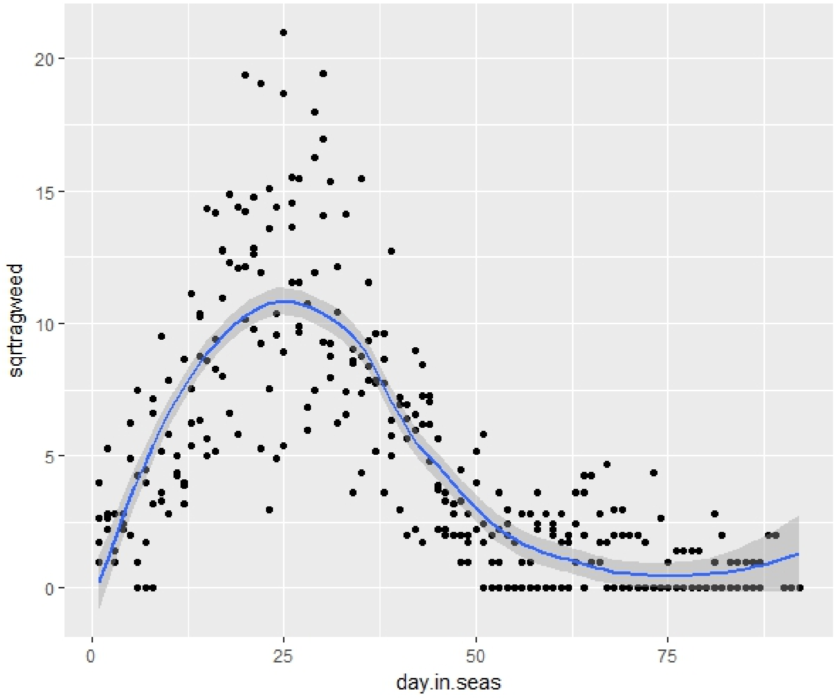

Figure 5.6: Relationships between the day in pollen season and the transformed response ($\sqrt{\texttt{Ragweed}}$)

The nonlinear relationship between the day in pollen season and the mean response is modelled using a restricted cubic spline function, with temperature, and wind speed as (linear) predictors and rain as a factor. The following code fits the model using the `ols` function in the `rms` package and the corresponding dynamic nomogram shown in Figure 5.7 is created.

```
> library(SemiPar)
> library(rms)
> library(DynNom)
> data(ragweed)
> ragweed$rain <- factor(ragweed$rain, 0:1, c('otherwise', '3 hours'))
> ragweed$sqrtragweed <- sqrt(ragweed$ragweed)
> dd <- datadist(ragweed)
> options(datadist = "dd")
> fit3 <- ols(sqrtragweed ~ rain + temperature + wind.speed +
+    rcs(day.in.seas, 5), data = ragweed)
> DynNom(fit3, ragweed)
```

The summary of the fitted model is given in Table 5.2. The need for a non-linear function for the day in season variable appears evident where the effect increases for up to 25 days in season and then decreases from that time point onwards. Interpreting the effect of this covariate through the model estimate, while adjusting for rain, temperature and wind speed, is hard to communicate.

	Estimate	Std. Error	t-value	p-value
Intercept	-9.4656	1.4014	-6.75	<0.0001
rain(otherwise)	-1.3593	0.4031	-3.37	0.0008
temperature	0.1003	0.0184	5.45	<0.0001
wind.speed	0.2431	0.0392	6.20	<0.0001
day.in.seas	0.6967	0.0352	19.78	<0.0001
day.in.seas'	-3.8046	0.2059	-18.48	<0.0001
day.in.seas"	9.3518	0.5999	15.59	<0.0001
day.in.seas"'	-7.8311	0.8159	-9.60	<0.0001

Table 5.2: The model summary for the Ragweed data

The effect of days in season is arguably more accessible using a dynamic nomogram (Figure 5.7) by choosing values of this covariate ranging from low to high and setting the remaining explanatory variables at their mode (for factors) and means (for covariates).

5.5.3 Time-to-event response

The classical summaries when modelling time-to-event data include plots of the estimated survival or hazard functions. When modelling the effect of covariates and factors on survival the Cox proportional hazards model is a popular choice as the underlying survival distribution does not need to be specified. The interpretation of the effect of explanatory variables on the 'risk' of the event of interest is often made using hazard ratios or by using a plot of the estimated survival function conditioning on the set of explanatory variables of interest. For example, consider the Cox proportional hazards model fitted to the Lung cancer data in Section 3.3.2 (the model summary is given in Table 3.5) and the corresponding dynamic nomogram which is created below to investigate the effect of these explanatory variables on time to death.

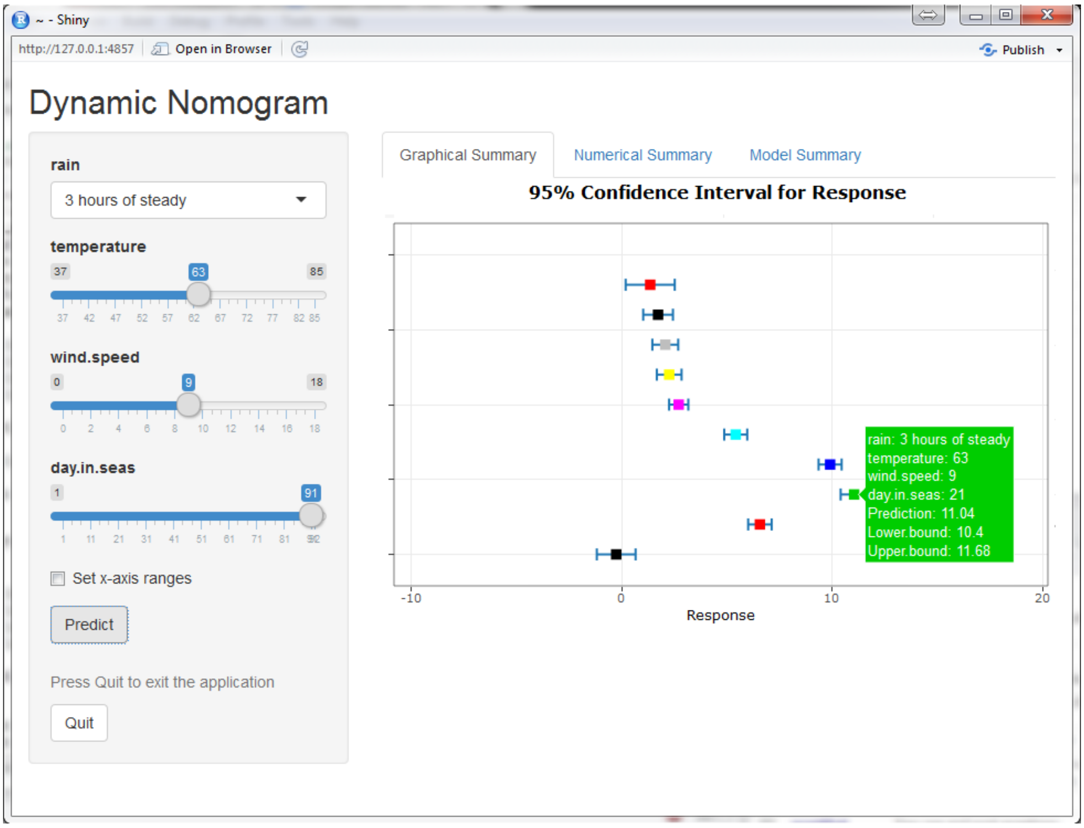

Figure 5.7: Dynamic nomogram for a Poisson regression model fitted to the Ragweed data. The plot displays the predicted response for day in season set at 1, 11, 21, 31, 41, 51, 61, 71, 81 and 91 respectively while setting when all other explanatory variables at the mode/mean as appropriate.

```
> library(survival)
> library(DynNom)
> data(lung)
> lung$sex <- factor(lung$sex, 1:2, c('male', 'female'))
> model <- coxph(Surv(time, status) ~ age + wt.loss + sex, data = lung)
> DynNom(model, lung)
```

The dynamic nomogram (Figure 5.8) displays the estimated survival function for a given set of values of the explanatory values. Presenting the results using estimated survival functions avoids the use of hazard ratios and interpretation of effects that act multiplicatively.

Rather than clutter the graph by including the corresponding interval estimates for each survival function, the degree of uncertainty in the estimate is highlighted

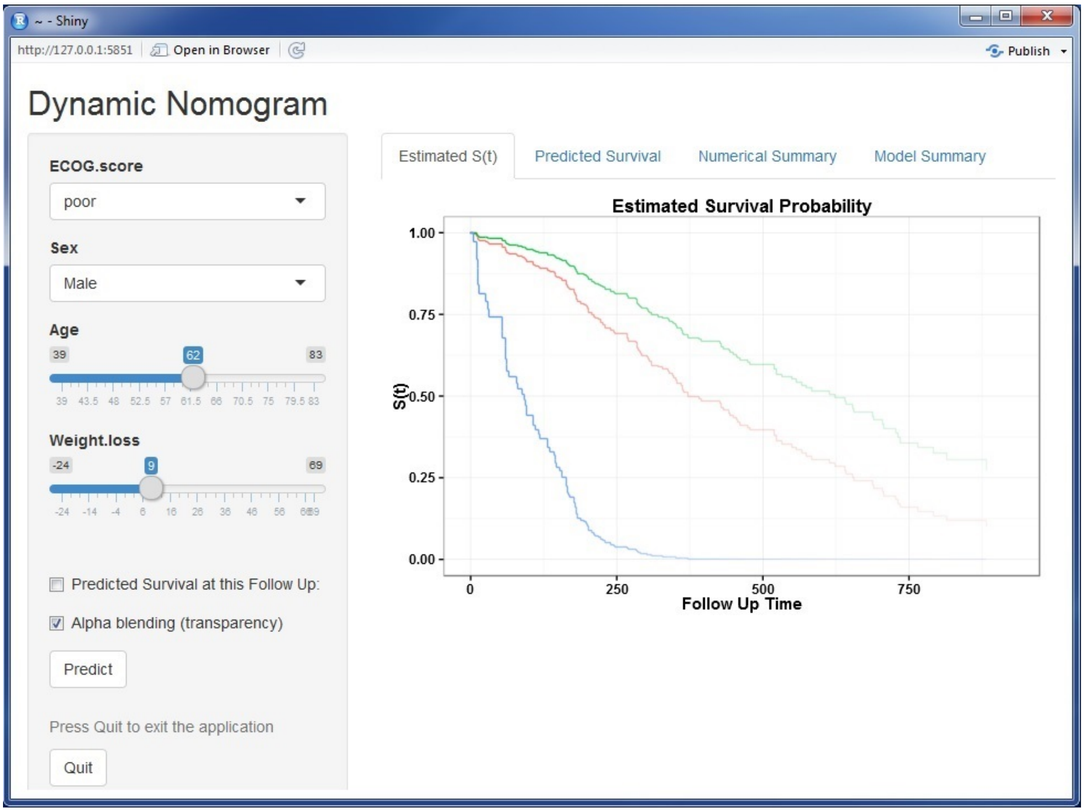

Figure 5.8: Dynamic nomogram for a Cox proportional hazards model using the Lung cancer data with alpha-blending.

graphically by adjusting the colour transparency (alpha-blending) to represent the number of subjects at risk over time. If required the predicted survival at any given follow-up time point can be generated also.

5.5.4 Publishing dynamic nomograms

Any dynamic nomogram created using the `DynNom` package can be displayed in an RStudio panel or a web browser as desired. In order to make a nomogram accessible to a wider audience the corresponding `ui.r`, `server.r` and `global.r` files needed to build the nomogram have to be uploaded to a server running `shiny` (e.g. https://www.shinyapps.io). To facilitate this, the `DNbuilder` function will generate the necessary files as illustrated below:

```
> library(PASWR)
> library(DynNom)
```

```
> data(titanic3)
> fit <- glm(survived ~ (age + pclass + sex) ^ 3, titanic3,
+     family = "binomial")
> DNbuilder(fit, titanic3)
```

The corresponding files can then be uploaded to an account which facilitates the deployment of Shiny applications. For example, the dynamic nomogram 'built' for the analysis of the Titanic data described above is available at https://amir-jalali.com/apps/dntitanic.

In theory, all models presented in the literature could have an accompanying web address to direct the reader to the corresponding dynamic nomogram allowing them to interact with the model and identify the exact nature of the model fitted. The model summary tab provides the (native) model summary from **R** allowing an examination of the precise model fitted and accompanying model summaries (e.g. R^2, AIC etc).

5.6 Summary

The use of appropriate visualisation plays a crucial role when translating raw data and the results of a data analysis. Examples of using modern software packages for visualising datasets have been illustrated in Chapter 2. Visualising time-to-event data and also the results of a statistical model are challenging problems. Accordingly, visualisation tools for these purposes as Shiny application and dynamic nomograms were created to use as translation tools.

The visualisation tool produces a collection of useful summaries for time-to-event data which depicts estimates of survival function, DPS function and MRL function. Although the empirical estimate of the MRL has been considered previously, this application is the first tool available to provide the MRL estimates using the estimator proposed by Alvarez et al. [11].

A dynamic nomogram is a useful visual representation of a statistical model and a translational tool for interacting with statistical models, regardless of the complexity of the model and assumed functional forms of the covariates. The importance of each variable, in terms of their effect size, becomes more evident through the assessment of the change in the predicted value. This is particularly useful for models containing modifiable risk factors.

As inferential statistical methods become more computational, the models arising are increasingly complex and are often hard to interpret and translate, in particular for non-statisticians. The dynamic nomogram tool created as part of this research is one such visual tool to help address this issue.

The tools presented in this thesis can be used to augment published research by including a hyperlink to a static graphic which directs the reader to a website where s/he can interact with the graph and create other views of the data not included in the original paper. The ability to accompany a model summary in a journal with a link to the corresponding dynamic nomogram has the potential to make all results more translational and all analyses more transparent as statistical best practice will be followed in terms of fulfilling the recommendations of making all research understandable and reproducible.

Chapter 6

Conclusions and Further Work

The body of this thesis was organised to mirror the stages of data analysis (post-design), i.e. visualise, model and translate.

The main aim of this thesis is to introduce the concept of 'translational statistics' and develop a framework by presenting new numerical, graphical and model summary tools to aid in translation. The use of the mean residual life function in the analysis of time-to-event data was a particular focus. In particular, methods to estimate of the uncertainty of the MRL estimator were presented and discussed.

The concept of translational medicine was introduced in Chapter 1 which aims to promote and advance the discovery of new diagnostic tools and treatments using a multidisciplinary, highly collaborative 'bench to bedside' approach. The idea of translational statistics was introduced to facilitate and enhance the integration of statistics within various disciplines by developing tools to communicate research findings in an accurate and accessible manner. Translational statistics aims not only to ensure that statistical best practice is adhered to but also to ensure that the results of the statistical analysis undertaken, regardless of its complexity, are understandable to a variety of stakeholders. The methods and translational tools proposed in this thesis are illustrated through the use of six datasets from a variety of research settings.

In Chapter 2, we briefly described the structure of static graphics using the Wilkinson's framework [147]. We paid particular attention to visualising time-to-event data and provide simple extensions to plots of the estimate of the survival function through the use of alpha-blending to highlight the uncertainty in the estimate. Interactive graphics were introduced as an alternative visualisation strategy to static graphics. Examples of the use of modern methods of generating dynamic graphics were given.

The necessity of using static graphics because of the reliance on the medium of print to disseminate results is undeniable at the current time. However, when moving to online makes increasingly more plausible to use interactive graphics, in particular, using marginal graphics and 'slicing and dicing' which are crucial to understanding big data. Modern web-based computing allows the simple development of interactive and dynamic tools for communicating and exploring research findings.

In Chapter 3, statistical modelling using the general linear modelling framework was introduced with a particular emphasis on interval estimation; examples of models for binary and Poisson responses presented. The remainder of the chapter was dedicated to modelling time-to-event data. Survival analysis is considered one of the oldest applications in statistics. Despite this, limited advances and attention has been paid to alternative summaries (numerical and graphical) for time-to-event responses. The key question of interest in studies where death is the outcome of interest is invariably "how much time do I have left?". The classical summary provided is typically an estimate of the survival function where an estimate of the probability of surviving beyond a particular time is reported rather than an estimate of the likely (e.g. mean) time remaining. Other summaries such as dynamic prediction for survival and residual life functions have been introduced and illustrated in Section 3.3.3 and Section 3.3.4 as valuable and informative alternatives.

In particular, the MRL is a promising summary of time-to-event data as the summary is provided in units of time rather than as a probability. Recent advances (such as Alvarez et al. [11]) have provided novel hybrid estimators of the MRL function and extensions to this method to generate uncertainty bands were explored in this thesis taking into account the complexity arising from the nature of hybrid estimators. Methods to provide a closed-form solution for this type of estimator were investigated in the literature and concerns regarding the corresponding restrictive assumptions discussed (see Gong [61]).

In Chapter 4, bootstrap approaches were introduced and compared as a reasonable alternative to compute uncertainty estimates for the MRL. From the results of the simulation study presented, the use of the percentile bootstrap technique was identified as a useful approach as it achieves adequate coverage and delivered adequate performance across a range of sample size and percentage censoring scenarios.

In Chapter 5, new tools for visualising time-to-event data, using all of the methods presented, were introduced. For some reason, the uptake of these approaches for summarising time-to-event data has been limited despite the usefulness of the summaries provided. It is hoped that the tools made available arising from this research (such as the visualisation application presented in Section 5.2) may address this by making them more accessible in a user-friendlier manner.

The standard way of representing a statistical model is often as a summary table, where the importance of explanatory variables is assessed by the magnitude of the estimated coefficients (which is sensitive to the covariate's scale) or by the p-value (which is sensitive to the sample size). It is relatively straightforward for statisticians to understand the model output, however, interpreting a model may be challenging for non-statisticians. One solution is to represent the model visually as nomograms as it provides an attractive graphical model summary and gives an indication of variable importance. (Static or dynamic) nomogram is a tool for model visualisation, not for model specification, and the responsibility is still on statistician to fit the most appropriate model which deals with overfitting, multicollinearity and variable selection for example. The use of (static) nomograms, however, can be cumbersome especially for 'complex' models involving higher order interactions and smoothing terms. For this reason, we propose the use of dynamic nomograms as a novel intuitive method to visualise statistical models. It appears to be a helpful tool based on user feedback and the number of package downloads (more than 13K downloaded by February 2108).

For example, the study on the risk of caesarean section was one of the examples introduced at the beginning of Chapter 1. The following succinct **R** code is all that is needed to produce the required dynamic nomogram along with the model summary:

```
> library(DynNom)
> diabetes <- read.csv("Diabetes.csv", header=T)
> fit <- glm(DiabeticType ~ Age + BMI + BabySex + PET + PIH,
```

```
+    data=diabetes,family=binomial)
> DynNom(fit, diabetes)
```

This nomogram can be embedded in the online version of the corresponding scientific publication and encourage authors to add a URLs directing the reader to the nomogram would undoubtedly aid in model translation.

From the academic, government, and industrial perspectives, the final result might be interactive graphics and dynamic nomograms (predictive models). Interactive graphics can be embedded in the online version of scientific publications as the medium is available now. Nowadays, users no longer have to buy their own domain necessarily, the introduction of the new cloud website such as 'RPubs', 'Shiny Server' or 'Tableau Public', provide an effortless way to publish web applications freely. Consequently, the journal policy can be modified in the way that they could encourage authors to submit the URLs which aid attractive and informative data visualisation (interactive graphics) and visualise the final model (dynamic nomogram). This allows users to slice-and-dice data to make easy comparisons or explore the final model and look into the role of each variable plays on the primary response.

6.1 Further work

The new ideas proposed in this thesis encompass two domains; modelling and communicating (in the means of visualisation and translation).

The need to use interactive graphics for data visualisation is well-discussed in Chapter 2. The introduction of user-friendly tools such as the `shiny` package and Tableau along with free cloud servers make it easier to create and share dynamic visualisations (e.g. dashboards). An example is presented in Section 2.3 using 'flexdashboard', 'shiny' and 'shinydashboard' to visualise data that are typical in a clinical trial involving the comparison of a change score over time between a treatment and control group.

An interesting extension in this context would be to create a Shiny application which allows a dataset to be uploaded and the 'ui' and 'server' files needed to generate a Shiny application similar to one presented for the clinical trial data are returned.

The variables in the uploaded data would be classified as categorical or continuous and where the user can divide them into primary or explanatory variables through the use of checkboxes. A selection of graphs can then be displayed once the roles of each variable are defined (e.g. scatterplots of the response by a covariate, labelled or sized by a factor or a boxplot of the response by treatment group). The Shiny app essentially creates another Shiny app (without knowledge of **R**), based on the information provided, which can then be used as a graphical tool to embed in an electronic publication for readers to interact with the data as required.

The graphics are best created using the ggplot2 framework as each element of graphics such as the geometric object (graphic type), data, mapping, statistical transformation and position adjustment can be specified and modified easily. The `ggplotly` function in the `plotly` package [126] can be incorporated to add interactivity into the graphics.

This user-friendly tool could be used by any researcher who needs to visualise data and share it on the web as part of the visualisation step of any data analysis.

The current version of `DynNom` package supports `lm`, `glm`, `coxph`, `ols`, `Glm`, `lrm` and `cph` model objects to generate dynamic nomograms. This means that it supports linear, generalized linear and Cox proportional hazards models, where the non-linearity structure is also allowed in models through smoothing for covariates. The natural extension in this context would be to extend the package to take a broader set of statistical model objects. They could classify as:

- Linear mixed models by `lmer` and `glmer` functions (`lme4` package) and `lme` function (`nlme` packages),
- Non-linear mixed models by the `nls` function (`stats` package), `nlmer` function (`lme4` package) and `nlme` function (`nlme` package),
- Generalized linear models with LASSO by `glmnet` function (`glmnet` package),
- Generalized additive models by `gam`, `gamm`, `mvn` and `cox.ph` functions (`gam` package),
- Parametric regression survival models by `servreg` function (`survival` package),

- Regression tree models by `tree function` (`tree` package), `rpart` function (`rpart` package), `tree` and `ctree` functions (`party` package) and `randomForest` function (`randomForest` package).

Adding these model objects would greatly enhance the package and is work in progress.

Further work in the area of time to event data could focus on the use of the median residual life function as another alternative (graphical) summary measure.

A review of recent developments in the use of the median (or any quantile) residual life (MDRL) function is given by Jeong [79]. The MDRL function also provides survival information in units of time unit where it is defined as the median additional time-to-event conditional on being event free at time t [125]. In other words, it shows the time that half of the population is expected to have remaining given surviving beyond a defined time. The MDRL function at time t defined by:

$$md(t) = Med(T - t|T > t) = S^{-1}[\frac{1}{2}S(t)] - t \tag{6.1}$$

The non-parametric estimate of the MDRL in the presence of censoring is straightforward by using the inverse survival formula in (6.1). However, this estimation is problematic in the case that there is a large amount of Type I censoring as at least half of those survived beyond time t must have the event time recorded.

The overall MDRL estimate for the lung cancer data in (1.2.3) is displayed in Figure 6.1 using the **R** function written to estimate the MDRL given in A.3. It shows a roughly similar trend to the MRL estimate in Figure 3.12 but shorter time expected where the median time remaining is about ten months at the start of follow-up. The median remaining lifetime decreases to about eight months if a patient is still alive after the first year of cancer. Surprisingly, the median remaining time shows a minor increase at the end of follow-up time. This increment is likely to be spurious and due to the smoothing and, as always, careful interpretation is needed when estimating the tail when little information available for this period.

The graphical summaries of the median residual lifetime for the lung cancer data grouped by gender or the between-group difference plot $(\hat{md}_f(t) - \hat{md}_m(t))$ are given

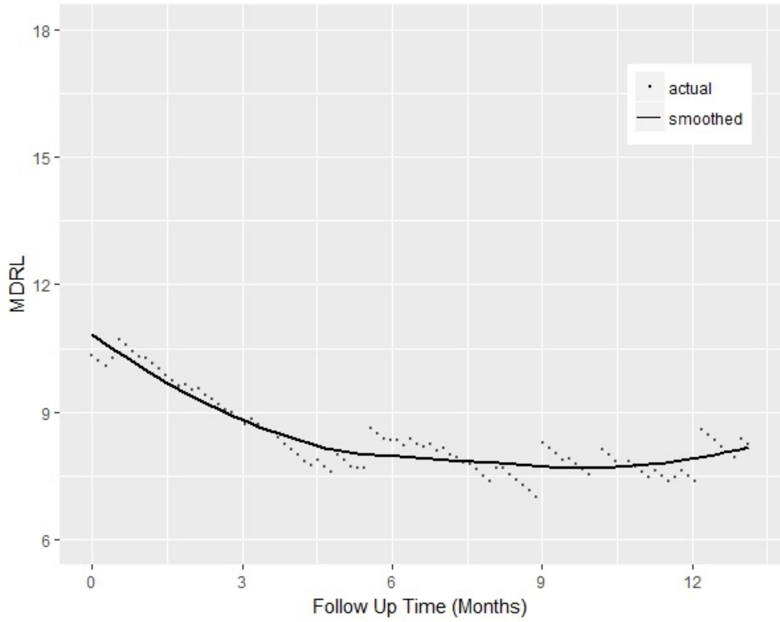

Figure 6.1: The MDRL estimates for the lung cancer data; points represent the actual estimations, and the line represents a smoothed estimation line

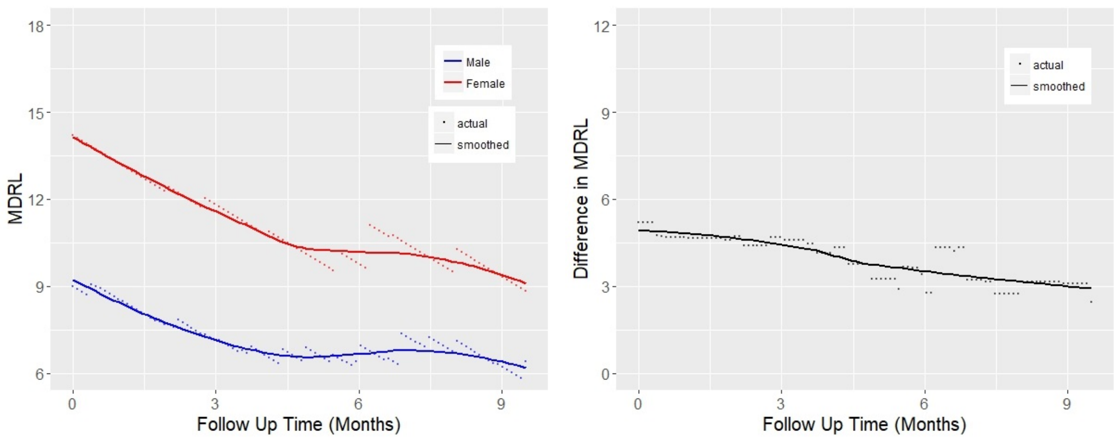

Figure 6.2: The MDRL estimates of the lung cancer data grouped by gender at left, and the estimate of MDRL difference at right; points represent the actual estimations, and lines represent smoothed estimation lines

in Figure 6.2. Once again both graphs suggest that women have better survival prospects. This difference, in units of time, is approximately five months at the start which will decrease to about three months for patients who still alive after about nine months.

The median residual life function does not uniquely characterise the time distribution while the mean residual life function does [80, 66]. As mentioned earlier it is not always possible to estimate the MDRL in the present of a large censoring proportion. Therefore, the main development in this area would be to develop an estimator which could incorporate Type-I censoring. One potential approach is to extrapolate the missing tail using the sample approach (based on extreme value theory) as proposed Alvarez et al. [11].

Recently, a novel method to impute right-censored observations using (parametric and non-parametric) Bayesian methods has been proposed by Moghaddam et al. [101]. By imputing censored observations, a pseudo-continuous response is generated for each censored observation allowing relevant and informative classical graphical and numerical summaries (e.g. density plots) to be used. A new method of estimating MRL and MDRL functions could be developed using Bayesian imputation as having 'complete' data would reduce the complexity in estimating a survival function in the presence of Type I censoring.

The code was written and is available in Appendix to estimate the MRL and MDRL functions in **R**; the development of an **R** package to implement these functions is ongoing.

Appendix A

A.1 R function for alpha-blending

The R code below is used to create the function 'dat.alpha'. This function receives time and censoring variables to fit a survival curve and return a data frame of the model summary including the number of subject at risk variable. This variable can be used later to make a survival curve with alpha-blending.In the case of giving a factor variable, function fits separate survival curves for each level of the factor.

```
dat.alpha <- function(times, status, factor=NULL){
if(is.null(factor)){
dat <- data.frame(times=times, status=status)
fit1<-survfit(Surv(times, status)~1, data=dat)
s.f0 <- as.data.frame(summary(fit1)[2:8])
if (s.f0$time[1] != 0){
sff2 <- s.f0[1, ]
sff2$time <- 0
sff2$n.risk <- fit1$n
sff2$surv <- 1
s.f <- rbind(sff2, s.f0)
}} else {
dat <- data.frame(times=times, status=status, factor=factor)
f.levels=levels(as.factor(factor))
```

```
s.f <- data.frame()
a <- 0
for (i in 1:length(f.levels)){
dat2 <- dat[dat$factor==f.levels[i],]
fit1 <- survfit(Surv(times, status)~1, data=dat2)
ss <- as.data.frame(summary(fit1)[2:8])
s.f0 <- cbind(ss, Factor=f.levels[i])
ind <- 1
if (a!=0){
ind=dim(s.f)[1] + 1
}
if (s.f0$time[1] != 0){
sff2 <- s.f0[ind, ]
sff2$time <- 0
sff2$n.risk <- fit1$n
sff2$surv <- 1
sff2$Factor <- f.levels[i]
s.f1 <- rbind(sff2, s.f0)
} else {
s.f1 <- rbind(s.f, cbind(ss, Factor=f.levels[i]))
}
s.f <- rbind(s.f, s.f1)
a <- a + 1
}}
return(s.f)
}
```

A.2 R function for estimating MRL

The R code below is used to create the function ‘mrl’. This function takes time, censoring status an arbitrary grid of time to return the estimation of MRL for the given time grids using the method introduced in Section 3.3.4.

```
mrl <- function(times, status, grid){
y <- data.frame(times=times, status=status)
fit1 <- survfit(Surv(times, status)~1)
KMarea <- function(fit, u){
if(sum(fit$time<=u, na.rm=T)==0){
survInU <- 1
} else {
survInU <- min(fit$surv[fit$time<=u])
}
X <- c(u, fit$time[fit$time>u])
Y <- c(survInU, fit$surv[fit$time>u])
sum((X[2:length(X)]-X[1:(length(X)-1)])*Y[1:(length(Y)-1)])
}
llik <- function(pa, y){
si <- rep(pa[1], dim(y)[1])
sh <- rep(pa[2], dim(y)[1])
if(any(1+sh*y$times/si <= 0) | si[1]<=0 | sh[1]>=0.2){
outp <- - 10 ^ 7
} else {
outp <- sum(y$status * log(1/si * (1 + sh * y$times / si)
^ ( - (1 / sh + 1))) + (1 - y$status)
* log((1 + sh * y$times / si) ^ ( - 1 / sh)))
}
-outp
}
Medi <- sort(fit1$time)[length(fit1$time) * 0.8]
or <- order(y$times, decreasing=FALSE)
y <- y[or,]
y <- y[y$times>=Medi,]
if(!length(y$times)==0){
u <- min(y$times, na.rm=TRUE)
```

```
y$times<-y$times-u
aa <- optim(c(0.1,0.1), llik, y=y, hessian = TRUE,
method = "Nelder-Mead", control=list(maxit=10000))
mlesi <- aa$par[1]
mlesh <- aa$par[2]
survInU <- fit1$surv[sum(fit1$time<=u)]
mlemu <- mlesi/(1-mlesh)
}
mrl_new <- function(ti){
if(sum(!fit1$time>=ti)==0){
survInX <- 1
} else {
survInX <- fit1$surv[sum(fit1$time<ti)]
}
(KMarea(fit1, ti) - KMarea(fit1, u) + mlemu*survInU) / survInX
}
mrlys <- 0
mrlxs <- grid
if (!length(y$times)==0){
mrlys <- sapply(mrlxs, mrl_new)
}
mrlys
}
```

A.3 R function for estimating MDRL

The R code below is used to create the function ‘mdrl’. This function takes time, censoring status an arbitrary grid of time to return the estimation of MDRL for the given time grids using the formula 6.1.

```
mdrl <- function(times, status, grid){
ngrid <- length(grid)
```

```
e.mdrl <- numeric(ngrid)
for (i in 1:ngrid){
t0 <- grid[i]
dat0 <- dat[times>t0,]
dat0$times <- times - t0
surv.t0 <- survfit(Surv(times, status)~1)
e.mdrl[i] <- summary(surv.t0)$table[[7]]
}
e.mdrl
}
```

A.4 Shiny demo R codes

The following R code is used to create the shiny application presented in Section 2.3.4.

```
library(shiny)
library(shinydashboard)
library(dplyr)
library(plotly)
dat <- read.csv("Continuous_response_RCT.csv", header=T)

## ui.R ##
ui <- shinyUI(fluidPage(
h2("Interactive graphic demo"),
fluidRow(sidebarLayout(
  sidebarPanel(selectInput('yaxis', 'Yaxis (continuous)',
    list("Primary response"=c("Response","Change.Score"),
    "Covariates"=c("Baseline","Age","BMI","Adherence","Resting.Pulse")),
    multiple=FALSE, selected="Response"),
  selectInput('xaxis', 'Xaxis (continuous)',
    list("Primary response"=c("Response","Change.Score"),
```

```
    "Covariates"=c("Baseline","Age","BMI","Adherence","Resting.Pulse")),
    multiple=FALSE, selected="Baseline"),
  checkboxInput('col', 'Colour', value=TRUE),
  conditionalPanel(condition='input.col==true',
    selectInput('factor', 'Colour Variable',
    list("Factors"=c("ID","Group","Smoker","Sex"),
      "Covariates"=c("Baseline","Age","BMI","Adherence")),
      multiple=FALSE, selected="Group")
  ),
  checkboxInput('size', 'Size'),
  conditionalPanel(condition='input.size==true',
    selectInput('cov', 'Size Variable',
    list("Factors"=c("ID","Group","Smoker","Sex"),
    "Covariates"=c("Baseline","Age","BMI","Adherence")),
    multiple=FALSE, selected="BMI")
  ),
  checkboxInput('panel', 'Panel (factor)'),
  conditionalPanel(condition='input.panel==true',
    selectInput('factor2', 'Panel', c("Group","Smoker","Sex"),
    multiple=FALSE, selected="Smoker")
  ),
  br(),
  checkboxInput("addsmooth", label="Add smoothing", value=FALSE),
  checkboxInput("addline", label="Add line of equality", value=TRUE)
),
mainPanel(plotlyOutput("plot0")
)))))

## server.R ##
server <- shinyServer(function(input, output){
  dData <- reactive({
    neVar <- c(input$yaxis, input$xaxis)
```

```
    if (input$col==TRUE){
      neVar <- c(neVar, input$factor)
    }
    if (input$size==TRUE){
      neVar <- c(neVar, input$cov)
    }
    if (input$panel==TRUE){
      neVar <- c(neVar, input$factor2)
    }
  dat[, neVar]
})
output$plot0 <- renderPlotly({
  p0 <- ggplot(dData(), aes_string(input$xaxis, input$yaxis)) +
  labs(x=input$xaxis, y=input$yaxis)
  if (input$col==TRUE){
    if (input$size==TRUE){
    p0 <- p0 + geom_point(aes_string(colour=input$factor,
      size=input$cov),
    labeller=labeller(.rows="label_both", .cols="label_value")) +
    guides(colour=guide_legend(title=NULL),
    size=guide_legend(title=NULL))
    } else {
    p0 <- p0 + geom_point(aes_string(colour=input$factor)) +
    guides(colour=guide_legend(title=NULL))
  }} else {
    if (input$size==TRUE){
    p0 <- p0 + geom_point(aes(size=dData()[,3])) +
    guides(size=guide_legend(title=NULL))
    } else {
    p0 <- p0 + geom_point()
  }}
  if (input$panel==TRUE){
```

```
    dimd <- ncol(dData())
    p0 <- p0 + facet_grid(.~dData()[,dimd])
  }
  if (input$addline){
    p0 <- p0 + geom_abline(aes(intercept=0, slope=1))
  }
  if (input$addsmooth){
    p0 <- p0 + geom_smooth()
  }
p0
})})

### run app
runApp(list(ui=ui, server=server))
```

A.5 Shinydashboard demo R codes

The following R code is used to create the shinydashboard application presented in Section 2.3.4.

```
library(shiny)
library(shinydashboard)
library(dplyr)
library(plotly)
dat <- read.csv("Continuous_response_RCT.csv", header=T)

## ui.R ##
ui = dashboardPage(
dashboardHeader(title="Shinydashboard demo", titleWidth=360),
dashboardSidebar(width=360, sidebarMenu(
  selectInput('yaxis', 'Yaxis (continuous)',
    list("Primary response"=c("Response","Change.Score"),
```

```
    "Covariates"=c("Baseline","Age","BMI","Adherence","Resting.Pulse")),
    multiple=FALSE, selected="Response"),
  selectInput('xaxis', 'Xaxis (continuous)',
    list("Primary response"=c("Response","Change.Score"),
    "Covariates"=c("Baseline","Age","BMI","Adherence","Resting.Pulse")),
    multiple=FALSE, selected="Baseline"),
  checkboxInput('col', 'Colour', value=TRUE),
  conditionalPanel(condition='input.col==true',
    selectInput('factor', 'Colour Variable',
    list("Factors"=c("ID","Group","Smoker","Sex"),
    "Covariates"=c("Baseline","Age","BMI","Adherence")),
    multiple=FALSE, selected="Group")
  ),
  checkboxInput('size', 'Size'),
  conditionalPanel(condition='input.size==true',
    selectInput('cov', 'Size Variable',
    list("Factors"=c("ID","Group","Smoker","Sex"),
    "Covariates"=c("Baseline","Age","BMI","Adherence")),
    multiple=FALSE, selected="BMI")
  ),
  checkboxInput('panel', 'Panel (factor)'),
  conditionalPanel(condition='input.panel==true',
  selectInput('factor2', 'Panel', c("Group","Smoker","Sex"),
  multiple=FALSE, selected="Smoker")
))),
dashboardBody(
  plotlyOutput("plot0"),
  box(title="Plotting control", status="primary",
    solidHeader=TRUE, collapsible=TRUE,
    checkboxInput("addsmooth", label="Add smoothing", value=FALSE),
    checkboxInput("addline", label="Add line of equality",
    value=TRUE)
```

```
)))


## server.R ##
server = shinyServer(function(input, output){
dData <- reactive({
  neVar <- c(input$yaxis, input$xaxis)
  if (input$col==TRUE){
    neVar <- c(neVar, input$factor)
  }
  if (input$size==TRUE){
    neVar <- c(neVar, input$cov)
  }
  if (input$panel==TRUE){
    neVar <- c(neVar, input$factor2)
  }
  dat[, neVar]
})
output$plot0 <- renderPlotly({
  p0 <- ggplot(dData(), aes_string(input$xaxis, input$yaxis)) +
  labs(x=input$xaxis, y=input$yaxis)
  if (input$col==TRUE){
    if (input$size==TRUE){
      p0 <- p0 + geom_point(aes_string(colour=input$factor,
        size=input$cov),
      labeller=labeller(.rows="label_both", .cols="label_value")) +
      guides(colour=guide_legend(title=NULL),
      size=guide_legend(title=NULL))
    } else {
      p0 <- p0 + geom_point(aes_string(colour=input$factor)) +
      guides(colour=guide_legend(title=NULL))
  }} else {
  if (input$size==TRUE){
```

```
      p0 <- p0 + geom_point(aes(size=dData()[,3])) +
      guides(size=guide_legend(title=NULL))
    } else {
      p0 <- p0 + geom_point()
  }}
  if (input$panel==TRUE){
    dimd <- ncol(dData())
    p0 <- p0 + facet_grid(.~dData()[,dimd])
  }
  if (input$addline){
    p0 <- p0 + geom_abline(aes(intercept=0, slope=1))
  }
  if (input$addsmooth){
    p0 <- p0 + geom_smooth()
  }
p0
})})

### run app
shinyApp(ui, server)
```

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